Leipzig University

Master Thesis

Integrated local energy decay estimates for solutions to the wave equation in the black hole exterior of sub-extremal Reissner-Nordström–de Sitter spacetimes

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Declaration of Authorship

I, Inés Vega González, declare that this thesis titled, "Integrated local energy decay estimates for solutions to the wave equation in the black hole exterior of sub-extremal Reissner-Nordström–de Sitter spacetimes" and the work presented in it are my own. I confirm that:

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Abstract

Institute of Theoretical Physics

Faculty of Physics

Master of Science

Integrated local energy decay estimates for solutions to the wave equation in the black hole exterior of sub-extremal Reissner-Nordström–de Sitter spacetimes

by Inés Vega González

We use a physical-space-based method for deriving the leading-order late-time behaviour of integrated local energy decay estimates of solutions to geometric wave equations on asymptotically flat, spherically symmetric spacetime backgrounds and apply it to the setting of wave equations with Klein-Gordon potentials on sub-extremal Reissner-Nordström–de Sitter (RNdS) black holes. This estimates could be used for deriving the precise leading-order late-time behaviour of asymptotics and energy decay. The method we use relies on exploiting the spatial decay properties of time integrals of solutions to derive the existence and precise genericity properties of energy of the solutions and obtain uniform decay estimates of local energy in time. This provides a toy model for studying how perturbations of sub-extremal RNdS initial data evolve in the context of the Einstein equations, which is related to the question of stability of the spacetime.

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Chapter 1

Introduction

Black holes are one of the fundamental predictions of general relativity. They are fascinating physical phenomena that has been studied by physicist and mathematicians for decades, using a wide range of methods and approaches. However, they still pose a variety of conceptual questions. This thesis intends to introduce the black hole concept through the analysis of waves on black hole backgrounds $(\mathbb{M}, g)$, by means of the example of the scalar wave equation:

$$\Box_g \phi = 0, \tag{1.1}$$

with $\Box_g$ denoting the geometric wave operator with respect to the sub-extremal Reissner-Nordström spacetime metric with $M > 0$ and a charge $Q \neq 0$, in a de Sitter background.

The *Reissner–Nordström–de Sitter family of spacetimes (RNdS)*, are the spherically symmetric solutions of the Einstein–Maxwell field equations in the presence of a positive cosmological constant Λ, i.e. they solve

$$G_{\nu\varrho} + \Lambda g_{\nu\varrho} = 8\pi \left(\mathcal{E}_{\nu\varrho} + T_{\nu\varrho}\right),$$
$$\nabla_\nu F^{\nu\varrho} = -4\pi J^\varrho,$$

with $T_{\nu\varrho} = J^\nu = 0$, and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic tensor, with A being the electromagnetic four-potential. Here, J^ν is the charge current of the matter and the stress-energy tensor of the Maxwell field is given by

$$\mathcal{E}_{\nu\varrho} = \frac{1}{4\pi}\left(F_{\nu\alpha}F_\varrho{}^\alpha - \frac{1}{4} g_{\nu\varrho} F_{\alpha\beta}F^{\alpha\beta}\right).$$

This solution models a non-rotating spherically symmetric charged black hole with mass $M > 0$ and a charge $Q \neq 0$, in a de Sitter background. A Reissner-Nordström (RN) spacetime can be divided in three different kinds, depending on the relation between the parameters M and Q: if $Q^2 < M^2$, the spacetime is called *sub-extremal*, if $Q^2 = M^2$, *extremal*, and if $Q^2 > M^2$, *super-extremal*. In this work, we are going to consider a sub-extremal spacetime.

There is already a rich body of heuristic work on this problem in the physics literature. We will follow the work of (Gajic, 2023), calculating integrated energy estimates for Schwarzschild spacetimes, (Angelopoulos, Aretakis, and Gajic, 2023), for sub-extremal Reissner–Nordström spacetimes, and (Dafermos and Rodnianski, 2007), for Schwarzschild-de Sitter spacetimes. The goal of the project is to follow their steps and fill the small gap in the literature doing part of the analysis for sub-extremal RNdS spacetimes.

The motivation for this study is diverse. In particular, as in the case of vanishing cosmological constant, studied in (Angelopoulos, Aretakis, and Gajic, 2023), we believe that considering a positive constant may be closer to the cosmological models of a black hole, and can act a a

bridge towards more complex theories.

The leading-order late-time behaviour of integrated local energy decay estimates of solutions to geometric wave equations of the form (1.1) is dictated by inverse-polynomially decaying tails in Schwarzschild spacetimes (Gajic, 2023). The same mechanism still applies when we consider geometric wave equations with a potential V,

$$\Box_g \phi = V\phi,$$

where V denotes a time-independent function satisfying $|V| \lesssim -\frac{1}{4}r^{-2}$ for $p > 2$ with respect to a suitable radial coordinate r. In the present thesis, we consider instead the Klein-Gordon equation, for potential $V = \mu^2$.

The main result is going to be a local integrated energy decay of the form:

$$\int_{\tau_1}^{\tau_2} \int_{\Sigma_\tau} \left[(Z\phi_{\leq L_0})^2 + (T\phi_{\leq L_0})^2 + \phi^2_{\leq L_0} \right] d\sigma \, dv \, dr \leq C \, E[\phi](\tau_1),$$

with $E[\phi](\tau_1)$ an initial energy and Σ_τ the constant time surfaces.

Chapter 2

Preliminaries

2.1 Geometry and foliations

In this section, we introduce the main geometric concepts and notation that will be used throughout the thesis.

2.1.1 The Reissner-Nordström-de Sitter spacetime

We consider the 4-dimensional manifold with boundary $(\mathcal{M}_{ext}, g)$, with:

$$\mathcal{M}_{ext} = \mathbb{R}_t \times (r_+, r_c)_r \times \mathbb{S}^2_{\theta,\varphi},$$

equipped with a coordinate chart (t, r, θ, φ) and the RNdS metric takes the form:

$$g = -D(r)dt^2 + \frac{1}{D(r)} + r^2 \not{g}_{\mathbb{S}^2}, \tag{2.1}$$

where $\not{g}_{\mathbb{S}^2} = d\theta^2 + \sin^2\theta d\varphi^2$ is the Euclidean metric on the 2-sphere, and the function $D(r)$ is given by:

$$D(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2} - \frac{\Lambda}{3}r^2. \tag{2.2}$$

The coordinates (θ, φ) are standard spherical coordinates on the 2-sphere $\mathbb{S}^2$ that degenerate at a line connecting the two poles. The spacetimes $(\mathcal{M}_{ext}, g)$ are called the RNdS exterior.

The RNdS metric in these coordinates appears to have singularities at $r = 0$ and at the zeros of $D(r)$. In the parameter range of (Λ, Q, M) that we are interested in, $D(r)$ has three real, positive and distinct roots $r_- < r_+ < r_c$, and one negative root $r_0 = -(r_- + r_+ + r_c) < 0$. The values r_- and r_+ correspond to the values of r at the Cauchy horizon $\mathcal{CH}$ and at the event horizon $\mathcal{H}$, respectively, while r_c represents the value at the cosmological horizon $\mathcal{C}$. The inner horizon is a Cauchy horizon $\mathcal{CH}^+_{\mathcal{R}}$ a boundary to the region of spacetime in which physics can be predicted from initial data prescribed on a surface Σ, as in Fig. 2.1, so solutions beyond the Cauchy horizon are not determined by initial data, that is why we will be only interested in the three positive roots of $D(r)$. There are infinitely many ways of smoothly extending the solution across the Cauchy horizon, and this problem is addressed by the *strong cosmic censorship conjecture* in general relativity; see for example (Luk and Oh, 2019) and (Dias et al., 2018).

With the help of the singularities, we can divide our RNdS spacetime in four regions: the exterior region up to the cosmological horizon, $\mathrm{I} = \mathbb{R}_t \times (r_+, r_c) \times \mathbb{S}^2$, the interior region up to the inner horizon, $\mathrm{II} = \mathbb{R}_t \times (r_-, r_+) \times \mathbb{S}^2$, the region beyond the cosmological horizon,

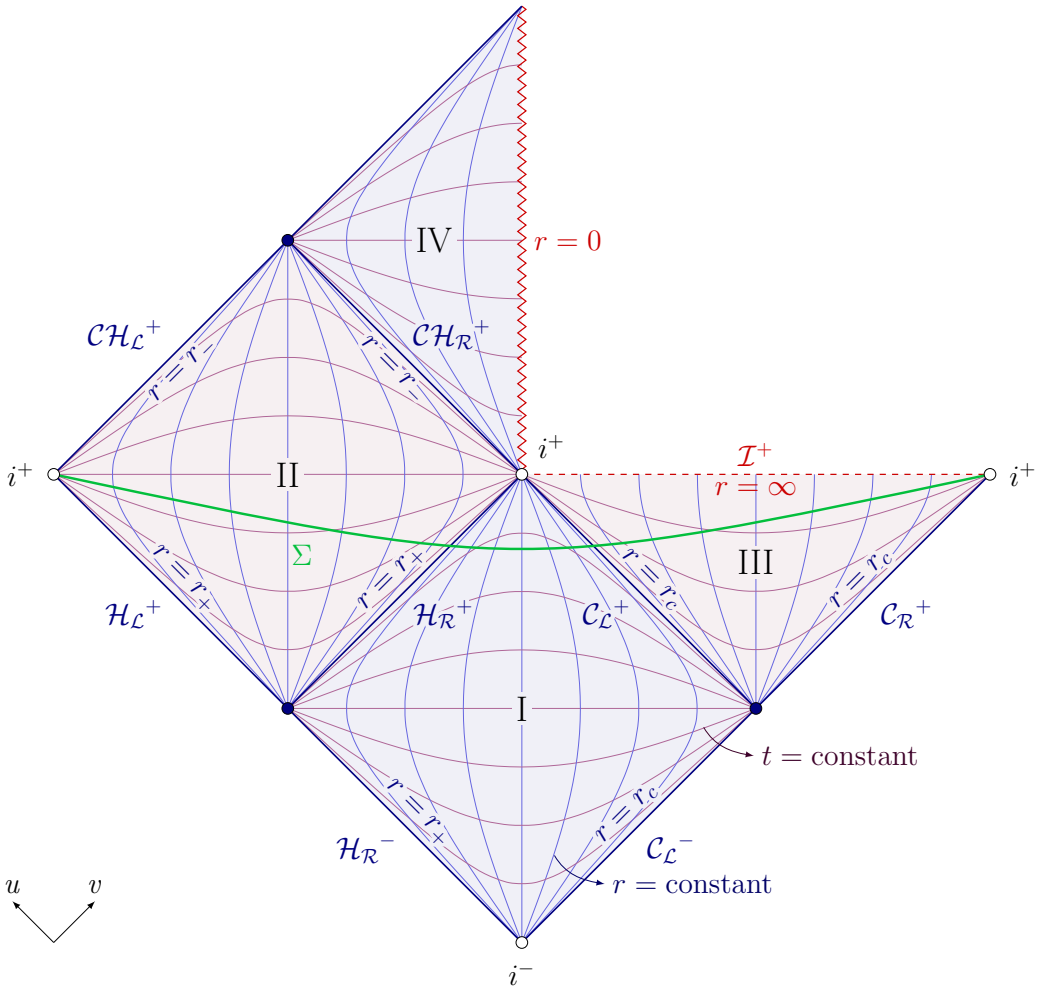


FIGURE 2.1: Part of the extended Penrose diagram of the RNdS spacetime. The wiggled red line represent the curvature singularity, the dashed red line correspond to conformal infinity. Lines at a 45º angle represent the different horizons, and all other lines represents hypersurfaces of constant t, in purple, or r, in blue. Filled dots stand for bifurcation surfaces, while empty dots indicate conformal points at infinity. The green line represents a possible Cauchy surface Σ for the region I∪II∪III. Each point in the picture represents a round 2-sphere of radius r.

$\mathrm{III} = \mathbb{R}_t \times (r_c, \infty) \times \mathbb{S}^2$, and the interior beyond the inner horizon, $\mathrm{IV} = \mathbb{R}_t \times (0, r_-) \times \mathbb{S}^2$. The Penrose diagram of this spacetime is given in Fig. 2.1.

Let us also note that the metric (2.1) is spherically symmetric and independent of t. Hence, the blocks I and IV, in which $f(r)$ is positive and ∂_t is a time-like Killing vector field, are static. The regions II and IV are dynamic regions. Moreover, the horizons are bifurcate Killing horizons generated by ∂_t.

We can rewrite the function $D(r)$ in terms of its positive roots,

$$D(r) = \frac{\Lambda}{3r}(r - r_-)(r - r_+)(r_c - r). \tag{2.3}$$

Inside of region I, $D(r)$ is positive and bounded. Only the singularity at $r = 0$ is a real geometric singularity, at which the curvature blows up. The apparent singularities at the zeros of $D(r)$ are artificial, and due to this particular choice of coordinates can be removed constructing Kruskal–type coordinates. Firstly, we introduce the tortoise radial coordinate r_*, which is defined

by:

$$\frac{dr_*}{dr} = \frac{1}{D(r)}. \tag{2.4}$$

The surface gravities of the Cauchy, event and cosmological horizons is defined by:

$$\kappa_i = \frac{1}{2}\left|\frac{dD}{dr}\right|_{r=r_i} \qquad i \in \{0, -, +, c\}.$$

One can choose the integration constant such that in each of the regions I, II, III and IV,

$$r_*(r) = -\frac{1}{2\kappa_-}\log|r - r_-| + \frac{1}{2\kappa_+}\log|r - r_+| - \frac{1}{2\kappa_c}\log|r - r_c| + \frac{1}{2\kappa_0}\log|r - r_0|. \tag{2.5}$$

We introduce a set of double null coordinates in each region,

$$\begin{cases} u = t - r_*(r), \\ v = t + r_*(r). \end{cases} \tag{2.6}$$

Note that the level sets of the coordinates functions v and u define null hypersurfaces. In these coordinates the metric takes the form:

$$g = -D(r)dudv + r^2 \not{g}_{\mathbb{S}^2}. \tag{2.7}$$

Note that the metric still degenerates when $D(r) = 0$. In the region I, that we are interested in, the event horizon $\mathcal{H}^+_{\mathcal{R}}$ corresponds to $u =\!\!\rightarrow +\infty$ and the cosmological horizon $\mathcal{C}^+_{\mathcal{L}}$ corresponds to $v =\!\!\rightarrow +\infty$. Since our double null coordinates are singular at these horizons, at the event horizon we change from (u, v, θ, φ) coordinates to Eddington–Finkelstein coordinates $(\underline{v}, r, \theta, \varphi)$, using:

$$\begin{cases} u = \underline{v} - 2r_*(r), \\ v = \underline{v}, \end{cases} \qquad \begin{cases} \partial_u = -\dfrac{D}{2}\partial_r, \\ \partial_v = \partial_{\underline{v}} + \dfrac{D}{2}\partial_r. \end{cases} \tag{2.8}$$

In these coordinates the metric becomes:

$$g = -D(r)d\underline{v}^2 + 2d\underline{v}dr + r^2 \not{g}_{\mathbb{S}^2},$$

At the cosmological horizon we change from (u, v, θ, φ) coordinates to Bondi-Sachs coordinates $(\underline{u}, r, \theta, \varphi)$, using:

$$\begin{cases} u = \underline{u}, \\ v = \underline{u} + 2r_*(r), \end{cases} \qquad \begin{cases} \partial_u = \partial_{\underline{u}} - \dfrac{D}{2}\partial_r, \\ \partial_v = \dfrac{D}{2}\partial_r. \end{cases} \tag{2.9}$$

In these coordinates the metric becomes:

$$g = -D(r)d\underline{u}^2 - 2d\underline{u}dr + r^2 \not{g}_{\mathbb{S}^2}.$$

By abuse of notation, we are going to refer to the Eddington–Finkelstein coordinates $(\underline{v}, r, \theta, \varphi)$ as (v, r, θ, φ), and the Bondi-Sachs coordinates $(\underline{u}, r, \theta, \varphi)$ as (u, r, θ, φ).

2.1.2 Spacetime foliation

Consider the time function τ,

$$\tau := \begin{cases} t & r_1 \leq r \leq R, \\ u + r_*(R) = t - r_* + r_*(R) & r \geq R, \\ v - r_*(r_1) = t + r_* - r_*(r_1) & r \leq r_1, \end{cases}$$

where r_1 is close to the event horizon, $r = r_+$, and $r = R$ is close to the cosmological horizon, $r = r_c$. Consider the following foliation by spacelike hypersurfaces Σ_τ of the region I, $(\mathcal{M}_{ext}, g)$, as show in Fig. 2.2.

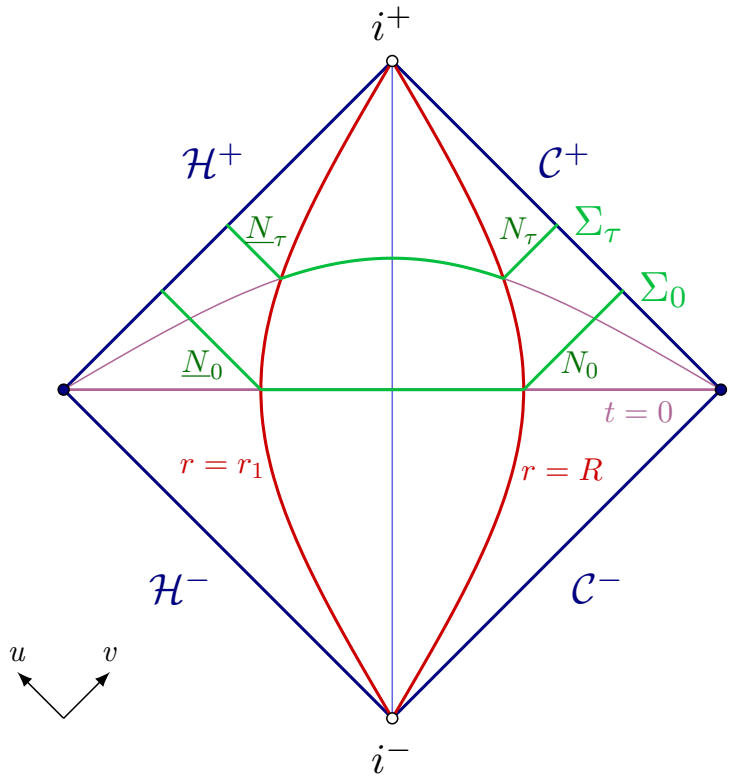


FIGURE 2.2: Penrose diagram of the region I, between the event horizon and the cosmological horizon, (r_+, r_c), for the RNdS spacetime. Two of the spacelike Cauchy hypersurfaces Σ_τ are represented by a green line, for a general value τ and for $\tau = 0$. The purple lines represent the corresponding spacelike Cauchy hypersurfaces, Σ_t, of constant t. The red lines are timelike hypersurfaces of constant r, one for each of the values close to the horizon that we have chosen, r_1 and R. The blue lines at a 45° angle represent the different horizons. $N_{\tau=0}$ and N_τ represent the null hypersurfaces of constant v for $r \geq R$, and $\underline{N}_{\tau=0}$ and $\underline{N}_\tau$ represent the null hypersurfaces of constant u for $r \leq r_1$. Filled dots stand for bifurcation surfaces, while empty dots indicate conformal points at infinity. Each point in the picture represents a round 2-sphere of radius r.

The hypersurfaces are,

$$\Sigma_{\tau'} := \{\tau = \tau'\} = \{t' = \tau'\} \cap \{r_1 \leq r \leq R\} \cup N_{\tau'} \cup \underline{N}_{\tau'}.$$

These coincide with constant t-slices, $t = \tau$ for $\{r_1 \leq r \leq R\}$. The null hypersurfaces N_τ, and $\underline{N}_\tau$,

$$N_{\tau'} := \{\tau = \tau'\} \cap \{r \geq R\},$$
$$\underline{N}_{\tau'} := \{\tau = \tau'\} \cap \{r \leq r_1\}.$$

We denote with

$$\mathcal{R} := \bigcup_{\tau \in [0,\infty)} \Sigma_\tau,$$

the spacetime subset of $\mathcal{M}_{ext}$ that will be relevant for the analysis in this thesis. We can also define the space region contained between two constant t-slices, for $0 \leq \tau_1 \leq \tau_2$,

$$\mathcal{R}_{\tau_1}^{\tau_2} := \bigcup_{\tau_1 \leq \tau \leq \tau_2} \Sigma_\tau = \{\tau_1 \leq \tau \leq \tau_2\}.$$

The intersections $\Sigma_\tau \cap \{r = r'\}$ are 2-spheres, which we will denote as $\mathbb{S}^2_{\tau,r'}$. The triple (r, θ, φ) defines a coordinate chart on the hypersurfaces Σ_τ, which can be complemented with the time function τ to obtain the coordinate chart $(\tau, r, \theta, \varphi)$ on $\mathcal{R}$. When we are working in the vicinity of the event Horizon $\mathcal{H}^+$, it would be suitable to use (v, r, θ, φ) coordinates, and on the other side, close to the cosmological horizon $\mathcal{C}^+$, we will use (u, r, θ, φ) coordinates.

We are going to introduce some useful vector fields. Firstly we are interested in defining a timelike vector field. We can take $T = \partial_t$, as the vector field formed by the tangent vectors to the timelike hypersurfaces of constant r. T is a Killing vector field, and this condition would be important since it would grant us the following condition,

$$T \text{ is a Killing vector field} \quad \Leftrightarrow \quad [\Box_g, T] = 0. \tag{2.10}$$

The problem with T is that it is not well defined near the horizon, where it tends to a null hypersurface. To solve this problem, we would like to have a vector field that is uniformly timelike but independent of the coordinate t. We choose a smooth vector field N, such that $N = T$ for $r_1 \leq r \leq R$, and $N = n_{\Sigma_\tau}$ in everywhere else in region I, where n_{Σ_τ} denotes the normal vector field to a time slicing Σ_τ. This vector field N is useful to define the energy associated to a geodesic $\gamma(s)$, $\epsilon = -g(N, \dot{\gamma})$. We can see this difference graphically on Fig. 2.3, where T is represented as blue arrows and N as black arrows.

Take a flow $\Phi_\tau^T : \mathcal{M}_{ext} \to \mathcal{M}_{ext}$ along the vector field $T = \partial_v$, in (v, r, θ, φ) coordinates. From the Killing property of T, (2.10), it follows that there is an isometry between all the hypersurfaces $(\Sigma_\tau, g|_{\Sigma_\tau})$ and $(\Sigma_0, g|_{\Sigma_0})$. For a point $p \in \mathcal{M}_{ext}$, there exists a geodesic $\gamma_{T,p}$, defined as the integral curve of the flow Φ_τ^T. Taking $p = (0, r_i, \theta_i, \varphi_i)$, then $\gamma_{T,p}(\tau) = (\tau, r_i, \theta_i, \varphi_i)$.

It will be convenient to define ingoing and outgoing null vector fields, respectively:

$$\begin{cases} \underline{L} = \partial_u, \\ L = \partial_v, \end{cases}$$

in (u, v, θ, φ) coordinates. We can also express them in (t, r, θ, φ) coordinates, and for this change of coordinates we have to use the definitions (2.6) of u and v, and also the definition (2.4) of r_*

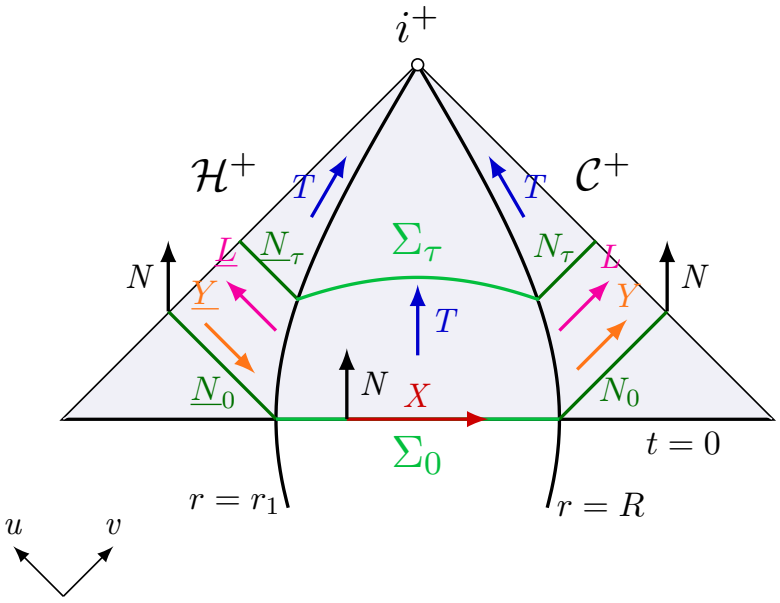


FIGURE 2.3: A 2-dimensional representation of the spacetime $\mathcal{R}$ as a shaded region, between the event horizon and the cosmological horizon, (r_+, r_c), for the RNdS spacetime. The green lines represent hypersurfaces Σ_τ, of constant τ. Each point in the picture represents a round 2-sphere of radius r.

with respect to $D(r)$,

$$\begin{cases} \underline{L} = \frac{1}{2}\left(\partial_t - \partial_{r_*}\right) = \frac{1}{2}\partial_t - \frac{D}{2}\partial_r, \\ L = \frac{1}{2}\left(\partial_t + \partial_{r_*}\right) = \frac{1}{2}\partial_t + \frac{D}{2}\partial_r. \end{cases} \tag{2.11}$$

Another useful set of outgoing null vector fields is:

$$\begin{cases} \underline{Y} = -\frac{2}{D}\underline{L}, \\ Y = \frac{2}{D}L. \end{cases} \tag{2.12}$$

These null vector fields are going to be useful to describe the areas of $r \leq r_1$ and $r \geq R$, where the most useful coordinates are (v, r, θ, φ), resp. (u, r, θ, φ). We can express $\underline{Y} = \partial_r$ in (v, r, θ, φ) coordinates, and $Y = \partial_r$ in (u, r, θ, φ) coordinates.

We will also use two spacelike vector fields, X and Z, such that

$$\begin{cases} T = L + \underline{L}, \\ X = \frac{1}{D}(L - \underline{L}), \\ Z = DX = L - \underline{L}. \end{cases} \tag{2.13}$$

In (t, r, θ, φ) coordinates: $T = \partial_t$, $X = \partial_r$ and $Z = \partial_{r_*}$.

2.1.3 Properties of functions on $\mathbb{S}^2$

Let $Y_{\ell m}(\theta, \varphi)$ denote spherical harmonics on $\mathbb{S}^2$, labelled by an angular momentum $\ell \in \mathbb{N}_0$ and an an azimuthal number $m \in \mathbb{Z}$, $|m| \leq \ell$. We can define projection operators $\pi_\ell : C^\infty(\Sigma_\tau) \to C^\infty(\Sigma_\tau)$ as follows (Gajic, 2023):

$$(\pi_\ell f)(r, \theta, \varphi) := \sum_{m=-\ell}^{\ell} \left[\int_{\mathbb{S}^2} f\left(r, \theta', \varphi'\right) Y_{\ell m}\left(\theta', \varphi'\right) \sin\theta' d\theta' d\varphi' \right] Y_{\ell m}(\theta, \varphi)$$

and extend π_ℓ as a bounded linear operator to $L^2(\Sigma_\tau)$ (with respect to the standard volume form on Σ_τ given the induced metric on Σ_τ).

We denote the $f_\ell = \pi_\ell f$ and refer to f_ℓ as a spherical harmonic mode. Since the functions $Y_{\ell m}$ make up an orthonormal basis on $L^2\left(\mathbb{S}^2\right)$, we can express:

$$f = \sum_{\ell=0}^{\infty} f_\ell.$$

We moreover denote for $\ell_0 \in \mathbb{N}_0$, $f_{\geq \ell_0} = \sum_{\ell=\ell_0}^{\infty} f_\ell$. We also define:

$$f_{\ell m} = \int_{\mathbb{S}^2} f\left(r, \theta', \varphi'\right) Y_{\ell m}\left(\theta', \varphi'\right) \sin\theta' d\theta' d\varphi'.$$

For an angular function with no dependence on the radial coordinate r, $g \in L^2(\mathbb{S}^2)$,

$$g(\theta, \varphi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} g_{\ell m} Y_{\ell m}$$

where $g_{\ell m} \in \mathbb{C}$.

We define the Laplacian on the unit round sphere $\mathbb{S}^2$ as:

$$\not{\Delta}_{\mathbb{S}^2}(\cdot) := \frac{1}{\sin\theta} \partial_\theta \left(\sin\theta \partial_\theta (\cdot)\right) + \frac{1}{\sin^2\theta} \partial_\varphi^2 (\cdot).$$

The derivatives of the spherical harmonics fulfil the following identities:

$$\begin{aligned} \not{\Delta}_{\mathbb{S}^2} Y_{\ell m} &= -\ell(\ell+1) Y_{\ell m}, \\ \partial_\varphi Y_{\ell m} &= i\, m Y_{\ell m}, \end{aligned}$$

where $i = \sqrt{-1}$. Consequently,

$$\not{\Delta}_{\mathbb{S}^2} f_\ell = -\ell(\ell+1) f_\ell. \tag{2.14}$$

We denote with $H^k\left(\mathbb{S}^2\right)$ the k-th order L^2-based Sobolev spaces on $\mathbb{S}^2$ with respect to the volume form $d\sigma := \sin\theta d\theta d\varphi$.

Lemma 2.1.1. *Let $k \in \mathbb{N}, \ell \in \mathbb{N}_0$ and $f \in H^k\left(\mathbb{S}^2\right)$. Then:*

$$\int\limits_{\mathbb{S}^2} \left|\nabla^k_{\mathbb{S}^2} f_{\geq \ell}\right|^2 d\sigma \geq (\ell(\ell+1))^k \int\limits_{\mathbb{S}^2} f^2_{\geq \ell} d\sigma, \tag{2.15}$$

$$\int\limits_{\mathbb{S}^2} \left|\nabla^k_{\mathbb{S}^2} f_{\ell}\right|^2 d\sigma = (\ell(\ell+1))^k \int\limits_{\mathbb{S}^2} f^2_{\ell} d\sigma, \tag{2.16}$$

Furthermore, if $f \in H^{2k}\left(\mathbb{S}^2\right)$, then

$$\int\limits_{\mathbb{S}^2} \left(\Delta^k_{\mathbb{S}^2} f\right)^2 d\sigma = \int\limits_{\mathbb{S}^2} \left|\nabla^{2k}_{\mathbb{S}^2} f_{\geq \ell}\right|^2 d\sigma,$$

and if $f \in H^2\left(\mathbb{S}^2\right)$, then there exists a numerical constant $C > 0$, such that

$$\|f\|_{L^\infty(\mathbb{S}^2)} \leq C \|f\|_{H^2(\mathbb{S}^2)}.$$

Proof. See (Gajic, 2023) □

It would be useful to consider *Young's inequality with $\epsilon > 0$* arbitrarily small:

Theorem 2.1.2. *If $a \geq 0$ and $b \geq 0$ are non-negative real numbers and if $\epsilon > 0$ is arbitrarily small, then,*

$$ab \leq \frac{a^2}{2\epsilon} + \frac{\epsilon b^2}{2}. \tag{2.17}$$

Proof. It is easy to derive. The square of every real number is zero or positive. Therefore, for every pair of real numbers a and b, we can write:

$$(a-b)^2 \geq 0, \qquad \Rightarrow \quad 0 \leq a^2 + b^2 - 2ab, \qquad \Rightarrow \quad 2ab \leq a^2 + b^2,$$

and we got the *Young's inequality* of order 2. Now consider a change of coordinates, $a' = a/\sqrt{\epsilon}$ and $b' = \sqrt{\epsilon} b$, and we get the expression we were looking for. □

2.1.4 Hardy inequality

Throughout the thesis, we will frequently make use of the following *Hardy inequality*:

Lemma 2.1.3. *Let $f \in C^1([a,b])$ with $a, b \in \mathbb{R}$ such that $0 \leq a < b$. Then for $q \in \mathbb{R} \setminus \{-1\}$:*

$$\int\limits_a^b x^q f^2(x)\, dx \leq \frac{2}{q+1}\left[b^{q+1} f^2(b) - a^{q+1} f^2(a)\right] + \frac{4}{(q+1)^2} \int\limits_a^b x^{q+2} \left(\frac{df}{dx}\right)^2 (x)\, dx. \tag{2.18}$$

Proof. Start the proof by integrating $\frac{d}{dx}\left(x^{q+1} f^2\right)$. By the Leibniz rule:

$$\int\limits_a^b \frac{d}{dx}\left(x^{q+1} f^2\right) dx = (q+1) \int\limits_a^b x^q f^2 dx + \int\limits_a^b x^{q+1} f \frac{df}{dx} dx. \tag{2.19}$$

Apply a Young's inequality ($2\alpha\beta \leq \alpha^2 + \beta^2$) on the integrand of the second term on the right-hand side of the equation above, taking

$$\alpha := \sqrt{\frac{q+1}{2}}\, x^{\frac{q}{2}}\, f, \qquad\qquad \beta := \sqrt{\frac{2}{q+1}}\, x^{\frac{q}{2}+1}\, \frac{df}{dx},$$

to estimate:

$$\left| x^{q+1} f \frac{df}{dx} \right| \geq -\frac{|q+1|}{2} x^q f^2 - \frac{2}{|q+1|} x^{q+2} \left(\frac{df}{dx} \right)^2.$$

Introducing this inequality into (2.19),

$$\int_a^b \frac{d}{dx}\left(x^{q+1} f^2\right) dx \geq \frac{q+1}{2} \int_a^b x^q f^2 dx - \frac{2}{q+1} \int_a^b x^{q+2} \left(\frac{df}{dx} \right)^2 dx, \tag{2.20}$$

By the fundamental theorem of calculus:

$$\int_a^b \frac{d}{dx}\left(x^{q+1} f^2\right) dx = b^{q+1} f^2(b) - a^{q+1} f^2(a). \tag{2.21}$$

Combining (2.20) and (2.21), we finally get to the desired expression,

$$\frac{q+1}{2} \int_a^b x^q f^2 dx \leq b^{q+1} f^2(b) - a^{q+1} f^2(a) + \frac{2}{q+1} \int_a^b x^{q+2} \left(\frac{df}{dx} \right)^2 dx.$$

□

We introduce a useful claim derived from the Hardy inequality.

Lemma 2.1.4. *Let $f \in C^1([a,b])$ with $a, b \in \mathbb{R}$ such that $0 \leq a < b$. Then,*

$$\int_a^b \frac{1}{x \log^2 x} f^2(x)\, dx \leq 2\left[-\frac{1}{\log b} f^2(b) + \frac{1}{\log a} f^2(a) \right] + 4 \int_a^b x \left(\frac{df}{dx} \right)^2 (x)\, dx. \tag{2.22}$$

Proof. We follow a process similar to the one from the proof of Hardy inequality. Applying the Leibniz rule into the following integral:

$$\int_a^b \frac{d}{dx}\left(-\frac{1}{\log x} f^2 \right) dx = \int_a^b \frac{1}{x \log^2 x} f^2\, dx - 2 \int_a^b \frac{1}{\log x}\, f\, \frac{df}{dx}\, dx. \tag{2.23}$$

Apply Young's inequality ($2\alpha\beta \leq \alpha^2 + \beta^2$) on the integrand of the second term on the right-hand side of the equation above, taking

$$\alpha := \frac{1}{\sqrt{2x}} \frac{f}{\log x}, \qquad\qquad \beta := \sqrt{2x}\, \frac{df}{dx},$$

to estimate:

$$2\int_a^b \frac{1}{\log x} f \frac{df}{dx}\, dx \leq \frac{1}{2}\int_a^b \frac{1}{x}\frac{f^2}{\log^2 x}\, dx + 2\int_a^b x\left(\frac{df}{dx}\right)^2 dx.$$

Introducing this inequality into (2.23),

$$\int_a^b \frac{d}{dx}\left(-\frac{1}{\log x} f^2\right) dx \geq \frac{1}{2}\int_a^b \frac{1}{x}\frac{f^2}{\log^2 x}\, dx - 2\int_a^b x\left(\frac{df}{dx}\right)^2 dx. \tag{2.24}$$

By the fundamental theorem of calculus:

$$\int_a^b \frac{d}{dx}\left(-\frac{1}{\log x} f^2\right) dx = -\frac{1}{\log b} f^2(b) + \frac{1}{\log a} f^2(a). \tag{2.25}$$

Combining (2.24) and (2.25), we finally get to the desired expression:

$$\frac{1}{2}\int_a^b \frac{1}{x}\frac{f^2}{\log^2 x}\, dx \leq -\frac{1}{\log b} f^2(b) + \frac{1}{\log a} f^2(a) + 2\int_a^b x\left(\frac{df}{dx}\right)^2 dx.$$

□

2.1.5 Photon sphere

We are going to go over some notions of Lorentzian geometry, for an introduction to Lorentzian geometry one can consult (Hawking and Ellis, 1973). Consider on our region I $(\mathcal{M}_{ext}, g)$. One defines the Christoffel symbols

$$\Gamma^{\mu}_{\nu\lambda} = \frac{1}{2} g^{\mu\alpha}\left(\partial_\nu g_{\alpha\lambda} + \partial_\lambda g_{\nu\alpha} - \partial_\alpha g_{\nu\lambda}\right).$$

Here $g_{\mu\nu}$ denote the components of g with respect to a local coordinate system $\{x^\mu\}$, $g^{\mu\nu}$ denotes the components of the inverse metric, and we are applying the Einstein summation convention where repeated upper and lower indices are summed. The *geodesics* $\gamma(s)$ are defined as smooth curves $\gamma : s \mapsto \gamma(s) \in \mathcal{M}_{ext}$, which in local coordinates on $\mathcal{M}_{ext}$, are solutions to

$$\ddot{\gamma}^\mu + \Gamma^{\mu}_{\nu\lambda}\dot{\gamma}^\nu\dot{\gamma}^\lambda = 0,$$

where $\cdot = \frac{d}{ds}$, s is an affine parameter, and with initial data:

$$\begin{cases} \dot{\gamma}(0) = \mathfrak{V} \in T_p\mathcal{M}_{ext}, \\ \gamma(0) = p \in \mathcal{M}_{ext}, \end{cases}$$

We can classify geodesics based on the metric of the spacetime that they are defined in:

$$\begin{cases} g(\dot{\gamma},\dot{\gamma}) = 1 & \text{spacelike geodesic,} \\ g(\dot{\gamma},\dot{\gamma}) = 0 & \text{null geodesic,} \\ g(\dot{\gamma},\dot{\gamma}) = -1 & \text{timelike geodesic.} \end{cases}$$

Timelike geodesics can be interpreted as massive particles trajectories, and null geodesics as the trajectories of a mass-less wave function. We only consider null geodesics: $g(\dot\gamma,\dot\gamma)=0$.

The Christoffel symbols allow us to define the covariant derivative on (k,l) tensor fields by

$$\nabla_\lambda A^{\nu_1\dots\nu_k}_{\mu_1\dots\mu_\ell}=\partial_\lambda A^{\nu_1\dots\nu_k}_{\mu_1\dots\mu_\ell}+\sum_{i=1}^{k}\Gamma^{\nu_i}_{\lambda\rho}A^{\nu_1\dots\nu_k}_{\mu_1\dots\mu_\ell}-\sum_{i=1}^{l}\Gamma^{\rho}_{\lambda\mu_i}A^{\nu_1\dots\nu_k}_{\mu_1\dots\rho\nu_\ell\dots\mu_\ell} \tag{2.26}$$

where it is understood that ρ replaces ν_i,μ_i, respectively in the two terms on the right. This defines $(k,l+1)$ tensor. As usual, if we contract this with a vector $\mathfrak{V}$ at p, then we will denote this operator as $\nabla_{\mathfrak{V}}$ and we note that this can be defined in the case that the tensor field is defined only on a curve tangent to $\mathfrak{V}$ at p. Let ∇ be the covariant derivative of $g_{\mu\nu}$. An *affinely parametrized geodesic* with an affine connection ∇ is defined as a curve $\gamma(s)$ such that parallel transport along the curve preserves the tangent vector to the curve, so

$$\nabla_{\dot\gamma}\dot\gamma=0 \quad\Leftrightarrow\quad \dot\gamma^\mu\nabla_\mu\dot\gamma^\nu=0.$$

Consider the local coordinate system (t,r,θ,φ) and the metric g defined in (2.1). Without loss of generality, use spherical symmetry to take $\theta=\frac{\pi}{2}$. Consider a null geodesic:

$$\gamma(s)=\left(t(s),r(s),\frac{\pi}{2},\varphi(s)\right).$$

One can introduce the following quantities,

$$\varepsilon_\gamma:=-g\left(T,\dot\gamma\right)=D\dot t\big|_\gamma\,,\qquad\qquad \lambda:=-g\left(\dot\gamma,\partial_\varphi\right)=r^2\dot\varphi\big|_\gamma\,.$$

These are conserved quantities associated to energy and angular momentum, respectively. From the definition of a null geodesic,

$$0=g(\dot\gamma,\dot\gamma)=-D\left(\dot t\big|_\gamma\right)^2+\frac{1}{D}\left(\dot r\big|_\gamma\right)^2+r^2\left(\dot\varphi\big|_\gamma\right)^2=-\frac{1}{D}\varepsilon_\gamma^2+\frac{1}{D}\left(\dot r\big|_\gamma\right)^2+\frac{1}{r^2}\lambda^2.$$

We can reduce this to the equation:

$$\varepsilon_\gamma^2=\left(\dot r\big|_\gamma\right)^2+\frac{D(r)}{r^2}\lambda^2.$$

This equation has a similar form to a Schrödinger equation, with a potential $V(r)=-\frac{D}{r^2}\lambda^2$. We can interpret it as a potential barrier generated by the black hole, with a maximum value to overcome by the geodesics. This maximum value is going to appear at a value of $r=r_{PS}$ what we will call the *photon sphere*, we will shortly see why. To calculate it we just got to solve:

$$\frac{d}{dr}V(r_{PS})=0. \tag{2.27}$$

The null geodesics that come from infinity will have different behaviours towards the back hole depending on their associated energy ε_γ.

The region with $r>2M$ of the Schwarzschild spacetime with a black hole of radius $2M$, contains a hypersurface at $r=3M$ that distinguish between different behaviours of null geodesics, and it is an important geometric feature of the spacetime. Any future endless null geodesic in the maximally extended Schwarzschild spacetime starting at some point with $r>3M$ and initially directed outwards will continue outwards and escape to infinity. While any null geodesic starting

at some point between $r = 2M$ and $r = 3M$ and initially directed inward will continue inward and fall into the black hole. The hypersurface at $r = 3M$ is called the photon sphere: any null geodesic starting at some point of the photon sphere and initially tangent to the photon sphere will remain in the photon sphere, with a well defined finite energy ε (since the vector field T associated to it is a Killing vector field). A similar photon sphere is present in the RNdS spacetime, where the photon can orbit the black hole.

It is shown in (Mokdad, 2017) that, in the case of three horizons, this photon sphere is located in the static exterior region given by $r \in (r_+, r_c)$. Define the following:

$$\Theta = \frac{1}{\sqrt{6\Lambda}}, \qquad \Delta = 1 - 12Q^2\Lambda, \qquad m_1 = \Theta\sqrt{1-\sqrt{\Delta}}, \qquad m_2 = \Theta\sqrt{1+\sqrt{\Delta}},$$
$$M_1 = m_1 - 2\Lambda m_1^3, \qquad M_2 = m_2 - 2\Lambda m_2^3.$$

Proposition 1 ((Three positive zeros and one photon sphere (Mokdad, 2017))**.** *The function $D(r)$ has exactly three positive distinct zeros if and only if*

$$Q \neq 0, \qquad 0 < \Lambda < \frac{1}{12Q^2}, \qquad M_1 < M < M_2 \tag{2.28}$$

holds. In this case, there is exactly one photon sphere in the static exterior region I of the black hole defined by the portion between the largest two zeros of $D(r)$.

If we take a non zero purely rotational null vector field along the angle φ it will be of the form $\mathcal{V} = \alpha T + \beta\partial_\varphi$. From the null condition, it is enough to examine the case where $\alpha = 1$. In this case,

$$\nabla_{\mathcal{V}}\mathcal{V} = D\left(\frac{D'}{2} - \frac{D}{r}\right)X,$$

and since we have $D(r) > 0$, we see that

$$\nabla_{\mathcal{V}}\mathcal{V} = 0 \Leftrightarrow D\left(\frac{D'}{2} - \frac{D}{r}\right) = 0 \quad \Leftrightarrow rD' - 2D = 0.$$

As we assumed a fixed $\theta = \frac{\pi}{2}$ from the beginning, one can see that the integral curves of $\mathcal{V}$ at the zeros of $rD'(r) - 2D(r)$ are geodesics. Hence, by the spherical symmetry, we get a full 'sphere' of null geodesics outside the black hole, this is why it is referred to as the photon sphere around the black hole. Since $rD'(r) - 2D(r) = \frac{6M}{r} - \frac{4Q^2}{r^2} - 2$, then by studying the polynomial $S(r) = -r^2 + 3Mr - 2Q^2$ we can determine the zeros. The discriminant of S is $\Delta_S = 9M^2 - 8Q^2 = (3M - 2\sqrt{2}Q)(3M + 2\sqrt{2}Q)$ which is positive if $M > \frac{2\sqrt{2}|Q|}{3}$.

The two roots, have the expressions

$$P_1 = \frac{3M - \sqrt{\Delta_S}}{2} \quad \text{and} \quad P_2 = \frac{3M + \sqrt{\Delta_S}}{2}.$$

Note that $\{r = P_1\}$ is not a photon sphere since $D(r)$ is negative on (r_-, r_+) and so the rotational vector $\mathcal{V}$ is necessarily spacelike. This means that there are no orbits inside the black hole horizon, which is consistent with the fact that this region is dynamic. We also note that in spite of the covering of the photon sphere by null geodesics it is not a null hypersurface, as a matter of fact, the spacelike vector X is normal to the photon sphere hypersurface, and therefore it is a timelike hypersurface.

The photon sphere for the RNdS spacetime is then situated at $\{r = P_2\}$, which we are going to rename to $\{r = r_{PS}\}$,

$$r_{PS} = \frac{3M + \sqrt{\Delta_S}}{2} = \frac{3M}{2} + \frac{\sqrt{9M^2 - 8Q^2}}{2}. \tag{2.29}$$

Trappings of null geodesics at the photon sphere are an obstruction to establishing integrated local energy decay, that is why it is going to be specially relevant for this thesis. For a more detailed description of the problem, see (Sbierski, 2015).

If N is future directed timelike, we call

$$E^N_\tau(\phi) := \int_{\Sigma_\tau} J^N(\phi) \cdot n_{\Sigma_\tau} \mathrm{vol}_g$$

the N-energy of ϕ at time τ, where vol_g denotes the volume element corresponding to the induced Riemannian metric g on Σ_τ, $J^N(\phi)$ the current.

Proposition 2 (Gaussian beam approximation (Sbierski, 2015))**.** *Let $(\mathcal{M}, g)$ be a time oriented globally hyperbolic Lorentzian manifold with time function t, foliated by the level sets $\Sigma_\tau = \{t = \tau\}$, where Σ_0 is a Cauchy hypersurface. Furthermore, let γ be a null geodesic that intersects Σ_0 and N a timelike, future directed vector field.*

For any neighbourhood $\mathcal{N}$ of γ, for any $\tau_1 > 0$ with $\Sigma_{\tau_1} \cup Im(\gamma) \neq \emptyset$, and for any $\eta > 0$ there exists a solution $\phi_1 \in C^\infty(\mathcal{M}, \mathbb{C})$ of the standard wave equation $\Box_g \phi = 0$ with $E^N_0(\phi_1) = 1$ and a $\phi_2 \in C^\infty(\mathcal{M}, \mathbb{C})$ with $supp(\phi_2) \subseteq \mathcal{N}$ such that

$$E^N_\tau(\phi_1 - \phi_2) < \eta, \qquad \forall 0 \leq \tau \tau_1. \tag{2.30}$$

provided that $c \leq -g(N, N) \leq C$ for $c, C > 0$.

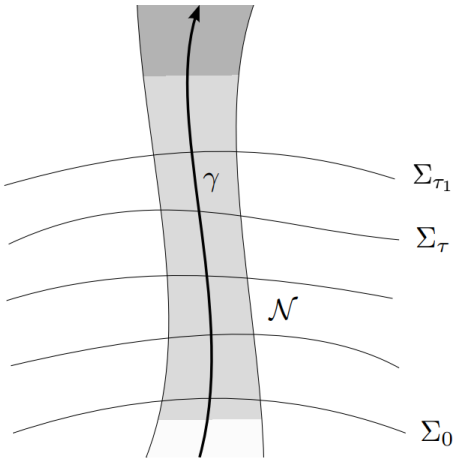


FIGURE 2.4: Graphical representation of the trapping by a null geodesic γ, with Σ_τ being Cauchy hypersurfaces of constant τ, and $\mathcal{N}$ a neighbourhood of γ (Sbierski, 2015).

Note that (2.30) can be reinterpreted to say that the energy of the wave packet is comparable to that of the geodesic $E^N_\tau(\phi_1) \sim \varepsilon_\gamma$, i.e.

$$\exists c, C > 0 \quad \text{such that} \quad c \cdot \varepsilon_\gamma \leq E^N_\tau(\phi_1) \leq C \cdot \varepsilon_\gamma.$$

This relation for the energy, together with $\text{supp}(\phi_2) \subseteq \mathcal{N}$ make the statement that the solution ϕ_1 hardly disperses up to time τ_1, rigorous. The energy of the solution ϕ_1 stays localised for finite time. See a graphical representation on Fig. 2.4, where ϕ_1 would stay close to the geodesic.

Remark. *Proposition 2 indicates that trapping of null geodesics at the photon sphere $\{r = r_{PS}\}$ is an obstruction to establishing integrated local energy decay. For obtaining it, we need to bound by the energy by initial data, but if the energy of the wave ϕ is always comparable to the energy of the geodesic, then it would be independent of τ. We will solve this by adding higher-order derivatives on the right-hand side of our integrated local energy estimates.*

2.1.6 Additional notation

Throughout the paper we will frequently use C or c to denote positive constants depending only on M, Q, Λ and μ. In order to make the notation more compact, we will not keep precise track of constants and we will instead make use of the following "algebra of constants" when constants c or C appear in an equality:

$$C + C = C \cdot C = C, \qquad c + c = c \cdot c = c.$$

We will moreover use the notations $f \lesssim g$ and $f \sim g$, with f, g non-negative functions, to mean respectively:

$$f \leq Cg, \qquad cg \leq f \leq Cg.$$

The notation $f \propto g$ means that f is proportional to g.

On the integral we are going to apply the following notation:

$$\int_{N_{\tau_2}} d\sigma du := \int_{u(r_1,\tau_2)}^{r_+} \int_{\mathbb{S}^2} d\sigma du.$$

Idem for the integral over the other hypersurfaces defined in Section 2.1.2.

A concept that will appear is the so-called *big and little-o notation*:

Definition 2.1.1 (Big-O notation)**.** A function $f(x)$ is big-O of $g(x)$, i.e. $f(x) = O(g(x))$ as $x \to \infty$, if there **exists** $\sigma > 0$, $x_0 \in \mathbb{R}$ such that $|f(x)| \leq \sigma g(x) \quad \forall x \geq x_0$.

Definition 2.1.2 (Little-o notation)**.** A function $f(x)$ is little-o of $g(x)$, i.e. $f(x) = o(g(x))$ as $x \to \infty$, if **for all** $\varepsilon > 0$, $\exists x_0 \in \mathbb{R}$ such that $|f(x)| \leq \varepsilon g(x) \quad \forall x \geq x_0$.

Little-o notation is a stronger statement than big-O notation. Here we applied the following variation of "big-O notation": we write $f = O_k(r^{-\alpha})$ for a suitably regular function $f : [r_+, r_c] \to \mathbb{R}$, $\alpha \in \mathbb{R}$ and $k \in \mathbb{N}_0$, if there exists a constant $C > 0$ such that the derivatives of order l fulfil $\left|\partial_r^l f\right| \leq C\, r^{-\alpha-l}$ for all $l = 0, \ldots, k$.

2.2 Wave equations

In this section, we introduce the wave equations that are studied in the thesis. We will focus on geometric wave equations, in particular the *Klein-Gordon equation*,

$$\Box_g \phi = \mu^2 \phi \qquad \text{for} \quad \mu \in \mathbb{R}, \tag{2.31}$$

with ϕ the Klein-Gordon scalar field, and with $\Box_g$ denoting the Laplace-Beltrami operator with respect to g,

$$\Box_g(\cdot) := \frac{1}{\sqrt{-\det g}}\partial_\alpha\left(\sqrt{-\det g}\left(g^{-1}\right)^{\alpha\beta}\partial_\beta(\cdot)\right).$$

For the metric g in (t, r, θ, φ) coordinates, (2.1), the Laplace-Beltrami operator takes the form:

$$\Box_g(\phi) = -\frac{1}{D}\partial_t^2\phi + \frac{1}{r^2}\partial_r\left(Dr^2\partial_r\phi\right) + \frac{1}{r^2}\not{\Delta}_{\mathbb{S}^2}\phi.$$

For the metric g in (u, v, θ, φ) coordinates (2.7),

$$\Box_g(\phi) = -\frac{4}{D}\partial_u\partial_v\phi + \frac{2}{r}\left(\partial_v\phi - \partial_u\phi\right) + \frac{1}{r^2}\not{\Delta}_{\mathbb{S}^2}\phi = -\frac{4}{D}\underline{L}L\phi + \frac{2}{r}\left(L\phi - \underline{L}\phi\right) + \frac{1}{r^2}\not{\Delta}_{\mathbb{S}^2}\phi. \quad (2.32)$$

Remark. *For a function that can be described as a sum of spherical harmonic modes, $\phi = \sum_{\ell=0}^{\infty}\phi_\ell$, if $\Box_g(\phi) = \mu^2\phi$, then $\Box_g(\phi_\ell) = \mu^2\phi_\ell$ for all $\ell \in \mathbb{N}_0$.*

Since we consider a Lorentzian metric g, then $\Box_g\phi = \mu^2\phi$ is a wave equation, i.e. describes a hyperbolic equation. If we had a Riemannian metric, then it would no longer be a wave equation, but an elliptic one, i.e. a Laplace equation. In that case, the initial value problem would not work, we would need a boundary value problem, without being able to consider one of the coordinates as time. The initial value problem is the only approach that makes sense for the wave equation.

We establish global existence and uniqueness for the initial value problem for (2.31) with initial data on $\Sigma_0 = [r_+, r_c] \times \mathbb{S}^2$.

Proposition 3. *Consider the initial data functions $(\Phi, \Phi') \in C_C^\infty(\Sigma_0) \times C_C^\infty(\Sigma_0)$. There exists a unique solution ϕ to (2.31) in $\mathcal{R}$, such that for all $T_* \geq 0$,*

$$\begin{aligned}\phi &\in C^\infty\left([0, T_*] \times \Sigma_0\right),\\ \phi|_{\Sigma_0} &= \Phi,\\ T\phi|_{\Sigma_0} &= \Phi'.\end{aligned}$$

Proof. It follows from standard global existence and uniqueness arguments for linear wave equations, see for example §1.3 of (Sogge, 2008). □

We introduce the following rescaled function that will play an important role in the thesis:

$$\psi := r\phi.$$

We can rewrite the Laplace-Beltrami operator (2.32) in terms of ψ:

$$\begin{aligned}\Box_g\phi = \Box_g\left(\frac{\psi}{r}\right) &= -\frac{4}{D}\underline{L}L\left(\frac{\psi}{r}\right) + \frac{2}{r}\left[L\left(\frac{\psi}{r}\right) - \underline{L}\left(\frac{\psi}{r}\right)\right] + \frac{1}{r^2}\not{\Delta}_{\mathbb{S}^2}\left(\frac{\psi}{r}\right)\\ = -\frac{4}{D}\underline{L}\left(\frac{rL\psi}{r^2} - \frac{D\psi}{2r^2}\right) + \frac{2}{r^3}\left(rL\psi - r\underline{L}\psi - D\psi\right) + \frac{1}{r^3}\not{\Delta}_{\mathbb{S}^2}\psi &= -\frac{4}{Dr}\underline{L}L\psi - \frac{D'}{r^2}\psi + \frac{1}{r^3}\not{\Delta}_{\mathbb{S}^2}\psi,\end{aligned} \quad (2.33)$$

where we used that $\underline{L}r = -\frac{D}{2}$, $Lr = \frac{D}{2}$.

For the Klein-Gordon field ϕ, functional differentiation with respect to ϕ of the action, $\mathcal{S}_{KG}$, obtained from the Lagrangian density (Wald, 2010),

$$\mathcal{L}_{KG} = -\frac{1}{2}\sqrt{-g}\left(g^{\alpha\beta}\nabla_\alpha\phi\nabla_\beta\phi + \mu^2\phi^2\right),$$

yields the Klein-Gordon equation (2.31) in a curved spacetime. The stress-energy tensor of the field is:

$$\mathbb{T}_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}\left(\left(g^{-1}\right)^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + \mu^2\phi^2\right), \tag{2.34}$$

This stress-energy tensor is conserved for fields ϕ that satisfy the Euler-Lagrange equations, as is the case with the KG field,

$$\nabla^\mu\mathbb{T}_{\mu\nu} = 0.$$

This conservation comes from the *Noether's theorem*.

Theorem 2.2.1. *Noether's theorem (José and Saletan, 1998) If a Lagrangian is invariant under a family of transformations, its dynamical system possesses a constant of the motion, and that constant can be found from a knowledge of the Lagrangian and the transformation.*

The condition of invariance can be used to derive an expression for a current associated with a conserved physical quantity. The conserved quantity is called the Noether charge, while the flow carrying that charge is called the Noether current.

2.2.1 Stationary wave function

We can consider a stationary wave $w(r,\theta,\varphi)$ as a smooth solution to the Klein-Gordon equation (2.31), such that:

$$\begin{cases} \Box_g w = \mu^2 w, \\ Tw = 0. \end{cases}$$

Since the RNdS spacetime is spherically symmetric, we can split the spatial waves w in spherical harmonic modes, as in §2.1.3. We are going to consider:

$$w = \sum_{\ell=0}^{\infty} w_\ell, \qquad\qquad w_\ell(r,\theta,\varphi) = \sum_{m=-\ell}^{\ell} w_{\ell m}(r)Y_{\ell m}(\theta,\varphi).$$

The functions $w_\ell(r,\theta,\varphi)$ with $|m| \leq \ell$ are stationary solutions to (2.31) (i.e. solutions with vanishing T-derivative). We will try to find the expression of $w_\ell(r,\theta,\varphi)$ that fulfils the wave equation $\Box_g w_\ell = \mu^2 w_\ell$. In (t,r,θ,φ) coordinates,

$$0 = \Box_g w_\ell - \mu^2 w_\ell = \frac{1}{r^2}\partial_r\left(Dr^2\partial_r w_\ell\right) + \frac{1}{r^2}\not{\Delta}_{\mathbb{S}^2} w_\ell - \mu^2 w_\ell.$$

Using the equality $\not{\Delta}_{\mathbb{S}^2} w_\ell = -\ell(\ell+1)w_\ell$,

$$D\partial_r^2 w_\ell + \left(D' + \frac{2D}{r}\right)\partial_r w_\ell - \left(\frac{\ell(\ell+1)}{r^2} + \mu^2\right)w_\ell = 0,$$

where we define $D' := dD/dr$.

Without loss of generality, we can fix the values of (θ, φ) for w_ℓ, and only consider the dependence on r, i.e. $w_\ell(r)$. Then, dividing by D on both sides:

$$\frac{d^2 w_\ell}{dr^2} + \left(\frac{D'}{D} + \frac{2}{r}\right)\frac{dw_\ell}{dr} - \left(\frac{\ell(\ell+1)}{Dr^2} + \frac{\mu^2}{D}\right) w_\ell = 0. \tag{2.35}$$

This is an ordinary differential equation (ODE), which are well studied and we can calculate its solutions around a singularity. See Appendix A for a description of some ODEs. We consider the polynomials:

$$f(r) = \frac{D'}{D} + \frac{2}{r}, \qquad g(r) = -\frac{\ell(\ell+1)}{Dr^2} - \frac{\mu^2}{D}. \tag{2.36}$$

In order to solve this ODE and find an expression for $w_\ell(r)$, we are going to use the procedures defined on Appendix A. First, we are going to classify the singularities of our spacetime, $r = r_+$ and $r = r_c$, with respect to the functions $f(r)$ and $g(r)$.

- *Close to* $r = r_+$.

Let us first consider $r = r_+$. This functions are not smooth around the singularity, i.e. for $\epsilon > 0$,

$$f(r), g(r) \notin C^\infty(r_+ - \epsilon, r_+ + \epsilon),$$

so $r = r_+$ is not an ordinary point. The functions $f(r)$ and $g(r)$ have factor of the $D(r)$ on the denominator. We saw the definition (2.3) in terms of $(r - r_+)$, and its derivative is

$$D'(r) = -\frac{\Lambda}{3r^2}(r-r_-)(r-r_+)(r_c-r) + \frac{\Lambda}{3r}(r-r_-)(r_c-r) + \frac{\Lambda}{3r}(r-r_+)(r_c-r) - \frac{\Lambda}{3r}(r-r_-)(r-r_+). \tag{2.37}$$

So then:

$$(r - r_+)\left.\frac{D'}{D}\right|_{r\downarrow r_+} = O(1).$$

Then our functions (2.36) dependence on the factor $(r - r_+)$ is

$$f(r)|_{r\downarrow r_+} = \frac{1}{r - r_+} + O(1), \tag{2.38}$$

$$g(r)|_{r\downarrow r_+} = -\frac{3}{\Lambda(r_+ - r_-)(r_c - r_+)}\left(\frac{\ell(\ell+1)}{r_+} + r_+\mu^2\right)\frac{1}{r - r_+} + O(1). \tag{2.39}$$

The functions $f(r)$ and $g(r)$ have a leading order factor of $(r - r_+)^{-1}$. Then,

$$(r - r_+)f(r), (r - r_+)^2 g(r) \in C^\infty(r_+ - \epsilon, r_+ + \epsilon),$$

for $\epsilon > 0$, so $r = r_+$ is a *regular singularity*. Our case is even less demanding, since $(r-r_+)g(r) \in C^\infty(r_+ - \epsilon, r_+ + \epsilon)$ already.

Consider the change of coordinates $z = r - r_+$ around the event horizon $\mathcal{H}^+$, so that $z \geq 0$, since we are in the region $r_+ \leq r \leq r_c$. We can do a Taylor expansion of $f(z)$ and $g(z)$ around the singularity $r = r_+$, i.e. around $z = 0$,

$$zf(z) = \sum_{n=0}^{\infty} f_{n-1} z^n, \qquad z^2 g(z) = \sum_{n=0}^{\infty} g_{n-2} z^n,$$

where at least one of the coefficients f_{-1}, g_{-2}, and g_{-1} is non-zero. If all of them were zero, that would mean that r_+ is an ordinary point. In our case, considering the expansions (2.38) and (2.39),

$$f_{-1}|_{r\downarrow r_+} = 1, \quad g_{-2}|_{r\downarrow r_+} = 0, \quad g_{-1}|_{r\downarrow r_+} = -\frac{3}{\Lambda(r_+ - r_-)(r_c - r_+)}\left(\frac{\ell(\ell+1)}{r_+} + r_+\mu^2\right). \tag{2.40}$$

Our equation now is

$$\frac{d^2w}{dr^2} + f(z)\frac{dw}{dz} + g(z)w = 0. \tag{2.41}$$

As an *ansatz*, we consider that the solution is of the form of a Frobenius series:

$$w(z) = z^\alpha \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} a_n z^{n+\alpha}. \tag{2.42}$$

Taking it into our equation (2.41):

$$\begin{aligned} 0 =& \sum_{n=0}^{\infty}(n+\alpha)(n+\alpha-1)a_n z^{n+\alpha-2} + \left(\frac{f_{-1}}{z} + f_0 + f_1 z + \cdots\right)\sum_{n=0}^{\infty}(n+\alpha)a_n z^{n+\alpha-1} \\ &+ \left(\frac{g_{-2}}{z^2} + \frac{g_{-1}}{z} + g_0 + g_1 z + \cdots\right)\sum_{n=0}^{\infty} a_n z^{n+\alpha} \\ =& \sum_{n=0}^{\infty}(n+\alpha)\left[(n+\alpha-1) + f_{-1} + g_{-2}\right]a_n z^{n+\alpha-2} + \sum_{n=0}^{\infty}(n+\alpha)a_n\left[f_0 + g_{-1}\right]z^{n+\alpha-1} + \cdots \end{aligned}$$

Since the $\{z^n\}$ are linearly independent, we get a system of equations:

$$\begin{aligned} \sum_{n=0}^{\infty}(n+\alpha)\left[(n+\alpha-1) + f_{-1} + g_{-2}\right]a_n = 0, \\ \sum_{n=0}^{\infty}(n+\alpha)a_n\left[f_0 + g_{-1}\right] = 0, \\ \cdots \end{aligned}$$

We need the value $a_0 = w(0) \neq 0$ to be prescribed, so for $n = 0$,

$$\alpha(\alpha-1) + f_{-1}\alpha + g_{-2} = 0.$$

This equation is known as the *indicial equation*. Consider α and β as the two roots of this equation, they are called the *exponents* of the singularity $r = r_+$. Substituting into the indicial equation our coefficients f_{-1}, g_{-2}, and g_{-1}, given in (2.40), we find $\alpha = 0$ and $\beta = 1 - f_{-1} = 0$. These exponents do not make sense when added into our *ansatz* (2.42), since they would both give us a general solution corresponding to an ordinary point:

$$w(z) = \sum_{n=0}^{\infty} a_n z^n.$$

In the case that both exponents are zero, the theory tells us that one of the solutions corresponds to a Frobenius series, and the other is not. For the first solution, we take a Frobenius

series, as (2.42), where $\alpha = 0$, so

$$w_1(z) = \sum_{n=0}^{\infty} a_n z^n. \tag{2.43}$$

To find an independent second solution $w_2(z)$, we follow the procedure described on §A.2.1. Substituting our exponent $\alpha = \beta = 0$ into (A.7),

$$w_2(z) = \frac{1}{a_0^2}\left(\sum_{n=0}^{\infty} a_n z^n\right)\log z + \sum_{n=0}^{\infty} b_n z^{n+1} \tag{2.44}$$

Since $1/a_0^2$ does not vanish, $w_2(z)$ has a logarithmic branch point at the singularity, and:

$$w_2(z) \sim \frac{\log z}{a_0} \qquad \text{as} \quad z \to 0.$$

Grouping together the two solutions $w_1(z)$ and $w_2(z)$, we find that the general solution for (2.35) around the singularity $r = r_+$,

$$w(z)|_{r\downarrow r_+} = A_+ \sum_{n=0}^{\infty} a_n z^n \; + \; B_+ \log z \sum_{n=0}^{\infty} b_n z^n, \tag{2.45}$$

where A_+, B_+, $\{a_n\}$, and $\{b_n\}$ are constants, with the condition:

$$B_+ \propto \frac{1}{b_0^2} \qquad \text{with} \quad b_0 \neq 0.$$

Rewriting (2.45) in terms of r,

$$w(r)|_{r\downarrow r_+} = A_+ \sum_{n=0}^{\infty} a_n (r - r_+)^n \; + \; B_+ \log(r - r_+) \sum_{n=0}^{\infty} b_n (r - r_+)^n.$$

- *Close to* $r = r_c$.

Until here we have considered the singularity of the event horizon $\mathcal{H}^+$, but we will consider what happens when we are around the cosmological horizon $\mathcal{C}^+$. First, we classify the singularity $r = r_c$.

The functions $f(r)$ and $g(r)$ are not smooth around the singularity, i.e. for $\epsilon > 0$,

$$f(r), g(r) \;\notin\; C^\infty(r_c - \epsilon, r_c + \epsilon),$$

so $r = r_c$ is not an ordinary point. Consider the values of $D(r)$ and $D'(r)$ of (2.3) and (2.37). The functions $f(r)$ and $g(r)$ in (2.36), depend on the factor $(r_c - r)$ as

$$f(r)|_{r\uparrow r_c} = \frac{1}{r_c - r} + O(1), \tag{2.46}$$

$$g(r)|_{r\uparrow r_c} = -\frac{3}{\Lambda(r_c - r_-)(r_c - r_+)}\left(\frac{\ell(\ell+1)}{r_c} + r_c\mu^2\right)\frac{1}{r_c - r} + O(1). \tag{2.47}$$

Both functions $f(r)$ and $g(r)$ have a leading order factor of $(r_c - r)^{-1}$. Then,

$$(r_c - r)f(r), (r_c - r)^2 g(r) \;\in\; C^\infty(r_c - \epsilon, r_c + \epsilon),$$

for $\epsilon > 0$, so $r = r_c$ is a *regular singularity*. Take the change of coordinates $z = r_c - r$, for z to be positive, since we are in the region $r_+ \leq r \leq r_c$. The process of finding solutions $w(z)$ for the ODE around the cosmological horizon is the same as the one we just did for $r = r_+$, the only difference is on the coefficient g_{-1},

$$g_{-1}|_{r\uparrow r_c} = -\frac{3}{\Lambda(r_c - r_-)(r_c - r_+)}\left(\frac{\ell(\ell+1)}{r_c} + r_c\mu^2\right), \tag{2.48}$$

and it will not affect the final result.

Around the singularity $r = r_c$ we would find a solution $w(z)$ to the equation (2.35) of the form:

$$w(z)|_{r\uparrow r_c} = A_c \sum_{n=0}^{\infty} \underline{a}_n z^n \; + \; B_c \log z \sum_{n=0}^{\infty} \underline{b}_n z^n, \tag{2.49}$$

where A_c, B_c, $\{\underline{a}_n\}$, and $\{\underline{b}_n\}$ are constants, with the condition:

$$B_c \propto \frac{1}{\underline{b}_0^2} \qquad \text{with} \quad \underline{b}_0 \neq 0.$$

Rewriting (2.49) in terms of r,

$$w(r)|_{r\uparrow r_c} = A_c \sum_{n=0}^{\infty} \underline{a}_n (r_c - r)^n \; + \; B_+ \log(r_c - r) \sum_{n=0}^{\infty} \underline{b}_n (r_c - r)^n.$$

We have the freedom to choose the values of the constants in the expression of $w(r)$. Near $r = r_+$ we can take $A_+ = 1/a_0$ and $B_+ = 0$, this way:

$$w(r)|_{r\downarrow r_+} = 1 + O(r - r_+). \tag{2.50}$$

If $\mu^2 r^2 + \ell(\ell+1) > 0$ and $w_\ell = 1$, then the Logarithmic branch near $r = r_c$ is always present, that is why we must have $B_c \neq 0$. If we took $B_c = 0$, then $w(r) = 1$ everywhere and we would have the constant function $l = 0$, and we are interested in having a finite value of $r\phi$ when r goes to infinity. Then, $B_c \neq 0$,

$$w(r)|_{r\uparrow r_c} = A_c\, \underline{a}_0 + B_c\, \underline{b}_0 \log(r_c - r) + O(r_c - r). \tag{2.51}$$

Properties of the stationary function

We define the operator $\mathcal{L}_{\mu,\ell}$ as follows:

$$\mathcal{L}_{\mu,\ell}\, u = \frac{d}{dr}\left(Dr^2 \frac{du}{dr}\right) - \left[\mu^2 r^2 + \ell(\ell+1)\right] u,$$

with $\mu^2 r^2 + \ell(\ell+1) > 0$, since we assume $\mu \neq 0$ or $\ell \neq 0$.

Proposition 4. *Let $u \in C^\infty([r_+, r_c])$, then there exists a $C = C(\mu, \ell) > 0$ such that*

$$\int_{r_+}^{r_c} Dr^2 \left(\frac{du}{dr}\right)^2 dr \leq C \int_{r_+}^{r_c} (\mathcal{L}_{\mu,\ell}\, u)^2\, dr. \tag{2.52}$$

Proof. We first multiply $\mathcal{L}_{\mu,\ell}\, u$ with $-u$ and integrate on $[r_+, r_c]$ to obtain:

$$\int_{r_+}^{r_c} -u\mathcal{L}_{\mu,\ell}\, u\, dr = \int_{r_+}^{r_c} \left[-u\frac{d}{dr}\left(Dr^2\frac{du}{dr}\right) + \left[\mu^2 r^2 + \ell\,(\ell+1)\right] u^2\right] dr.$$

We integrate the first integral on the right-hand side by parts:

$$\int_{r_+}^{r_c} -u\frac{d}{dr}\left(Dr^2\frac{du}{dr}\right) dr = -uDr^2\,\frac{du}{dr}\bigg|_{r=r_c} + uDr^2\,\frac{du}{dr}\bigg|_{r=r_+} + \int_{r_+}^{r_c} Dr^2\left(\frac{du}{dr}\right)^2 dr.$$

The boundary terms are zero since $D(r_+) = 0 = D(r_c)$. Then,

$$\int_{r_+}^{r_c} -u\mathcal{L}_{\mu,\ell}\, u\, dr = \int_{r_+}^{r_c} \left[Dr^2\left(\frac{du}{dr}\right)^2 + \left[\mu^2 r^2 + \ell\,(\ell+1)\right] u^2\right] dr. \tag{2.53}$$

We apply Young's inequality (2.17) to estimate:

$$|u\mathcal{L}_{\mu,\ell}| \leq \frac{\epsilon}{2}u^2 + \frac{1}{2\epsilon}\left(\mathcal{L}_{\mu,\ell}\, u\right)^2.$$

We can absorb the term $\frac{\epsilon}{2}u^2$ by the term u^2 in (2.53) if we consider $\epsilon \leq 2\left[\mu^2 r^2 + \ell\,(\ell+1)\right]$, i.e.

$$\int_{r_+}^{r_c} \left[Dr^2\left(\frac{du}{dr}\right)^2 + \left[\mu^2 r^2 + \ell\,(\ell+1)\right] u^2\right] dr \leq C\int_{r_+}^{r_c} \left(\mathcal{L}_{\mu,\ell}\, u\right)^2 dr.$$

We retrieve the result (2.52) we were looking for. □

For a function u to be $u \equiv 0$ the following two conditions have to be fulfilled at the same time:

$$\begin{cases} \mathcal{L}_{\mu,\ell}\, u = 0, \\ uDr^2\,\dfrac{du}{dr}\bigg|_{r=r_+,r_c} = 0. \end{cases} \tag{2.54}$$

For our stationary function w_ℓ, the first condition is already fulfilled:

$$\mathcal{L}_{\mu,\ell}\, w_\ell = 0 \qquad \Leftrightarrow \qquad \Box_g w_\ell = \mu^2 w_\ell.$$

Strict positivity of w_ℓ can be shown to hold via a contradiction argument. Suppose there exists $r_n \in [r_+, r_c]$ such that $w_\ell(r_n) = 0$. Consider the interval $[r_+, r_n]$, the conditions (2.54) are both fulfilled,

$$\mathcal{L}_{\mu,\ell}\, w_\ell = 0, \qquad\qquad w_\ell Dr^2\,\frac{dw_\ell}{dr}\bigg|_{r=r_+,r_n} = 0.$$

Then we can apply the estimates from the proof of Proposition 4 with r_c replaced by r_n,

$$\int_{r_+}^{r_c} \left[Dr^2\left(\frac{dw_\ell}{dr}\right)^2 + \left[\mu^2 r^2 + \ell\,(\ell+1)\right] w_\ell^2\right] dr \leq C\int_{r_+}^{r_c} \left(\mathcal{L}_{\mu,\ell}\, w_\ell\right)^2 dr = 0.$$

We obtain L^2-bounds for w_ℓ in $[r_+, r_n]$, from which it follows that $w_\ell \equiv 0$ in $[r_+, r_n]$. The same argument can be done for the interval $[r_n, r_c]$, so this way we find $w_\ell \equiv 0$ in $[r_+, r_c]$, which is a contradiction with $B_c \neq 0$: we are not interested in the null function. This way, w_ℓ has to be a *smooth, non-vanishing function* in $[r_+, r_c]$.

Chapter 3

Energy boundedness

In this chapter, we will derive estimates for energies with are non-negative definite integrals. We will show that E naturally comes up when integrating $\partial_t\phi\left(\Box_g\phi - \mu^2\phi\right) = 0$ in spacetime $(\mathcal{M}_{ext}, g)$, and that it is conserved in time as a consequence of the t-invariance of the coefficients of the KG equation.

We define the *degenerate energy* of a Klein-Gordon field ϕ as follows:

$$\begin{aligned} E_{deg}[\phi](\tau) := &\int\limits_{\Sigma_\tau\cap\{r_1\leq r\leq R\}} 2\left[Dr^2(X\phi)^2 + \frac{1}{D}r^2(T\phi)^2 + \left|\not\nabla_{\mathbb{S}^2}\phi\right|^2 + \mu^2r^2\phi^2\right]d\sigma\,dr \\ &+ \int\limits_{N_\tau}\left[2r^2(L\phi)^2 + \frac{D}{2}\left|\not\nabla_{\mathbb{S}^2}\phi\right|^2 + \frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,dv \\ &+ \int\limits_{\underline{N}_\tau}\left[2r^2(\underline{L}\phi)^2 + \frac{D}{2}\left|\not\nabla_{\mathbb{S}^2}\phi\right|^2 + \frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,du. \end{aligned} \tag{3.1}$$

On the integral terms in N_τ and $\underline{N}_\tau$ we find a factor of $D(r)$ inside the integrands. By its definition in (2.3), $D(r) \propto (r - r_+)(r_c - r)$ tends to zero as r goes towards the horizons. This implies that for all values of τ the value of the integrals in N_τ and $\underline{N}_\tau$ at the horizon are the same, this is the reason why we call this energy degenerate.

Proposition 5. *Let $\phi \in C^\infty(\mathcal{R}) \cap C^1(\hat{\mathcal{R}})$ be a solution to (2.31) in the spacetime $(\mathcal{M}_{ext}, g)$. Then for all $\tau_1, \tau_2 \in \mathbb{R}$ such that $0 \leq \tau_1 < \tau_2$,*

$$E_{deg}[\phi](\tau_2) + \int\limits_{\mathcal{H}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2r^2(L\phi)^2 d\sigma dv + \int\limits_{\mathcal{C}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2r^2(\underline{L}\phi)^2 d\sigma du = E_{deg}[\phi](\tau_1). \tag{3.2}$$

Furthermore, since those two integrals are positive definite, there exist a constant $C > 0$ such that

$$E_{deg}[\phi](\tau_2) \leq C\; E_{deg}[\phi](\tau_1).$$

This can be interpreted as there being a positive energy flux through the horizons $\mathcal{H}^+$ and $\mathcal{C}^+$.

Proof. We are going to see where this energy expression comes from. There exists a conservation relation for the stress-energy tensor, $\nabla^\mu\mathbb{T}_{\mu\nu} = 0$, which for the stress energy tensor associated

to the Klein-Gordon field ϕ (2.34), takes the form:

$$\begin{aligned}0 =&\nabla^\mu \mathbb{T}_{\mu\nu} = \nabla^\mu \left(\nabla_\mu\phi\nabla_\nu\phi\right) - \frac{1}{2}g_{\mu\nu}\left(\left(g^{-1}\right)^{\alpha\beta}\nabla^\mu\left(\nabla_\alpha\phi\nabla_\beta\phi\right)\right) - \frac{1}{2}\mu^2\nabla^\mu\left(g_{\mu\nu}\phi^2\right)\\ =&\Box_g\phi\nabla_\nu\phi + \nabla_\mu\phi\nabla^\mu\nabla_\nu\phi - g_{\mu\nu}\left(\left(g^{-1}\right)^{\alpha\beta}\left(\nabla^\mu\nabla_\alpha\phi\right)\nabla_\beta\phi\right) - \frac{1}{2}\mu^2\nabla_\nu\left(\phi^2\right)\\ =&\Box_g\phi\nabla_\nu\phi - \left(\nabla_\nu\nabla_\alpha\phi\right)\nabla^\alpha\phi + \left(\nabla_\mu\nabla_\nu\phi\right)\nabla^\mu\phi - \mu^2\left(\nabla_\nu\phi^2\right)\\ =&\Box_g\phi\nabla_\nu\phi - \mu^2\left(\nabla_\nu\phi^2\right),\end{aligned}$$

where we made use of the Leibniz rule. Since ∇^μ is a covariant derivative, it fulfils the property $\nabla g = 0$. The second and third terms on the third line cancel one another, we can see that using the covariant derivative definition by Christoffel symbols in (2.26).

Considering the Killing vector field $T = \partial_t$, the Noether's current associated to T is $J = \mathbb{T}_{\mu\nu}T^\nu$. Since T is a Killing vector field, J is conserved, that is

$$\text{div } J = \nabla^\mu\left(\mathbb{T}_{\mu\nu}T^\nu\right) = \left(\nabla^\mu\mathbb{T}_{\mu\nu}\right)T^\nu + \mathbb{T}_{\mu\nu}\nabla^\mu T^\nu = 0,$$

Since T is a Killing vector field, $\nabla^\mu T^\nu$ in antisymmetric, and the expression of $\mathbb{T}_{\mu\nu}$ in (2.34), is also antisymmetric, so the term $\mathbb{T}_{\mu\nu}\nabla^\mu T^\nu$ is zero. Then combining this last two equations:

$$0 = \text{div } J = \left(\nabla^\mu\mathbb{T}_{\mu\nu}\right)T^\nu = \Box_g\phi\nabla_\nu\phi T^\nu - \mu^2\left(\nabla_\nu\phi\right)^2 T^\nu = T\phi\left(\Box_g\phi - \mu^2\phi\right).$$

We want to integrate $T\phi\left(\Box_g\phi - \mu^2\phi\right)$ by parts on a suitable time slab. First we multiply on both sides by $-Dr^2$:

$$\begin{aligned}0 =&-Dr^2T\phi\left(\Box_g\phi - \mu^2\phi\right) = -Dr^2\left(L\phi + \underline{L}\phi\right)\left(-\frac{4}{D}\underline{L}L\phi + \frac{2}{r}\left(L\phi - \underline{L}\phi\right) + \frac{1}{r^2}\not{\Delta}_{\mathbb{S}^2}\phi - \mu^2\phi\right)\\ =&4r^2L\phi\underline{L}L\phi + 4r^2\underline{L}\phi\underline{L}L\phi - 2Dr\left(\left(L\phi\right)^2 - \left(\underline{L}\phi\right)^2\right) - D\not{\Delta}_{\mathbb{S}^2}\phi\left(L + \underline{L}\right)\phi + \mu^2Dr^2\phi\left(L + \underline{L}\right)\phi\\ =&2r^2\underline{L}\left(\left(L\phi\right)^2\right) + 2r^2L\left(\left(\underline{L}\phi\right)^2\right) - 2Dr\left(\left(L\phi\right)^2 - \left(\underline{L}\phi\right)^2\right) - D\not{\nabla}_{\mathbb{S}^2}\left(\not{\nabla}_{\mathbb{S}^2}\phi\left(L + \underline{L}\right)\phi\right)\\ &+ D\not{\nabla}_{\mathbb{S}^2}\phi\not{\nabla}_{\mathbb{S}^2}\left(\left(L + \underline{L}\right)\phi\right) + \frac{1}{2}\mu^2\left(L + \underline{L}\right)\left(Dr^2\phi^2\right)\\ =&\underline{L}\left(2r^2\left(L\phi\right)^2 + \frac{D}{2}\left|\not{\nabla}_{\mathbb{S}^2}\phi\right|^2 + \frac{D}{2}\mu^2r^2\phi^2\right) + L\left(2r^2\left(\underline{L}\phi\right)^2 + \frac{D}{2}\left|\not{\nabla}_{\mathbb{S}^2}\phi\right|^2 + \frac{D}{2}\mu^2r^2\phi^2\right)\\ &- \not{\nabla}_{\mathbb{S}^2}\left(D\not{\nabla}_{\mathbb{S}^2}\phi\left(L + \underline{L}\right)\phi\right),\end{aligned}$$

where we used the Leibniz rule and the definitions of $\underline{L}$ and L. The last term on the right hand side corresponds to divergences with respect to $\mathbb{S}^2$, which vanish after integration over $\mathbb{S}^2$, via the divergence theorem.

Integrating by parts over $d\sigma\, du\, dv$ on $\mathcal{R}^{\tau_2}_{\tau_1}$,

$$\begin{aligned}0 =& \int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}}\left[2r^2\left(L\phi\right)^2 + \frac{D}{2}\left|\not{\nabla}_{\mathbb{S}^2}\phi\right|^2 + \frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\, dv\\ &- \int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}}\left[2r^2\left(L\phi\right)^2 + \frac{D}{2}\left|\not{\nabla}_{\mathbb{S}^2}\phi\right|^2 + \frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\, dv\\ &+ \int\limits_{N_{\tau_2}}\left[2r^2\left(L\phi\right)^2 + \frac{D}{2}\left|\not{\nabla}_{\mathbb{S}^2}\phi\right|^2 + \frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\, dv - \int\limits_{N_{\tau_1}}\left[2r^2\left(L\phi\right)^2 + \frac{D}{2}\left|\not{\nabla}_{\mathbb{S}^2}\phi\right|^2 + \frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\, dv\end{aligned}$$

$$
\begin{aligned}
&+\int_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}}\left[2r^2\left(\underline{L}\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,du\\
&+\int_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}}\left[2r^2\left(L\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,dv\\
&+\int_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}}\left[2r^2\left(\underline{L}\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,du\\
&-\int_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}}\left[2r^2\left(\underline{L}\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,du\\
&+\int_{\underline{N}_{\tau_2}}\left[2r^2\left(\underline{L}\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,du-\int_{\underline{N}_{\tau_1}}\left[2r^2\left(\underline{L}\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,du
\end{aligned}
$$

For the integration surfaces $\Sigma_\tau\cap\{r_1\le r\le R\}$, it is more convenient to have an integral in dr rather that du or dv. These are surfaces of constant t, so there we used the following change of coordinates of the differential form,

$$du=\frac{2}{D}dr,\qquad\qquad dv=\frac{2}{D}dr.$$

Then,

$$
\begin{aligned}
&\int_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}}\left[4r^2\left(\left(L\phi\right)^2+\left(\underline{L}\phi\right)^2\right)+\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\mu^2r^2\phi^2\right]d\sigma\,dr\\
&+\int_{N_{\tau_2}}\left[2r^2\left(L\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,dv+\int_{\underline{N}_{\tau_2}}\left[2r^2\left(\underline{L}\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,du\\
&\quad+\int_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}}\left[2r^2\left(\underline{L}\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,du\\
&\quad+\int_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}}\left[2r^2\left(L\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,dv\\
&=\int_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}}\left[4r^2\left(\left(L\phi\right)^2+\left(\underline{L}\phi\right)^2\right)+\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\mu^2r^2\phi^2\right]d\sigma\,dr\\
&+\int_{N_{\tau_1}}\left[2r^2\left(L\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,dv+\int_{\underline{N}_{\tau_1}}\left[2r^2\left(\underline{L}\phi\right)^2+\frac{D}{2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\phi\right|^2+\frac{D}{2}\mu^2r^2\phi^2\right]d\sigma\,du
\end{aligned}
$$

We can rewrite the integrand for the integrals over $\Sigma_\tau\cap\{r_1\le r\le R\}$ in terms of the vector fields X and T, using the relations (2.13),

$$\left(L\phi\right)^2+\left(\underline{L}\phi\right)^2=\frac{1}{2}\left(\left(T\phi\right)^2+D^2\left(X\phi\right)^2\right).$$

From this equation and the definition of the degenerate energy (3.1), we obtain (3.2). □

We introduce the rescaled function $\psi = r\phi$. We can write the Klein-Gordon equation (2.31), in terms of ψ:

$$0 = \Box_g\phi - \mu^2\phi = \Box_g\left(\frac{\psi}{r}\right) - \mu^2\left(\frac{\psi}{r}\right) = -\frac{4}{Dr}\underline{L}L\psi - \frac{D'}{r^2}\psi + \frac{1}{r^3}\not{\Delta}_{\mathbb{S}^2}\psi - \frac{\mu^2}{r}\psi,$$

where we used (2.33). Multiplying both sides by Dr:

$$0 = -4\underline{L}L\psi + \frac{D}{r^2}\not{\Delta}_{\mathbb{S}^2}\psi - D\left(\mu^2 + \frac{D'}{r}\right)\psi. \tag{3.3}$$

We can rewrite the degenerate energy $E_{deg}[\phi](\tau)$ in terms of this function ψ, using the vector fields $\underline{L}$ and L,

$$\begin{aligned} E[\psi](\tau) := &\int_{\Sigma_\tau\cap\{r_1\le r\le R\}} \left[\frac{4}{D}\left((L\psi)^2 + (\underline{L}\psi)^2\right) + \frac{2}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + 2\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, dr \\ &+ \int_{N_\tau}\left[2(L\psi)^2 + \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, dv \\ &+ \int_{\underline{N}_\tau}\left[2(\underline{L}\psi)^2 + \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, du. \end{aligned} \tag{3.4}$$

We can find a boundedness condition for this energy, the same way as in Proposition 5.

Proposition 6. *Let $\phi \in C^\infty(\mathcal{R}) \cap C^1(\hat{\mathcal{R}})$ be a solution to (2.31) in the spacetime $(\mathcal{M}_{ext}, g)$. Let $\psi = r\phi$. Then for all $\tau_1, \tau_2 \in \mathbb{R}$ such that $0 \le \tau_1 < \tau_2$,*

$$E[\psi](\tau_2) + \int_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} 2(L\psi)^2\, d\sigma dv + \int_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}} 2(\underline{L}\psi)^2\, d\sigma du = E[\psi](\tau_1). \tag{3.5}$$

Furthermore, since those two integrals are positive definite, there exist a constant $C > 0$ such that

$$E[\psi](\tau_2) \le C\, E[\psi](\tau_1). \tag{3.6}$$

This can be interpreted as there being a positive energy flux through the horizons $\mathcal{H}^+$ and $\mathcal{C}^+$.

Proof. By definition, the vector field T goes parallel to the hypersurfaces $\mathcal{C}^+$ and $\{r = \tilde{R}\}$. We integrate $-T\psi\left(\Box_g\phi - \mu^2\phi\right)$ by parts on a suitable time slab. Using (3.3) and the Leibniz rule,

$$\begin{aligned} 0 =& -T\psi\left(\Box_g\phi - \mu^2\phi\right) = -\left(L\psi + \underline{L}\psi\right)\left[-4\underline{L}L\psi + \frac{D}{r^2}\not{\Delta}_{\mathbb{S}^2}\psi - D\left(\mu^2 + \frac{D'}{r}\right)\psi\right] \\ =& 4L\psi\underline{L}L\psi + 4\underline{L}\psi\underline{L}L\psi - \frac{D}{r^2}\left(L\psi + \underline{L}\psi\right)\not{\Delta}_{\mathbb{S}^2}\psi + D\left(\mu^2 + \frac{D'}{r}\right)\left(L\psi + \underline{L}\psi\right)\psi \\ =& 2\underline{L}\left((L\psi)^2\right) + 2L\left((\underline{L}\psi)^2\right) - \nabla\!\!\!/_{\mathbb{S}^2}\left(\frac{D}{r^2}\left(L\psi + \underline{L}\psi\right)\nabla\!\!\!/_{\mathbb{S}^2}\psi\right) \\ &+ \frac{1}{2}(L + \underline{L})\left(\frac{D}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2\right) + \frac{1}{2}(L + \underline{L})\left(D\left[\mu^2 + \frac{D'}{r}\right]\psi^2\right) \\ =& \underline{L}\left(2\left(L\psi\right)^2 + \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{D}{2}\left[\mu^2 + \frac{D'}{r}\right]\psi^2\right) - \nabla\!\!\!/_{\mathbb{S}^2}\left(\frac{D}{r^2}\left(L\psi + \underline{L}\psi\right)\nabla\!\!\!/_{\mathbb{S}^2}\psi\right) \end{aligned}$$

$$+ L\left(2\left(\underline{L}\psi\right)^2 + \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{D}{2}\left[\mu^2 + \frac{D'}{r}\right]\psi^2\right).$$

The term $\nabla\!\!\!/_{\mathbb{S}^2}(\cdots)$ corresponds to divergences with respect to $\mathbb{S}^2$, which vanish after integration over $\mathbb{S}^2$, via the divergence theorem. Integrating by parts over $d\sigma\, du\, dv$ on $\mathcal{R}^{\tau_2}_{\tau_1}$,

$$\begin{aligned}
0 = &\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}} \left[\frac{4}{D}\left((L\psi)^2 + (\underline{L}\psi)^2\right) + \frac{2}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + 2\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, dr \\
&- \int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}} \left[\frac{4}{D}\left((L\psi)^2 + (\underline{L}\psi)^2\right) + \frac{2}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + 2\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, dr \\
&\quad + \int\limits_{N_{\tau_2}} \left[2(L\psi)^2 + \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, dv \\
&\quad - \int\limits_{N_{\tau_1}} \left[2(L\psi)^2 + \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, dv \\
&\quad + \int\limits_{\underline{N}_{\tau_2}} \left[2(\underline{L}\psi)^2 + \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, du \\
&\quad - \int\limits_{\underline{N}_{\tau_1}} \left[2(\underline{L}\psi)^2 + \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, du \\
&\quad + \int\limits_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} 2(L\psi)^2\, d\sigma dv + \int\limits_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}} 2(\underline{L}\psi)^2\, d\sigma du
\end{aligned}$$

where we used that $D(r_+) = 0$ and $D(r_c) = 0$. For some of the integrals, we applied a change of coordinates over the differential forms such that, for constant t surfaces, $du = -\frac{2}{D}dr$ and $dv = \frac{2}{D}dr$, and for constant r surfaces, $du = 2dt = dv$. Identifying the energy terms with the definition of $E[\psi](\tau)$ in (3.4), we find (3.5). □

3.1 Energy boundedness for the renormalized wave function

Consider the spherical harmonic modes ϕ_ℓ of the solution to the Klein-Gordon equation (2.31),

$$\phi = \sum_{\ell=0}^{\infty}\phi_\ell, \qquad\qquad \phi_\ell(t,r,\theta,\varphi) = \sum_{m=-\ell}^{\ell}\phi_{\ell m}(t,r)Y_{\ell m}(\theta,\varphi),$$

such that, for all $\ell \in \mathbb{N}_0$,

$$\Box_g\phi_\ell = \mu^2\phi_\ell \qquad \text{for} \quad \mu \in \mathbb{R}. \tag{3.7}$$

We introduce the following *renormalization* of the spherical harmonic modes ϕ_ℓ:

$$\check{\phi}_\ell(t,r,\theta,\varphi) := \frac{\phi_\ell}{w_\ell},$$

where $w_\ell(r)$ is the spherical harmonic mode of the stationary wave $w(r,\theta,\varphi)$. We obtain equations for the renormalized functions $\check{\phi}_\ell$ in the lemma below.

Lemma 3.1.1. *Let ϕ be a solution to the Klein-Gordon equation (2.31). Then $\check{\phi}_\ell$ satisfies the equation:*

$$0 = Z^2\check{\phi}_\ell + 2D\left(\frac{1}{r} + \frac{1}{w_\ell}\frac{dw_\ell}{dr}\right)Z\check{\phi}_\ell - T^2\check{\phi}_\ell. \tag{3.8}$$

Proof. We substitute $\phi_\ell = w_\ell\check{\phi}_\ell$ into the Klein-Gordon equation (3.7), where we use the equality $\not{\Delta}_{\mathbb{S}^2}\phi_\ell = -\ell(\ell+1)\phi_\ell$,

$$\begin{aligned}
0 =& \Box_g\phi_\ell - \mu^2\phi_\ell = \Box_g\left(w_\ell\check{\phi}_\ell\right) - \mu^2 w_\ell\check{\phi}_\ell \\
=& -\frac{1}{D}\partial_t^2\left(w_\ell\check{\phi}_\ell\right) + \frac{1}{r^2}\partial_r\left(Dr^2\partial_r\left(w_\ell\check{\phi}_\ell\right)\right) + \frac{1}{r^2}\not{\Delta}_{\mathbb{S}^2}\left(w_\ell\check{\phi}_\ell\right) - \mu^2 w_\ell\check{\phi}_\ell \\
=& -\frac{1}{D}w_\ell\partial_t^2\check{\phi}_\ell + \frac{1}{r^2}\partial_r\left(Dr^2\partial_r\left[\left(\partial_r w_\ell\right)\check{\phi}_\ell + w_\ell\left(\partial_r\check{\phi}_\ell\right)\right]\right) - \left(\frac{\ell(\ell+1)}{r^2} + \mu^2\right)w_\ell\check{\phi}_\ell \\
=& -\frac{1}{D}w_\ell\partial_t^2\check{\phi}_\ell + \left(D' + \frac{2D}{r}\right)\check{\phi}_\ell\partial_r w_\ell + 2D\partial_r\check{\phi}_\ell\partial_r w_\ell + D\check{\phi}_\ell\partial_r^2 w_\ell \\
&+ \left(D' + \frac{2D}{r}\right)w_\ell\partial_r\check{\phi}_\ell + Dw_\ell\partial_r^2\check{\phi}_\ell - \left(\frac{\ell(\ell+1)}{r^2} + \mu^2\right)w_\ell\check{\phi}_\ell.
\end{aligned}$$

Put the linear terms in $\check{\phi}_\ell$ together,

$$\begin{aligned}
0 =& -\frac{1}{D}w_\ell\partial_t^2\check{\phi}_\ell + \left(2D\partial_r w_\ell + D'w_\ell + \frac{2D}{r}w_\ell\right)\partial_r\check{\phi}_\ell + Dw_\ell\partial_r^2\check{\phi}_\ell \\
&+ \left[D\partial_r^2 w_\ell + \left(D' + \frac{2D}{r}\right)\partial_r w_\ell - \frac{\ell(\ell+1)}{r^2}w_\ell - \mu^2 w_\ell\right]\check{\phi}_\ell
\end{aligned}$$

The last term on the right-hand side is zero, using the equation (2.35). Multiplying by D/w_ℓ,

$$0 = -\partial_t^2\check{\phi}_\ell + \left(2D^2\frac{\partial_r w_\ell}{w_\ell} + DD' + \frac{2D^2}{r}\right)\partial_r\check{\phi}_\ell + D^2\partial_r^2\check{\phi}_\ell.$$

and changing to the vector field notation (2.13), with $Z^2 = D^2\partial_r^2 + DD'\partial_r$,

$$0 = -T^2\check{\phi}_\ell + 2D\left(\frac{\partial_r w_\ell}{w_\ell} + \frac{1}{r}\right)Z\check{\phi}_\ell + Z^2\check{\phi}_\ell.$$

□

Remark. *The choice of renormalization $\check{\phi}_\ell$ is made so that the equations satisfied by $\check{\phi}_\ell$ contain only T- and Z-derivatives of $\check{\phi}_\ell$, i.e. they do not feature zeroth-order terms in $\check{\phi}_\ell$.*

We can find an energy estimate for this renormalized function. Firstly, we need to define the *renormalized energy* of the renormalized Klein-Gordon field $\check{\phi}_\ell$ as follows:

$$\begin{aligned}
\check{E}\left[\check{\phi}_\ell\right](\tau) :=& \int\limits_{\Sigma_\tau\cap\{r_1\le r\le R\}} \frac{1}{D}r^2 w_\ell^2\left(\left(T\check{\phi}_\ell\right)^2 + \left(Z\check{\phi}_\ell\right)^2\right)d\sigma\,dr \\
&+ \int\limits_{N_\tau} r^2 w_\ell^2\left(L\check{\phi}_\ell\right)^2 d\sigma\,dv + \int\limits_{\underline{N}_\tau} r^2 w_\ell^2\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\,du.
\end{aligned} \tag{3.9}$$

The integrated energy boundedness estimates for $\check{\phi}_\ell$:

Proposition 7. *Let $\phi \in C^\infty(\mathcal{R}) \cap C^1(\hat{\mathcal{R}})$ be a solution to (2.31) in the spacetime $(\mathcal{M}_{ext}, g)$, and $\check{\phi}_\ell$ the renormalization of the spherical harmonic modes ϕ_ℓ. Then for all $\tau_1, \tau_2 \in \mathbb{R}$ such that $0 \leq \tau_1 < \tau_2$,*

$$\check{E}\left[\check{\phi}_\ell\right](\tau_2) + \int\limits_{\mathcal{C}^+ \cap \{\tau_1 \leq \tau \leq \tau_2\}} r^2 w_\ell^2 (\underline{L}\check{\phi}_\ell)^2 d\sigma\, du + \int\limits_{\mathcal{H}^+ \cap \{\tau_1 \leq \tau \leq \tau_2\}} r^2 w_\ell^2 (L\check{\phi}_\ell)^2 d\sigma\, dv = \check{E}\left[\check{\phi}_\ell\right](\tau_1). \tag{3.10}$$

Furthermore, since the integrals over $\mathcal{C}^+$ and $\mathcal{H}^+$ are positive definite, there exist a constant $C > 0$ such that

$$\check{E}\left[\check{\phi}_\ell\right](\tau_2) \leq C\, \check{E}\left[\check{\phi}_\ell\right](\tau_1). \tag{3.11}$$

This can be interpreted as there being a positive energy flux through the horizons $\mathcal{H}^+$ and $\mathcal{C}^+$.

Proof. Multiply (3.8) by $f(r)T\check{\phi}_\ell$, and applying the Leibniz rule we to obtain:

$$\begin{aligned}
0 =& f(r)T\check{\phi}_\ell Z^2\check{\phi}_\ell + 2Df(r)\left[\frac{1}{r} + \frac{1}{w_\ell}\frac{dw_\ell}{dr}\right] T\check{\phi}_\ell Z\check{\phi}_\ell - f(r)T\check{\phi}_\ell T^2\check{\phi}_\ell \\
=& Z\left(fT\check{\phi}_\ell Z\check{\phi}_\ell\right) - (Zf)T\check{\phi}_\ell Z\check{\phi}_\ell - \frac{1}{2}T\left(f\left(Z\check{\phi}_\ell\right)^2\right) + 2Df\left[\frac{1}{r} + \frac{1}{w_\ell}\frac{dw_\ell}{dr}\right] T\check{\phi}_\ell Z\check{\phi}_\ell - \frac{1}{2}T\left(f\left(T\check{\phi}_\ell\right)^2\right) \\
=& Z\left(fT\check{\phi}_\ell Z\check{\phi}_\ell\right) - T\left(\frac{f}{2}\left(\left(Z\check{\phi}_\ell\right)^2 + \left(T\check{\phi}_\ell\right)^2\right)\right) + D\left[\frac{2f}{r} + \frac{2f}{w_\ell}\frac{dw_\ell}{dr} - \frac{df}{dr}\right] T\check{\phi}_\ell Z\check{\phi}_\ell.
\end{aligned}$$

It is more convenient to write this equation in terms of the vector fields L and $\underline{L}$, in (u, v, θ, φ) coordinates. Using the relations (2.13),

$$\begin{aligned}
0 =& (L - \underline{L})\left(f\left[L\check{\phi}_\ell + \underline{L}\check{\phi}_\ell\right]\left[L\check{\phi}_\ell - \underline{L}\check{\phi}_\ell\right]\right) - (L + \underline{L})\left(\frac{f}{2}\left[\left(L\check{\phi}_\ell - \underline{L}\check{\phi}_\ell\right)^2 + \left(L\check{\phi}_\ell + \underline{L}\check{\phi}_\ell\right)^2\right]\right) \\
&+ D\left[\frac{2f}{r} + \frac{2f}{w_\ell}\frac{dw_\ell}{dr} - \frac{df}{dr}\right](L\check{\phi}_\ell + \underline{L}\check{\phi}_\ell)(L\check{\phi}_\ell - \underline{L}\check{\phi}_\ell) \\
=& (L - \underline{L})\left(f\left[\left(L\check{\phi}_\ell\right)^2 - \left(\underline{L}\check{\phi}_\ell\right)^2\right]\right) - (L + \underline{L})\left(f\left[\left(L\check{\phi}_\ell\right)^2 + \left(\underline{L}\check{\phi}_\ell\right)^2\right]\right) \\
&+ D\left[\frac{2f}{r} + \frac{2f}{w_\ell}\frac{dw_\ell}{dr} - \frac{df}{dr}\right]\left(\left(L\check{\phi}_\ell\right)^2 + \left(\underline{L}\check{\phi}_\ell\right)^2\right) \\
=& 2L\left(f(r)\left(\underline{L}\check{\phi}_\ell\right)^2\right) - 2\underline{L}\left(f(r)\left(L\check{\phi}_\ell\right)^2\right) + D\left[\frac{2f}{r} + \frac{2f}{w_\ell}\frac{dw_\ell}{dr} - \frac{df}{dr}\right]\left(\left(L\check{\phi}_\ell\right)^2 - \left(\underline{L}\check{\phi}_\ell\right)^2\right).
\end{aligned}$$

We want to get rid of the term in $T\check{\phi}_\ell Z\check{\phi}_\ell$, the one that is going to gives us the bulk term when we integrate this equation. The value of $f(r)$ here does not have to coincide with the one in (4.4), so we take $f(r) = r^2 w_\ell^2$, and substitute it on the previous equation. Thus, dividing both sides by 2,

$$0 = L\left(r^2 w_\ell^2\left(\underline{L}\check{\phi}_\ell\right)^2\right) - \underline{L}\left(r^2 w_\ell^2\left(L\check{\phi}_\ell\right)^2\right).$$

Integrating by parts over $d\sigma\, du\, dv$ on $\mathcal{R}^{\tau_2}_{\tau_1}$,

$$\begin{aligned}0 = &\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}} \frac{2}{D}r^2w_\ell^2\left(\left(\underline{L}\check{\phi}_\ell\right)^2+\left(L\check{\phi}_\ell\right)^2\right)d\sigma\, dr+\int\limits_{\underline{N}_{\tau_2}} r^2w_\ell^2\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du+\int\limits_{N_{\tau_2}} r^2w_\ell^2\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv\\ &-\int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}} \frac{2}{D}r^2w_\ell^2\left(\left(\underline{L}\check{\phi}_\ell\right)^2+\left(L\check{\phi}_\ell\right)^2\right)d\sigma\, dr-\int\limits_{\underline{N}_{\tau_1}} r^2w_\ell^2\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du-\int\limits_{N_{\tau_1}} r^2w_\ell^2\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv\\ &+\int\limits_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}} r^2w_\ell^2(\underline{L}\check{\phi}_\ell)^2 d\sigma\, du+\int\limits_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} r^2w_\ell^2(L\check{\phi}_\ell)^2 d\sigma\, dv.\end{aligned}$$

For the integration surfaces $\Sigma_\tau\cap\{r_1\le r\le R\}$, which are surfaces of constant t, we used the change of coordinates $du=-\frac{2}{D}dr$, and $dv=\frac{2}{D}dr$. Furthermore, we rewrite their integrands as in terms of the vector fields Z, T:

$$(T\check{\phi}_\ell)^2+(Z\check{\phi}_\ell)^2=(L\check{\phi}_\ell+\underline{L}\check{\phi}_\ell)^2+(L\check{\phi}_\ell-\underline{L}\check{\phi}_\ell)^2=2\left(\left(\underline{L}\check{\phi}_\ell\right)^2+\left(L\check{\phi}_\ell\right)^2\right).$$

Then:

$$\begin{aligned}&\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}} \frac{1}{D}r^2w_\ell^2\left(\left(T\check{\phi}_\ell\right)^2+\left(Z\check{\phi}_\ell\right)^2\right)d\sigma\, dr+\int\limits_{\underline{N}_{\tau_2}} r^2w_\ell^2\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du+\int\limits_{N_{\tau_2}} r^2w_\ell^2\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv\\ &\qquad+\int\limits_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}} r^2w_\ell^2(\underline{L}\check{\phi}_\ell)^2 d\sigma\, du+\int\limits_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} r^2w_\ell^2(L\check{\phi}_\ell)^2 d\sigma\, dv\\ =&\int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}} \frac{1}{D}r^2w_\ell^2\left(\left(T\check{\phi}_\ell\right)^2+\left(Z\check{\phi}_\ell\right)^2\right)d\sigma\, dr+\int\limits_{\underline{N}_{\tau_1}} r^2w_\ell^2\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du+\int\limits_{N_{\tau_1}} r^2w_\ell^2\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv.\end{aligned}$$

Identifying the expression of the renormalized energy (3.9) into this equation, we get the expression (3.10). □

Let us consider a hypersurface of constant $\{r=\tilde{R}\}$, which is really close to r_c. It should be independent of R, because R and r_1 could depend on ℓ, and we want $\tilde{R}$ to be fixed for all $\ell\in\mathbb{N}_0$, but with out loss of generality we take $\tilde{R}>R$. We will repeat the previous calculations for the energy boundedness, but taking our integration surface only until $r=\tilde{R}$, instead of r_c, i.e.

$$\mathcal{P}^{\tau_2}_{\tau_1}:=\bigcup_{\tau_1\le\tau\le\tau_2}\Sigma_\tau\cap\{r_+\le r\le\tilde{R}\}.$$

It is represented in Fig. 3.1. This implies that we will have extra terms, since the hypersurface $\{r=\tilde{R}\}$ is timelike, where previously we had a null hypersurface $\{r=r_c\}$.

We can define an integrated quantity that corresponds to the expression of the renormalized energy (3.9), but with the integral over the null hypersurface N_τ only until the value of $\{r=\tilde{R}\}$:

$$\begin{aligned}\check{E}_{\tilde{R}}\left[\check{\phi}_\ell\right](\tau):=&\int\limits_{\Sigma_\tau\cap\{r_1\le r\le R\}} \frac{1}{D}r^2w_\ell^2\left(\left(T\check{\phi}_\ell\right)^2+\left(Z\check{\phi}_\ell\right)^2\right)d\sigma\, dr\\ &+\int\limits_{N_\tau\cap\{r\le\tilde{R}\}} r^2w_\ell^2\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv+\int\limits_{\underline{N}\tau} r^2w_\ell^2\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du.\end{aligned}\tag{3.12}$$

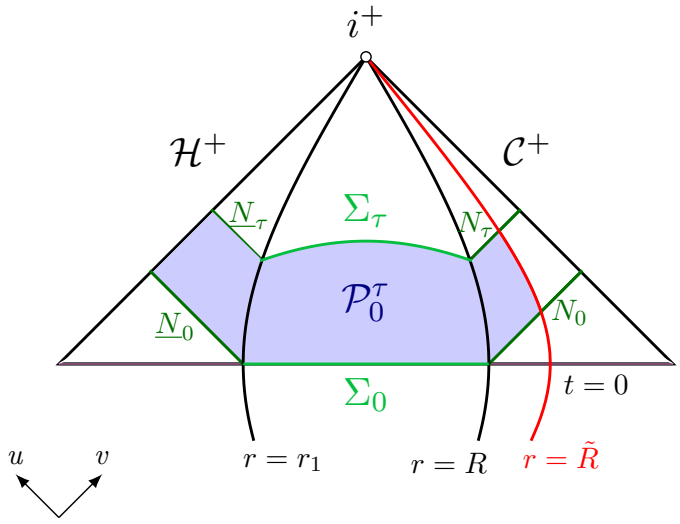


FIGURE 3.1: A 2-dimensional representation of the spacetime subset $\mathcal{R}$ of the RNdS spacetime, for values of $\tau \geq 0$. Two of the spacelike Cauchy hypersurfaces Σ_τ are represented by a green line, for a general value τ and for $\tau = 0$. The black lines are timelike hypersurfaces of constant r, one for each of the values close to the horizon that we have chosen, r_1 and R, and in red we have the timelike hypersurfaces of $\{r = \tilde{R}\}$. The blue area is the integration surface $\mathcal{P}_0^\tau$. Each point in the picture represents a round 2-sphere of radius r.

Since the integral over $N_\tau \cap \{r \geq \tilde{R}\}$ in (3.9) is positive,

$$\check{E}_{\tilde{R}}\left[\check{\phi}_\ell\right](\tau) \leq C\, \check{E}\left[\check{\phi}_\ell\right](\tau). \tag{3.13}$$

We can find the boundedness of the new energy $\check{E}_{\tilde{R}}\left[\check{\phi}_\ell\right](\tau)$ in the space of $\{r \leq \tilde{R}\}$.

Proposition 8. *Let $\phi \in C^\infty(\mathcal{R}) \cap C^1(\hat{\mathcal{R}})$ be a solution to (2.31) in the spacetime $(\mathcal{M}_{ext}, g)$, and $\check{\phi}_\ell$ the renormalization of the spherical harmonic modes ϕ_ℓ. Then for all $\tau_1, \tau_2 \in \mathbb{R}$ such that $0 \leq \tau_1 < \tau_2$,*

$$\begin{aligned}\check{E}_{\tilde{R}}\left[\check{\phi}_\ell\right](\tau_2) + \int_{\tau_1}^{\tau_2} r^2 w_\ell^2 \left(\left(\underline{L}\check{\phi}_\ell\right)^2 - \left(L\check{\phi}_\ell\right)^2\right)\Big|_{r=\tilde{R}} d\sigma\, du + \int_{\mathcal{H}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} r^2 w_\ell^2 (L\check{\phi}_\ell)^2 d\sigma\, dv \\ \leq C\, \check{E}\left[\check{\phi}_\ell\right](\tau_1).\end{aligned} \tag{3.14}$$

Proof. Following the same steps as we did for Proposition 7, we arrive at:

$$0 = L\left(r^2 w_\ell^2 \left(\underline{L}\check{\phi}_\ell\right)^2\right) - \underline{L}\left(r^2 w_\ell^2 \left(L\check{\phi}_\ell\right)^2\right).$$

Integrate it by parts over $d\sigma\, du\, dv$, on our new integration surface $\mathcal{P}_{\tau_1}^{\tau_2}$,

$$\begin{aligned}0 = &\int_{\Sigma_{\tau_2}\cap\{r_1\leq r\leq R\}} \frac{2}{D} r^2 w_\ell^2 \left(\left(\underline{L}\check{\phi}_\ell\right)^2 + \left(L\check{\phi}_\ell\right)^2\right) d\sigma\, dr + \int_{\underline{N}_{\tau_2}} r^2 w_\ell^2 \left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du \\ &+ \int_{N_{\tau_2}\cap\{r\leq\tilde{R}\}} r^2 w_\ell^2 \left(L\check{\phi}_\ell\right)^2 d\sigma\, dv - \int_{\Sigma_{\tau_1}\cap\{r_1\leq r\leq R\}} \frac{2}{D} r^2 w_\ell^2 \left(\left(\underline{L}\check{\phi}_\ell\right)^2 + \left(L\check{\phi}_\ell\right)^2\right) d\sigma\, dr\end{aligned}$$

$$-\int_{\underline{N}_{\tau_1}} r^2 w_\ell^2 \left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du - \int_{N_{\tau_1}\cap\{r\le\tilde{R}\}} r^2 w_\ell^2 \left(L\check{\phi}_\ell\right)^2 d\sigma\, dv$$
$$+\int_{\tau_1}^{\tau_2} r^2 w_\ell^2 \left(\left(\underline{L}\check{\phi}_\ell\right)^2 - \left(L\check{\phi}_\ell\right)^2\right)\Big|_{r=\tilde{R}} d\sigma\, du + \int_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} r^2 w_\ell^2 (L\check{\phi}_\ell)^2 d\sigma\, dv,$$

where we applied a change of coordinates over the terms $\Sigma_\tau \cap \{r_1 \le r \le R\}$, into coordinates (t, r, θ, φ), $du = 2dt - \frac{2}{D}dr$ and $dv = 2dt + \frac{2}{D}dr$. For the terms $\{r = \tilde{R}\}$, it is more convenient to express them in Bondi coordinates (u, r, θ, φ), $du = du$ and $dv = du + \frac{2}{D}dr$.

We rewrite their integrands of the terms $\Sigma_\tau \cap \{r_1 \le r \le R\}$,

$$(T\check{\phi}_\ell)^2 + (Z\check{\phi}_\ell)^2 = (L\check{\phi}_\ell + \underline{L}\check{\phi}_\ell)^2 + (L\check{\phi}_\ell - \underline{L}\check{\phi}_\ell)^2 = 2\left(\left(\underline{L}\check{\phi}_\ell\right)^2 + \left(L\check{\phi}_\ell\right)^2\right).$$

Then we can identify the expression of the renormalized energy for $\{r \le \tilde{R}\}$, (3.12), into this equation and we get the expression:

$$\check{E}_{\tilde{R}}\left[\check{\phi}_\ell\right](\tau_2) + \int_{\tau_1}^{\tau_2} r^2 w_\ell^2 \left(\left(\underline{L}\check{\phi}_\ell\right)^2 - \left(L\check{\phi}_\ell\right)^2\right)\Big|_{r=\tilde{R}} d\sigma\, du + \int_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} r^2 w_\ell^2 (L\check{\phi}_\ell)^2 d\sigma\, dv$$
$$= \check{E}_{\tilde{R}}\left[\check{\phi}_\ell\right](\tau_1).$$

Using (3.13), we arrive at the result (3.14). □

Chapter 4

Integrated energy estimates

In this section, we derive the integrated energy estimates that are necessary for establishing pointwise energy decay estimates in time. With the energy estimates from the last section, we find that the energy is bounded, but with the integrated energy estimates we can go further and find that the energy always decays, but it is out of the scope of this thesis. We start by deriving local integrated energy estimates for renormalized quantities $\check{\phi}_\ell$.

4.1 Integrated energy estimates for the renormalized wave function

The local integrated energy estimates for the renormalized quantities $\check{\phi}_\ell$ can be split into two parts. The first one is:

Proposition 9. *Let $\phi \in C^\infty(\mathcal{R}) \cap C^1(\hat{\mathcal{R}})$ be a solution to (2.31) in the spacetime $(\mathcal{M}_{ext}, g)$, and $\check{\phi}_\ell$ the renormalization of the spherical harmonic modes ϕ_ℓ. Then there exists a constant $C = C(\ell, \mu) > 0$ such that for all $\tau_1, \tau_2 \in \mathbb{R}$ with $0 \leq \tau_1 < \tau_2$,*

$$\int_{\tau_1}^{\tau_2} \int_{\Sigma_\tau} D \left\{ \left[\left(2 + \frac{q}{2}\right) r^{-q-1} + 2r^{-q} \frac{1}{w_\ell} \frac{dw_\ell}{dr} \right] (Z\check{\phi}_\ell)^2 + \frac{q}{2} r^{-q-1} (T\check{\phi}_\ell)^2 \right\} d\sigma \, du \, dv \leq C \, \check{E}\left[\check{\phi}_\ell\right](\tau_1). \tag{4.1}$$

Proof. Let $f : [r_+, r_c] \to \mathbb{R}$ be a differentiable function. We multiply (3.8) by $f(r) Z\check{\phi}_\ell$, and apply the Leibniz rule to obtain:

$$\begin{aligned}
0 =& f(r) Z\check{\phi}_\ell Z^2 \check{\phi}_\ell + 2Df(r) \left(\frac{1}{r} + \frac{1}{w_\ell}\frac{dw_\ell}{dr}\right) \left(Z\check{\phi}_\ell\right)^2 - f(r) Z\check{\phi}_\ell T^2 \check{\phi}_\ell \\
=& \frac{1}{2} Z\left(f \left(Z\check{\phi}_\ell\right)^2\right) - \frac{1}{2}(Zf)(Z\check{\phi}_\ell)^2 + 2Df \left(\frac{1}{r} + \frac{1}{w_\ell}\frac{dw_\ell}{dr}\right) \left(Z\check{\phi}_\ell\right)^2 - T\left(f Z\check{\phi}_\ell T\check{\phi}_\ell\right) \\
&+ \frac{1}{2} Z\left(f \left(T\check{\phi}_\ell\right)^2\right) - \frac{1}{2}(Zf)(T\check{\phi}_\ell)^2 \\
=& Z\left(\frac{f}{2}(Z\check{\phi}_\ell)^2 + \frac{f}{2}(T\check{\phi}_\ell)^2\right) - T\left(f Z\check{\phi}_\ell T\check{\phi}_\ell\right) + D\left[\frac{2f}{r} + \frac{2f}{w_\ell}\frac{dw_\ell}{dr} - \frac{1}{2}\frac{df}{dr}\right](Z\check{\phi}_\ell)^2 - \frac{D}{2}\frac{df}{dr}(T\check{\phi}_\ell)^2,
\end{aligned}$$

where $Zf(r) = Df'(r)$. Dividing by D,

$$0 = X\left(\frac{f}{2}\left(\left(Z\check{\phi}_\ell\right)^2 + \left(T\check{\phi}_\ell\right)^2\right)\right) - T\left(\frac{f}{D} Z\check{\phi}_\ell T\check{\phi}_\ell\right) + \left[\frac{2f}{r} + \frac{2f}{w_\ell}\frac{dw_\ell}{dr} - \frac{1}{2}\frac{df}{dr}\right](Z\check{\phi}_\ell)^2 - \frac{1}{2}\frac{df}{dr}(T\check{\phi}_\ell)^2. \tag{4.2}$$

We are going to impose that the last two terms, with $(Z\check{\phi}_\ell)^2$ and $(T\check{\phi}_\ell)^2$, have to be positive. Later we will see why. Then, we want to find a function $f(r)$ such that it fulfils the conditions

$$\begin{cases} \left[\dfrac{2f}{r} + \dfrac{2f}{w_\ell}\dfrac{dw_\ell}{dr} - \dfrac{1}{2}\dfrac{df}{dr}\right] \geq 0, \\ \dfrac{df}{dr} \leq 0. \end{cases} \tag{4.3}$$

We take:

$$f(r) = r^{-q} \qquad \text{with} \quad q > 0, \tag{4.4}$$

with q sufficiently large, and depending on w_ℓ. Now we check if the conditions (4.3) are fulfilled for this $f(r)$. The second condition is fulfilled by this $f(r)$,

$$\frac{df}{dr} = -qr^{-q-1} \leq 0.$$

Looking at the first condition, $w_\ell(r)$ is relevant for finding $f(r)$. We can divide our spacetime $\mathcal{R}$ in three distinct areas: area 1 for $r_c < r \leq r_1$, area 2 for $r_1 \leq r \leq R$, with r_1 being sufficiently close to r_+, and area 3 $R \leq r < r_c$, with R being sufficiently close to r_c. We can see a graphic representation of them in Fig. 4.1.

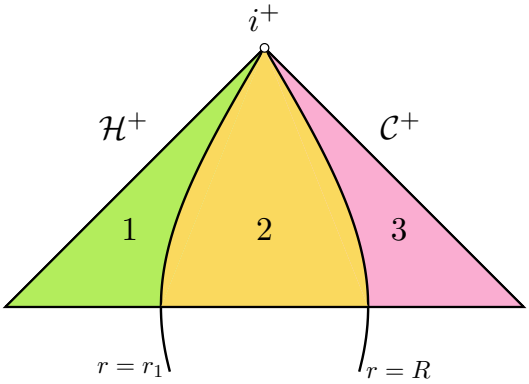


FIGURE 4.1: A 2-dimensional representation of the spacetime subset $\mathcal{R}$ of the RNdS spacetime, for values of $\tau \geq 0$. The green area or area 1 corresponds to values of $r_c < r \leq r_1$, the yellow one is area 2, for $r_1 \leq r \leq R$, and the pink one or area 3, for $R \leq r < r_c$. Each point in the picture represents a round 2-sphere of radius r.

In the area 1, the singularity affecting the function $w(r)$ is $r = r_+$, and it takes the form (2.50). Its derivative would be of the form:

$$\left.\frac{dw_\ell}{dr}\right|_{r\downarrow r_+} = A_+a_1 + \frac{B_+b_0}{r - r_+} + B_+b_1 \log(r - r_+) + \cdots = \frac{a_1}{a_0} + O(r - r_+), \tag{4.5}$$

since $A_+ = 1/a_0$ and $B_+ = 0$. Then,

$$\left.\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right|_{r\downarrow r_+} = \frac{a_1}{a_0} + O(r - r_+) \geq 0.$$

So the first condition of (4.3),

$$\left[\frac{2f}{r}+\frac{2f}{w_\ell}\frac{dw_\ell}{dr}-\frac{1}{2}\frac{df}{dr}\right]\Big|_{r\downarrow r_+}=2r^{-q-1}+2r^{-q}\,\frac{1}{w_\ell}\frac{dw_\ell}{dr}\Big|_{r\downarrow r_+}+\frac{q}{2}r^{-q-1}\geq 0,$$

is fulfilled in area 1.

For the area 3, the singularity affecting the function $w(r)$ is $r=r_+$, and it takes the form (2.51). Its derivative would be of the form:

$$\frac{dw_\ell}{dr}\Big|_{r\uparrow r_c}=A_c\,\underline{a}_1+\frac{B_c\,\underline{b}_0}{r_c-r}+B_c\,\underline{b}_1\log(r_c-r)+\cdots=\frac{B_c\,\underline{b}_0}{r_c-r}+O(r_c-r). \tag{4.6}$$

On §2.2.1 we proved that $B_c\neq 0$ holds, then,

$$\frac{1}{w_\ell}\frac{dw_\ell}{dr}\Big|_{r\uparrow r_c}=-\frac{\log(r_c-r)}{r_c-r}+O(1)\geq 0.$$

In the area 3, r is really close to r_c, so $\log(r_c-r)\leq 0$. Here the first condition of (4.3) is also fulfilled.

We are only left with observing what happens in area 2, where the expression of $w(r)$ would be a combination of the other two expressions for the singularities. The first condition of (4.3) turns into:

$$\frac{2f}{r}+\frac{2f}{w_\ell}\frac{dw_\ell}{dr}-\frac{1}{2}\frac{df}{dr}=2r^{-q-1}+2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}+\frac{q}{2}r^{-q-1}\geq 0.$$

Dividing this by $2r^{-q}$,

$$\left(1+\frac{q}{4}\right)r^{-1}>L,\qquad\qquad \text{with } L:=\max_{[r_1,R]}\left|\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right|.$$

We collect that for condition (4.3) to be fulfilled in area 2, we must set a condition for q,

$$q>4r_cL-1.$$

Now that we have found a suitable $f(r)$, we take $f(r)=r^{-q}$ and substitute it into equation (4.2), to obtain,

$$\begin{aligned}0=&X\left(\frac{r^{-q}}{2}\left(\left(Z\check{\phi}_\ell\right)^2+\left(T\check{\phi}_\ell\right)^2\right)\right)-T\left(\frac{r^{-q}}{D}Z\check{\phi}_\ell T\check{\phi}_\ell\right)\\&+\left[\left(2+\frac{q}{2}\right)r^{-q-1}+2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](Z\check{\phi}_\ell)^2+\frac{q}{2}r^{-q-1}(T\check{\phi}_\ell)^2.\end{aligned}$$

It is more convenient to write this equation in terms of the vector fields L and $\underline{L}$, in (u,v,θ,φ) coordinates. Using the relations (2.13),

$$\begin{aligned}0=&\frac{1}{D}(L-\underline{L})\left(\frac{r^{-q}}{2}\left[\left(L\check{\phi}_\ell-\underline{L}\check{\phi}_\ell\right)^2+\left(L\check{\phi}_\ell+\underline{L}\check{\phi}_\ell\right)^2\right]\right)-(L+\underline{L})\left(\frac{r^{-q}}{D}\left(L\check{\phi}_\ell-\underline{L}\check{\phi}_\ell\right)\left(L\check{\phi}_\ell+\underline{L}\check{\phi}_\ell\right)\right)\\&+\left[\left(2+\frac{q}{2}\right)r^{-q-1}+2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right]\left(L\check{\phi}_\ell-\underline{L}\check{\phi}_\ell\right)^2+\frac{q}{2}r^{-q-1}\left(L\check{\phi}_\ell+\underline{L}\check{\phi}_\ell\right)^2.\end{aligned}$$

Multiplying the previous equation by D,

$$\begin{aligned}0 =&(L-\underline{L})\left(r^{-q}\left[\left(L\check{\phi}_\ell\right)^2+\left(\underline{L}\check{\phi}_\ell\right)^2\right]\right)-(L+\underline{L})\left(r^{-q}\left[\left(L\check{\phi}_\ell\right)^2-\left(\underline{L}\check{\phi}_\ell\right)^2\right]\right)\\&+D\left[\left(2+\frac{q}{2}\right)r^{-q-1}+2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right]\left(L\check{\phi}_\ell-\underline{L}\check{\phi}_\ell\right)^2+\frac{q}{2}Dr^{-q-1}\left(L\check{\phi}_\ell+\underline{L}\check{\phi}_\ell\right)^2\\=&L\left(2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2\right)-\underline{L}\left(2r^{-q}\left(L\check{\phi}_\ell\right)^2\right)\\&+D\left[\left(2+\frac{q}{2}\right)r^{-q-1}+2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](L\check{\phi}_\ell-\underline{L}\check{\phi}_\ell)^2+\frac{q}{2}Dr^{-q-1}(L\check{\phi}_\ell+\underline{L}\check{\phi}_\ell)^2.\end{aligned}$$

The expression simplified quite a lot. Integrating by parts over $d\sigma\,du\,dv$ on $\mathcal{R}^{\tau_2}_{\tau_1}$,

$$\begin{aligned}0=&\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}}2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2d\sigma\,du-\int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}}2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2d\sigma\,du+\int\limits_{\underline{N}_{\tau_2}}2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2d\sigma\,du\\&-\int\limits_{\underline{N}_{\tau_1}}2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2d\sigma\,du+\int\limits_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}}2r^{-q}(\underline{L}\check{\phi}_\ell)^2d\sigma\,du\\&-\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}}2r^{-q}\left(L\check{\phi}_\ell\right)^2d\sigma\,du+\int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}}2r^{-q}\left(L\check{\phi}_\ell\right)^2d\sigma\,du-\int\limits_{N_{\tau_2}}2r^{-q}\left(L\check{\phi}_\ell\right)^2d\sigma\,dv\\&+\int\limits_{N_{\tau_1}}2r^{-q}\left(L\check{\phi}_\ell\right)^2d\sigma\,dv-\int\limits_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}}2r^{-q}(L\check{\phi}_\ell)^2d\sigma\,dv\\&+\int_{\tau_1}^{\tau_2}\int_{\Sigma_\tau}D\left\{\left[\left(2+\frac{q}{2}\right)r^{-q-1}+2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](L\check{\phi}_\ell-\underline{L}\check{\phi}_\ell)^2+\frac{q}{2}r^{-q-1}(L\check{\phi}_\ell+\underline{L}\check{\phi}_\ell)^2\right\}d\sigma\,du\,dv.\end{aligned}$$

For the integration surfaces $\Sigma_\tau\cap\{r_1\le r\le R\}$ the most convenient coordinate system is (t,r,θ,φ), since these are surfaces of constant t. We can apply a change of coordinates over the differential forms such that, $du=2dt-\frac{2}{D}dr$ and $dv=2dt+\frac{2}{D}dr$,

$$\begin{aligned}0=&\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}}\frac{1}{D}4r^{-q}\left[\left(\underline{L}\check{\phi}_\ell\right)^2-\left(L\check{\phi}_\ell\right)^2\right]d\sigma\,dr+\int\limits_{\underline{N}_{\tau_2}}2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2d\sigma\,du-\int\limits_{N_{\tau_2}}2r^{-q}\left(L\check{\phi}_\ell\right)^2d\sigma\,dv\\&-\int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}}\frac{1}{D}4r^{-q}\left[\left(\underline{L}\check{\phi}_\ell\right)^2-\left(L\check{\phi}_\ell\right)^2\right]d\sigma\,dr-\int\limits_{\underline{N}_{\tau_1}}2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2d\sigma\,du+\int\limits_{N_{\tau_1}}2r^{-q}\left(L\check{\phi}_\ell\right)^2d\sigma\,dv\\&+\int\limits_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}}2r^{-q}(\underline{L}\check{\phi}_\ell)^2d\sigma\,du-\int\limits_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}}2r^{-q}(L\check{\phi}_\ell)^2d\sigma\,dv\\&+\int_{\tau_1}^{\tau_2}\int_{\Sigma_\tau}D\left\{\left[\left(2+\frac{q}{2}\right)r^{-q-1}+2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](Z\check{\phi}_\ell)^2+\frac{q}{2}r^{-q-1}(T\check{\phi}_\ell)^2\right\}d\sigma\,du\,dv.\end{aligned}$$

We can rewrite the integrand for the integrals over $\Sigma_\tau\cap\{r_1\le r\le R\}$ in terms of the vector fields Z and T, using the relations (2.13),

$$T\check{\phi}_\ell Z\check{\phi}_\ell=(L\check{\phi}_\ell+\underline{L}\check{\phi}_\ell)(L\check{\phi}_\ell-\underline{L}\check{\phi}_\ell)=(L\check{\phi}_\ell)^2-(\underline{L}\check{\phi}_\ell)^2.$$

Then finally,

$$
\begin{aligned}
&-\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}} \frac{1}{D}4r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right)d\sigma\,dr+\int\limits_{\underline{N}_{\tau_2}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\,du-\int\limits_{N_{\tau_2}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\,dv\\
&\qquad+\int\limits_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}} 2r^{-q}(\underline{L}\check{\phi}_\ell)^2 d\sigma\,du-\int\limits_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} 2r^{-q}(L\check{\phi}_\ell)^2 d\sigma\,dv\\
&\quad+\int_{\tau_1}^{\tau_2}\int_{\Sigma_\tau} D\left\{\left[\left(2+\frac{q}{2}\right)r^{-q-1}+2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](Z\check{\phi}_\ell)^2+\frac{q}{2}r^{-q-1}(T\check{\phi}_\ell)^2\right\}d\sigma\,du\,dv\\
&=-\int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}} \frac{1}{D}4r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right)d\sigma\,dr+\int\limits_{\underline{N}_{\tau_1}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\,du-\int\limits_{N_{\tau_1}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\,dv.
\end{aligned}
\tag{4.7}
$$

We have to combine this expression with the renormalized energy boundedness that we found in Proposition 7. Let us bound the left-hand side of (4.7) by the initial renormalized energy $\check{E}\left[\check{\phi}_\ell\right](\tau_1)$. In (4.7) we find some energy-like terms, but with inadequate signs. Consider first the ones with $\tau=\tau_1$, i.e., we want to prove:

$$
\begin{aligned}
&-\int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}} \frac{1}{D}4r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right)d\sigma\,dr+\int\limits_{\underline{N}_{\tau_1}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\,du-\int\limits_{N_{\tau_1}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\,dv\\
&\qquad\le C\,\check{E}\left[\check{\phi}_\ell\right](\tau_1).
\end{aligned}
\tag{4.8}
$$

Inside of the area $r_1\le r\le R$, the factor $1/D$ does not give us trouble, since we are away from the horizons. We can then make use of the following inequality:

$$
|T\check{\phi}_\ell Z\check{\phi}_\ell|\le\frac{1}{2}\left(\left(T\check{\phi}_\ell\right)^2+\left(Z\check{\phi}_\ell\right)^2\right).
$$

This way, no matter if the integrand is positive or negative, the integral over $\Sigma_{\tau_1}\cap\{r_1\le r\le R\}$ is bounded by its counterpart in $\check{E}\left[\check{\phi}_\ell\right](\tau_1)$. The integral over N_{τ_1} is negative, so we can always bound it by something positive.

This whole time we have considered $w_\ell(r)$ to be a smooth stationary non-zero solution of the Klein-Gordon equation, i.e. $0<C\le w_\ell(r)$, for a constant C. Since r is bounded in $r\in[r_+,r_c]$, and $q>0$ is quite large,

$$
r^{-q}\le C\,r^2\le C\,r^2w_\ell^2(r).
$$

Even though the integral over $\underline{N}_{\tau_1}$ is positive but we can choose the constant such that $r^{-q}\lesssim r^2w_\ell^2$. Thus, we have proven that (4.8) holds. By the same reasoning, the same holds for $\tau=\tau_2$. Combining it with (3.11),

$$
\begin{aligned}
&-\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}} \frac{1}{D}4r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right)d\sigma\,dr+\int\limits_{\underline{N}_{\tau_2}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\,du-\int\limits_{N_{\tau_2}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\,dv\\
&\qquad\le C\,\check{E}\left[\check{\phi}_\ell\right](\tau_1).
\end{aligned}
$$

We check the boundary terms now. At the cosmological horizon:

$$
\underline{L}\check{\phi}_\ell\big|_{r=r_c}=\underline{L}\left(\frac{\phi_\ell}{w_\ell}\right)\bigg|_{r=r_c}=\frac{\underline{L}\phi_\ell}{w_\ell}+\frac{D}{2w_\ell^2}\frac{dw_\ell}{dr}\phi_\ell=0.
$$

since $D(r_c) = 0$ and $\underline{L} = -D/2$. The function w_ℓ in the area around the cosmological horizon is given by (2.51), and exactly at the horizon it would blow up logarithmically, i.e. $w_\ell(r_c) = -\infty$, due to the factor of $\log(r_c - r)$. Then,

$$\int\limits_{\mathcal{C}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2r^{-q}(\underline{L}\check{\phi}_\ell)^2 d\sigma\, du = 0.$$

At the event horizon the integral is not zero, since by our chosen w_ℓ in (2.50), $w_\ell(r_+) = 1$,

$$L\check{\phi}_\ell\big|_{r=r_+} = L\left(\frac{\phi_\ell}{w_\ell}\right)\bigg|_{r=r_+} = \frac{L\phi_\ell}{w_\ell} - \frac{D}{2w_\ell^2}\frac{dw_\ell}{dr}\phi_\ell = L\phi_\ell|_{r=r_+}.$$

since $D(r_+) = 0$ and $L = D/2$. Same way as for the integral over N_{τ_1}, we can bound this term,

$$-\int\limits_{\mathcal{H}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2r^{-q}(L\check{\phi}_\ell)^2 d\sigma\, dv \leq C\int\limits_{\mathcal{H}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} r^2 w_\ell^2 (L\check{\phi}_\ell)^2 d\sigma\, dv \tag{4.9}$$

Making use of the proposition (3.10), it implies that:

$$\int\limits_{\mathcal{H}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} r^2 w_\ell^2 (L\check{\phi}_\ell)^2 d\sigma\, dv \leq C\,\check{E}\left[\check{\phi}_\ell\right](\tau_1).$$

Which allows us to bound the integral over $\mathcal{H}^+$ even further. We chose the function $f(r)$ so that the bulk terms in (4.7) are positive, then using all of the inequalities above, we find that the left-hand side of (4.7) is bounded by the initial renormalized energy,

$$\begin{aligned}
&-\int\limits_{\Sigma_{\tau_2}\cap\{r_1\leq r\leq R\}} \frac{1}{D}4r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right)d\sigma\, dr + \int\limits_{\underline{N}_{\tau_2}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du - \int\limits_{N_{\tau_2}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv \\
&\qquad + \int\limits_{\mathcal{C}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2r^{-q}(\underline{L}\check{\phi}_\ell)^2 d\sigma\, du - \int\limits_{\mathcal{H}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2r^{-q}(L\check{\phi}_\ell)^2 d\sigma\, dv \\
&+\int_{\tau_1}^{\tau_2}\int\limits_{\Sigma_\tau} D\left\{\left[\left(2+\frac{q}{2}\right)r^{-q-1} + 2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](Z\check{\phi}_\ell)^2 + \frac{q}{2}r^{-q-1}(T\check{\phi}_\ell)^2\right\} d\sigma\, du\, dv \\
&\qquad\qquad \leq C\,\check{E}\left[\check{\phi}_\ell\right](\tau_1).
\end{aligned}$$

From this equation, we can find the integrated energy estimates (4.1), for the renormalized function $\check{\phi}_\ell$. □

There is a second part for this proposition.

Proposition 10. *Let $\phi \in C^\infty(\mathcal{R}) \cap C^1(\hat{\mathcal{R}})$ be a solution to (2.31) in the spacetime $(\mathcal{M}_{ext}, g)$, and $\check{\phi}_\ell$ the renormalization of the spherical harmonic modes ϕ_ℓ. Let $\tilde{R} > R$ be arbitrarily large. Then there exists a constant $C = C(\ell, \mu, \tilde{R}) > 0$ such that for all $\tau_1, \tau_2 \in \mathbb{R}$ with $0 \leq \tau_1 < \tau_2$,*

$$\int_{\tau_1}^{\tau_2}\int_{r_+}^{\tilde{R}} D\left\{\left[\left(2+\frac{q}{2}\right)r^{-q-1} + 2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](Z\check{\phi}_\ell)^2 + \frac{q}{2}r^{-q-1}(T\check{\phi}_\ell)^2\right\} d\sigma\, du\, dv \\ + \int_{\tau_1}^{\tau_2} 2r^{-q}\left(\left(\underline{L}\check{\phi}_\ell\right)^2 + \left(L\check{\phi}_\ell\right)^2\right)\Big|_{r=\tilde{R}} d\sigma\, du \leq C\, \check{E}\left[\check{\phi}_\ell\right](\tau_1). \tag{4.10}$$

Proof. The process we follow now is parallel to the one we used for proving Proposition 9. Considering the equation (4.2), taking $f(r) = r^{-q}$ and changing the vector fields to $\underline{L}$ and L:

$$0 = L\left(2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2\right) - \underline{L}\left(2r^{-q}\left(L\check{\phi}_\ell\right)^2\right) + D\left[\left(2+\frac{q}{2}\right)r^{-q-1} + 2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](L\check{\phi}_\ell - \underline{L}\check{\phi}_\ell)^2 \\ -\frac{q}{2}Dr^{-q-1}(L\check{\phi}_\ell + \underline{L}\check{\phi}_\ell)^2.$$

We integrate this equation by parts over $d\sigma\, du\, dv$ on $\mathcal{P}_{\tau_1}^{\tau_2}$,

$$\begin{aligned}
0 = &\int_{\Sigma_{\tau_2}\cap\{r_1\leq r\leq R\}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du - \int_{\Sigma_{\tau_1}\cap\{r_1\leq r\leq R\}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du + \int_{\underline{N}_{\tau_2}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du \\
&- \int_{\underline{N}_{\tau_1}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du + \int_{u=\tau_1-\tilde{R}}^{u=\tau_2-\tilde{R}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2\Big|_{r=\tilde{R}} d\sigma\, du - \int_{\Sigma_{\tau_2}\cap\{r_1\leq r\leq R\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv \\
&+ \int_{\Sigma_{\tau_1}\cap\{r_1\leq r\leq R\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv - \int_{N_{\tau_2}\cap\{r\leq\tilde{R}\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv \\
&+ \int_{N_{\tau_1}\cap\{r\leq\tilde{R}\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv + \int_{v=\tau_1+\tilde{R}}^{v=\tau_2+\tilde{R}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2\Big|_{r=\tilde{R}} d\sigma\, dv - \int_{\mathcal{H}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2r^{-q}(L\check{\phi}_\ell)^2 d\sigma\, dv \\
&+ \int_{\tau_1}^{\tau_2}\int_{r_+}^{\tilde{R}} D\left\{\left[\left(2+\frac{q}{2}\right)r^{-q-1} + 2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](L\check{\phi}_\ell - \underline{L}\check{\phi}_\ell)^2 + \frac{q}{2}r^{-q-1}(L\check{\phi}_\ell + \underline{L}\check{\phi}_\ell)^2\right\} d\sigma\, du\, dv.
\end{aligned}$$

For the terms $\{r = \tilde{R}\}$, it is more convenient to express them in Bondi coordinates (u, r, θ, φ). We change coordinates using (2.9), then $du = du$, and $dv = du + \frac{2}{D}dr$. However, for the surfaces $\Sigma_\tau \cap \{r_1 \leq r \leq R\}$ the most convenient coordinate system is (t, r, θ, φ). We can apply a change of coordinates over the differential forms such that, $du = 2dt - \frac{2}{D}dr$ and $dv = 2dt + \frac{2}{D}dr$. Then,

$$\begin{aligned}
0 = &\int_{\Sigma_{\tau_2}\cap\{r_1\leq r\leq R\}} \frac{4}{D}r^{-q}\left(\left(\underline{L}\check{\phi}_\ell\right)^2 - \left(L\check{\phi}_\ell\right)^2\right) d\sigma\, dr + \int_{\underline{N}_{\tau_2}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du \\
&- \int_{N_{\tau_2}\cap\{r\leq\tilde{R}\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv - \int_{\Sigma_{\tau_1}\cap\{r_1\leq r\leq R\}} \frac{4}{D}r^{-q}\left(\left(\underline{L}\check{\phi}_\ell\right)^2 - \left(L\check{\phi}_\ell\right)^2\right) d\sigma\, dr
\end{aligned}$$

$$
\begin{aligned}
&- \int\limits_{\underline{N}_{\tau_1}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du + \int\limits_{N_{\tau_1}\cap\{r\leq\tilde{R}\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv \\
&+ \int\limits_{\tau_1}^{\tau_2} 2r^{-q}\left(\left(\underline{L}\check{\phi}_\ell\right)^2 + \left(L\check{\phi}_\ell\right)^2\right)\Big|_{r=\tilde{R}} d\sigma\, du - \int\limits_{\mathcal{H}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2r^{-q}(L\check{\phi}_\ell)^2 d\sigma\, dv \\
&+ \int\limits_{\tau_1}^{\tau_2}\int\limits_{r_+}^{\tilde{R}} D\left\{\left[\left(2+\frac{q}{2}\right)r^{-q-1} + 2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](Z\check{\phi}_\ell)^2 + \frac{q}{2}r^{-q-1}(T\check{\phi}_\ell)^2\right\} d\sigma\, du\, dv.
\end{aligned}
$$

We can rewrite the integrand for the integrals over $\Sigma_\tau \cap \{r_1 \leq r \leq R\}$ in terms of the vector fields Z and T, using the relations (2.13),

$$
T\check{\phi}_\ell Z\check{\phi}_\ell = (L\check{\phi}_\ell + \underline{L}\check{\phi}_\ell)(L\check{\phi}_\ell - \underline{L}\check{\phi}_\ell) = (L\check{\phi}_\ell)^2 - (\underline{L}\check{\phi}_\ell)^2.
$$

Then finally,

$$
\begin{aligned}
&- \int\limits_{\Sigma_{\tau_2}\cap\{r_1\leq r\leq R\}} \frac{4}{D}r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right) d\sigma\, dr + \int\limits_{\underline{N}_{\tau_2}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du - \int\limits_{N_{\tau_2}\cap\{r\leq\tilde{R}\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv \\
&\qquad + \int\limits_{\tau_1}^{\tau_2} 2r^{-q}\left(\left(\underline{L}\check{\phi}_\ell\right)^2 + \left(L\check{\phi}_\ell\right)^2\right)\Big|_{r=\tilde{R}} d\sigma\, du - \int\limits_{\mathcal{H}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2r^{-q}(L\check{\phi}_\ell)^2 d\sigma\, dv \\
&\qquad + \int\limits_{\tau_1}^{\tau_2}\int\limits_{r_+}^{\tilde{R}} D\left\{\left[\left(2+\frac{q}{2}\right)r^{-q-1} + 2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](Z\check{\phi}_\ell)^2 + \frac{q}{2}r^{-q-1}(T\check{\phi}_\ell)^2\right\} d\sigma\, du\, dv \\
&= - \int\limits_{\Sigma_{\tau_1}\cap\{r_1\leq r\leq R\}} \frac{4}{D}r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right) d\sigma\, dr + \int\limits_{\underline{N}_{\tau_1}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du - \int\limits_{N_{\tau_1}\cap\{r\leq\tilde{R}\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv
\end{aligned}
\tag{4.11}
$$

This is the first equation that we are interested in. We want to bound the left-hand side of (4.11) by the initial renormalized energy $\check{E}\left[\check{\phi}_\ell\right](\tau_1)$. There are some energy-like terms in it, but with inadequate signs. First consider the ones with $\tau = \tau_1$, i.e., we want to prove:

$$
\begin{aligned}
&- \int\limits_{\Sigma_{\tau_1}\cap\{r_1\leq r\leq R\}} \frac{4}{D}r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right) d\sigma\, dr + \int\limits_{\underline{N}_{\tau_1}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du - \int\limits_{N_{\tau_1}\cap\{r\leq\tilde{R}\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv \\
&\qquad\qquad \leq C\,\check{E}\left[\check{\phi}_\ell\right](\tau_1).
\end{aligned}
\tag{4.12}
$$

Inside of the area $\{r_1 \leq r \leq R\}$, the factor $1/D$ does not give us trouble, since we are away from the horizons. We can then make use of the following inequality:

$$
|T\check{\phi}_\ell Z\check{\phi}_\ell| \leq \frac{1}{2}\left(\left(T\check{\phi}_\ell\right)^2 + \left(Z\check{\phi}_\ell\right)^2\right).
$$

This way, no matter if the integrand is positive or negative, the integral over $\Sigma_{\tau_1}\cap\{r_1 \leq r \leq R\}$ is bounded by its counterpart in (3.12). The integral over $N_{\tau_1}\cap\{r \leq \tilde{R}\}$ is negative, so we can always bound it by something positive. Since $r^{-q} \lesssim r^2 w_\ell^2$, we can choose the constant C such that the integral over $\underline{N}_{\tau_1}$ is positive. Thus, we have proven that (4.12) holds. By the same

reasoning, the same holds for $\tau = \tau_2$:

$$-\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}} \frac{4}{D} r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right) d\sigma\, dr + \int\limits_{\underline{N}_{\tau_2}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du - \int\limits_{N_{\tau_2}\cap\{r\le\tilde{R}\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv \\ \le C\,\check{E}\left[\check{\phi}_\ell\right](\tau_1).$$

The boundary terms in (4.11) do not cancel, but using (3.14) we can bound them:

$$\int\limits_{\tau_1}^{\tau_2} r^2 w_\ell^2\left(\left(\underline{L}\check{\phi}_\ell\right)^2 - \left(L\check{\phi}_\ell\right)^2\right)\Big|_{r=\tilde{R}} d\sigma\, du + \int\limits_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} r^2 w_\ell^2 (L\check{\phi}_\ell)^2 d\sigma\, dv \le C\,\check{E}\left[\check{\phi}_\ell\right](\tau_1).$$

Combining all of this, we can bound the left-hand side of (4.11),

$$-\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}} \frac{4}{D} r^{-q}\left(T\check{\phi}_\ell Z\check{\phi}_\ell\right) d\sigma\, dr + \int\limits_{\underline{N}_{\tau_2}} 2r^{-q}\left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du - \int\limits_{N_{\tau_2}\cap\{r\le\tilde{R}\}} 2r^{-q}\left(L\check{\phi}_\ell\right)^2 d\sigma\, dv \\ + \int\limits_{\tau_1}^{\tau_2} 2r^{-q}\left(\left(\underline{L}\check{\phi}_\ell\right)^2 + \left(L\check{\phi}_\ell\right)^2\right)\Big|_{r=\tilde{R}} d\sigma\, du - \int\limits_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} 2r^{-q}(L\check{\phi}_\ell)^2 d\sigma\, dv \\ + \int\limits_{\tau_1}^{\tau_2}\int\limits_{r_+}^{\tilde{R}} D\left\{\left[\left(2+\frac{q}{2}\right) r^{-q-1} + 2r^{-q}\frac{1}{w_\ell}\frac{dw_\ell}{dr}\right](Z\check{\phi}_\ell)^2 + \frac{q}{2} r^{-q-1}(T\check{\phi}_\ell)^2\right\} d\sigma\, du\, dv \\ \le C\,\check{E}\left[\check{\phi}_\ell\right](\tau_1).$$

Finally, this allows us to get to the we get (4.10). □

4.2 Renormalized energy bounded by the degenerate energy

In order to go from the integrated energy estimates for $\check{\phi}_\ell$ in Proposition 9 and 10 to an integrated energy estimate for ϕ, first we need to bound the initial energy $\check{E}\left[\check{\phi}_\ell\right](\tau_1)$ by the initial energy associated to ϕ_ℓ. We already defined the energy of ϕ in (3.1), but for a certain value of ℓ it takes the form:

$$\begin{aligned} E[\phi_\ell](\tau) = &\int\limits_{\Sigma_\tau\cap\{r_1\le r\le R\}} 2\left[Dr^2(X\phi_\ell)^2 + \frac{1}{D} r^2(T\phi_\ell)^2 + \left(\ell(\ell+1) + \mu^2 r^2\right)\phi_\ell^2\right] d\sigma\, dr \\ &+ \int\limits_{N_\tau}\left[2r^2(L\phi_\ell)^2 + \frac{D}{2}\left(\ell(\ell+1)+\mu^2 r^2\right)\phi_\ell^2\right] d\sigma\, dv \\ &+ \int\limits_{\underline{N}_\tau}\left[2r^2(\underline{L}\phi_\ell)^2 + \frac{D}{2}\left(\ell(\ell+1)+\mu^2 r^2\right)\phi_\ell^2\right] d\sigma\, du. \end{aligned} \tag{4.13}$$

Here we used the equality $\not{\Delta}_{\mathbb{S}^2}\phi_\ell = -\ell(\ell+1)\phi_\ell$. We can make use of the following positivity property:

$$\ell(\ell+1) + \mu^2 > 0.$$

We will additionally split ϕ as follows:

$$\phi = \phi_{\leq L_0} + \phi_{>L_0}.$$

where L_0 will be taken suitably large, depending on μ. Define:

$$\phi_{\leq L_0} := \sum_{\ell \leq L_0} \phi_\ell, \qquad\qquad \phi_{>L_0} := \sum_{\ell=L_0+1}^{\infty} \phi_\ell.$$

For now we will consider a finite $\ell \leq L_0$.

The renormalized energy (3.9) is the sum of three integrals, one for each of the distinct areas of our foliation, as we saw in Fig. 4.1. The energy (4.13) also is defined like this, so we will try to bound each integral of the same area with each other. For the area 1, we want to show:

$$\int_{\underline{N}_{\tau_1}} r^2 w_\ell^2 \left(\underline{L}\check{\phi}_\ell\right)^2 d\sigma\, du \leq C \int_{\underline{N}_{\tau_1}} \left[2r^2(\underline{L}\phi_\ell)^2 + \frac{D}{2}\left(\ell(\ell+1) + \mu^2 r^2\right)\phi_\ell^2\right] d\sigma\, du. \tag{4.14}$$

In this area we know that $w(r)$ takes the form (2.50), and its derivative is (4.5), then:

$$\underline{L}\check{\phi}_\ell\big|_{r\downarrow r_+} = \underline{L}\left(\frac{\phi_\ell}{w_\ell}\right)\bigg|_{r\downarrow r_+} = \frac{\underline{L}\phi_\ell}{w_\ell} + \phi_\ell \frac{D}{2w_\ell^2}\frac{dw_\ell}{dr} = \underline{L}\phi_\ell|_{r\downarrow r_+},$$

since on the second term, the leading order is that of $D(r)$, and in this area it is almost zero. Then the condition (4.14) is fulfilled.

The only term where we run into difficulties when bounding it is the integral over N_{τ_1} of the renormalized energy (3.9), i.e. area 3 of Fig. 4.1. We want to prove that:

$$\int_{N_{\tau_1}} r^2 w_\ell^2 \left(L\check{\phi}_\ell\right)^2 d\sigma\, dv \leq C \int_{N_{\tau_1}} \left[2r^2(L\phi_\ell)^2 + \frac{D}{2}\left(\ell(\ell+1) + \mu^2 r^2\right)\phi_\ell^2\right] d\sigma\, dv. \tag{4.15}$$

Consider:

$$w_\ell\, L\check{\phi}_\ell = w_\ell\, L\left(\frac{\phi_\ell}{w_\ell}\right) = L\phi_\ell + \phi_\ell \frac{Lw_\ell}{w_\ell} = L\phi_\ell + \phi_\ell \frac{D}{2w_\ell}\frac{dw_\ell}{dr}.$$

Using the property $(a-b)^2 \leq 2a^2 + 2b^2$,

$$r^2 w_\ell^2 (L\check{\phi}_\ell)^2 \leq 2r^2(L\phi_\ell)^2 + \frac{D^2 r^2}{2w_\ell^2}\left(\frac{dw_\ell}{dr}\right)^2 \phi_\ell^2.$$

We now have to check if the term in ϕ_ℓ^2 on the right-hand side can be identified with the term in ϕ_ℓ^2 of the energy (4.13). Since we are in the vicinity of the cosmological horizon, we know that $w(r)$ takes the form (2.51), and its derivative is (4.6), then:

$$\frac{1}{w_\ell^2}\left(\frac{dw_\ell}{dr}\right)^2\bigg|_{r\uparrow r_c} \approx \frac{1}{(r_c - r)^2}\frac{1}{\log^2(r_c - r)} + O(1),$$

Then we can bound:

$$\frac{D^2 r^2}{2w_\ell^2}\left(\frac{dw_\ell}{dr}\right)^2 \phi_\ell^2 \leq C_{L_0}\frac{D}{(r_c - r)}\frac{1}{\log^2(r_c - r)}\phi_\ell^2,$$

This inequality gives us a boundary condition for $D(r)$ in $r \geq R$,

$$D^2 \leq C(r_+, r_c)\,(r_c - r)^2. \tag{4.16}$$

We are left with showing:

$$\int\limits_{N_{\tau_1}} \frac{D}{(r_c - r)\log^2(r_c - r)}\phi_\ell^2\, d\sigma\, dv \leq C \int\limits_{N_{\tau_1}} \left[2r^2(L\phi_\ell)^2 + \frac{D}{2}\left(\ell(\ell+1) + \mu^2 r^2\right)\phi_\ell^2\right] d\sigma\, dv.$$

In this region, it is better to work in Bondi coordinates (u, r, θ, φ), since N_{τ_1} is a surface of constant u. We change coordinate using (2.9),

$$\int\limits_{N_{\tau_1}} \frac{D}{(r_c - r)\log^2(r_c - r)}\phi_\ell^2\, d\sigma\, dv = \int\limits_R^{r_c} \frac{2}{(r_c - r)\log^2(r_c - r)}\phi_\ell^2\, d\sigma\, dr$$

To solve this integral, we will use the claim (2.22), derived from the Hardy inequality. We consider a change of variable:

$$\phi_\ell(r_c - x) \equiv \phi_\ell(r), \qquad x \equiv r_c - r, \qquad dx \equiv -dr.$$

Then,

$$\begin{aligned}\int\limits_R^{r_c} \frac{1}{(r_c - r)\log^2(r_c - r)}\phi_\ell^2\, d\sigma\, dr &= -\int\limits_{x=r_c-R}^{x=0} \frac{1}{x\log^2 x}\phi_\ell^2\, d\sigma\, dx = \int\limits_0^{r_c-R} \frac{1}{x\log^2 x}\phi_\ell^2\, d\sigma\, dx \\ &\leq -\frac{2}{\log(r_c - R)}\phi^2(r_c - R) + 4\int\limits_0^{r_c-R} x(Y\phi_\ell)^2\, d\sigma\, dx.\end{aligned}$$

where we applied (2.22), for $a = 0$, $b = r_c - R$. The vector field Y defined in (2.12) is used here since $Y = \partial_r$ in (u, r, θ, φ) coordinates. The term in $\phi^2(r_c - R)$ has a negative sign, but since $r_c - R$ are chosen to be very close, then $\log(r_c - R) < 0$, so it is a positive term.

Changing back from x to r:

$$\int\limits_R^{r_c} \frac{1}{(r_c - r)\log^2(r_c - r)}\phi_\ell^2\, d\sigma\, dr \leq C\left[\int\limits_R^{r_c}(r_c - r)(Y\phi_\ell)^2\, d\sigma\, dr + \left|\frac{1}{\log(r_c - R)}\right|\phi^2(R)\right]. \tag{4.17}$$

Our goal is to only be left with the integral term on the right-hand side, and get rid of the term with the logarithm. For that, we introduce a step function.

Definition 4.2.1. We define a *step function* $\chi(r)$ as vanishing for values of $0 \leq r \leq R$, and of value one for $r \geq R + 1$, however, we want $\chi(r)$ to be a continuous and differentiable function in r.

We can see a graphic representation of a possible step function $\chi(r)$ in Fig. 4.2.

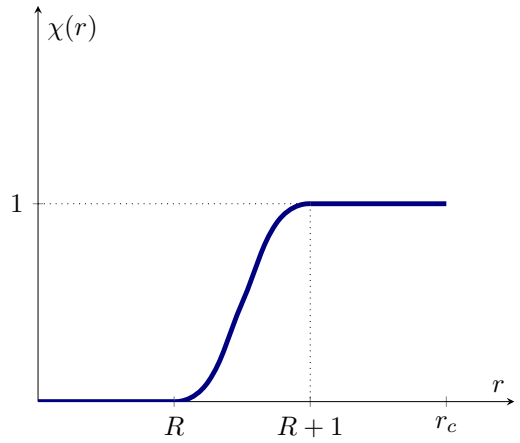


FIGURE 4.2: Smooth step function $\chi(r)$, dependent on R.

On the claim (2.22) we now choose $f = \chi\,\phi_\ell$. This way, we will only consider the interval $(R+1, r_c)$, where the boundary term in (4.17) disappears,

$$\int_R^{r_c} \frac{1}{(r_c-r)\log^2(r_c-r)}\chi^2\phi_\ell^2\,d\sigma\,dr \le C\int_R^{r_c}(r_c-r)\left(Y\left(\chi\phi_\ell\right)\right)^2\,d\sigma\,dr + C\left|\frac{1}{\log(r_c-R)}\right|\chi^2\phi^2(R)$$
$$\le C\int_R^{r_c}(r_c-r)\left(Y\left(\chi\phi_\ell\right)\right)^2\,d\sigma\,dr,$$

since $\chi\phi_\ell(R) = 0$.

Separating the integral over (R, r_c) in a sum of integrals over $(R, R+1)$ and $(R+1, r_c)$, and considering the definition of $\chi(r)$,

$$\int_{R+1}^{r_c} \frac{1}{(r_c-r)\log^2(r_c-r)}\phi_\ell^2\,d\sigma\,dr \le \int_R^{r_c} \frac{1}{(r_c-r)\log^2(r_c-r)}\chi^2\phi_\ell^2\,d\sigma\,dr$$
$$\le C\int_R^{r_c}(r_c-r)\left(Y\left(\chi\phi_\ell\right)\right)^2\,d\sigma\,dr \le C\left[\int_R^{r_c}(r_c-r)\chi^2\left(Y\phi_\ell\right)^2\,dr + \int_R^{R+1}\left(\chi'\right)^2\phi_\ell^2\,dr\right],$$

using the property $(a-b)^2 \le 2a^2 + 2b^2$. The derivative of $\chi(r)$ is $\chi'(r) \ge 0$ everywhere, so:

$$\int_{R+1}^{r_c} \frac{1}{(r_c-r)\log^2(r_c-r)}\phi_\ell^2\,d\sigma\,dr \le C\int_R^{r_c}(r_c-r)\left(Y\phi_\ell\right)^2\,dr + \tilde{C}\int_R^{R+1}\phi_\ell^2\,dr.$$

Since we are considering the positivity property $\ell(\ell+1)+\mu^2 > 0$, we can absorb the term with ϕ_ℓ^2 on the right-hand of this inequality by the term with ϕ_ℓ^2 in the energy integral (4.15). Then we have found a lower bound for $D(r)$ when $\{r \ge R\}$, that we can add to the upper bound (4.16),

$$\tilde{C}(r_c-r)^2 \le D^2 \le C(r_+, r_c)\,(r_c-r)^2.$$

We just proved the condition we were looking for, (4.15), but for the interval $(R+1, r_c)$. We do not loose information in $(R, R+1)$, because we can cover it, as we will see later.

We are left with proving the same thing, but for the area 2, i.e. we need to show:

$$\begin{aligned}&\int\limits_{\Sigma_{\tau_1}\cap\{r_1\leq r\leq R\}} \frac{1}{D}r^2 w_\ell^2\left(\left(T\check{\phi}_\ell\right)^2+\left(Z\check{\phi}_\ell\right)^2\right)d\sigma\,dr\\ \leq C&\int\limits_{\Sigma_{\tau_1}\cap\{r_1\leq r\leq R\}} 2\left[Dr^2(X\phi_\ell)^2+\frac{1}{D}r^2(T\phi_\ell)^2+\left(\ell(\ell+1)+\mu^2r^2\right)\phi_\ell^2\right]d\sigma\,dr.\end{aligned} \tag{4.18}$$

This is possible for two reasons: the first is that we have proven that the function w_ℓ is non-zero for finite values of r; the second is related to the way the vector field Z acts over $\check{\phi}_\ell$,

$$Z\check{\phi}_\ell = Z\left(\frac{\phi_\ell}{w_\ell}\right)=\frac{Z\phi_\ell}{w_\ell}-\frac{Z(w_\ell)}{w_\ell^2}\phi_\ell.$$

Using the property $(a-b)^2\leq 2a^2+2b^2$,

$$\left(Z\check{\phi}_\ell\right)^2\leq\frac{2}{w_\ell^2}\left(Z\phi_\ell\right)^2+\frac{2}{w_\ell^4}\left(Zw_\ell\right)^2\phi_\ell^2.$$

Since we are considering the positivity property $\ell(\ell+1)+\mu^2>0$, there is no problem finding a constant C such that (4.18) is fulfilled.

Finally, we have been able to prove that the renormalized initial energy is bounded by the energy of ϕ_ℓ,

$$\check{E}\left[\check{\phi}_\ell\right](\tau_1)\leq C\,E[\phi_\ell](\tau_1). \tag{4.19}$$

4.3 Integrated energy estimates for $\ell\leq L_0$

In order to go from the integrated energy estimates for $\check{\phi}_\ell$ in (3.10) to an integrated energy estimate for ϕ, we will first consider the case $\ell\leq L_0$. We can make use of the positivity property: $\ell(\ell+1)+\mu^2>0$. Consider the rescaled function $\psi=r\phi$.

Proposition 11. *Let $\phi\in C^\infty(\mathcal{R})\cap C^1(\hat{\mathcal{R}})$ be a solution to (2.31) in the spacetime $(\mathcal{M}_{ext},g)$. Let L_0 be a finite number. Then there exists a constant $C=C(M,Q,\Lambda,\mu,\tilde{R},L_0)$ such that for all $0\leq\tau_1\leq\tau_2\leq\infty$:*

$$\begin{aligned}&\int_{\tau_1}^{\tau_2}\int_{r_+}^{r_1}\left[\left(Z\phi_{\leq L_0}\right)^2+\left(T\phi_{\leq L_0}\right)^2+\phi_{\leq L_0}^2\right]d\sigma\,dv\,dr+\int_{\tau_1}^{\tau_2}\int_{r_1}^{R}\left[\left(Z\phi_{\leq L_0}\right)^2+\left(T\phi_{\leq L_0}\right)^2+\phi_{\leq L_0}^2\right]d\sigma\,dt\,dr\\ &+\int_{\tau_1}^{\tau_2}\int_{R}^{\tilde{R}}\left[\left(Z\phi_{\leq L_0}\right)^2+\left(T_{\leq L_0}\right)^2+\phi_{\leq L_0}^2\right]d\sigma\,du\,dr\leq C\;E[\phi](\tau_1).\end{aligned} \tag{4.20}$$

Proof. Let $f:[r_+,r_c]\to\mathbb{R}$ be a differentiable function, which will be specified later. We multiply both sides of (3.3) by $f(r)Z\psi=fL\psi-f\underline{L}\psi$, and using the Leibniz rule,

$$0=(fL\psi-f\underline{L}\psi)\left\{-4\underline{L}L\psi+\frac{D}{r^2}\not{\Delta}_{\mathbb{S}^2}\psi-D\left(\mu^2+\frac{D'}{r}\right)\psi\right\}$$

$$
\begin{aligned}
=&-4fL\psi\underline{L}L\psi+4f\underline{L}\psi\underline{L}L\psi+\frac{fD}{r^2}Z\psi\not{\Delta}_{\mathbb{S}^2}\psi-fD\left(\mu^2+\frac{D'}{r}\right)\psi Z\psi\\
=&\underline{L}\left(-2f\left(L\psi\right)^2\right)+L\left(2f\left(\underline{L}\psi\right)^2\right)-D\frac{df}{dr}\left[(L\psi)^2+(\underline{L}\psi)^2\right]+\not{\nabla}_{\mathbb{S}^2}\left(\frac{fD}{r^2}Z\psi\not{\nabla}_{\mathbb{S}^2}\psi\right)\\
&-\frac{1}{2}Z\left(\frac{fD}{r^2}\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2\right)+\frac{1}{2}Z\left(\frac{fD}{r^2}\right)\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2-Z\left(\frac{fD}{2}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right)\\
&+Z\left(\frac{fD}{2}\left[\mu^2+\frac{D'}{r}\right]\right)\psi^2\\
=&\underline{L}\left(-2f\left(L\psi\right)^2\right)+L\left(2f\left(\underline{L}\psi\right)^2\right)-Z\left(\frac{fD}{2r^2}\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2+\frac{fD}{2}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right)\\
&-\frac{df}{dr}D\left((L\psi)^2+(\underline{L}\psi)^2\right)+\frac{D}{2}\frac{d}{dr}\left(\frac{fD}{r^2}\right)\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2+\frac{D}{2}\frac{d}{dr}\left(fD\left[\mu^2+\frac{D'}{r}\right]\right)\psi^2\\
&+\not{\nabla}_{\mathbb{S}^2}\left(\frac{fD}{r^2}Z\psi\not{\nabla}_{\mathbb{S}^2}\psi\right).
\end{aligned}
$$

where we used that $Z=\partial_{r_*}=D\partial_r$. Rewriting it it terms of the vector fields Z and T, using (2.13),

$$
\begin{aligned}
0=&Z\left(f(L\psi)^2+f(\underline{L}\psi)^2-\frac{fD}{2r^2}\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2-\frac{fD}{2}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right)+T\left(-f\left(L\psi\right)^2+f(\underline{L}\psi)^2\right)\\
&-\frac{df}{dr}D\left((L\psi)^2+(\underline{L}\psi)^2\right)+\frac{D}{2}\frac{d}{dr}\left(\frac{fD}{r^2}\right)\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2+\frac{D}{2}\frac{d}{dr}\left(fD\left[\mu^2+\frac{D'}{r}\right]\right)\psi^2+\not{\nabla}_{\mathbb{S}^2}\left(\frac{fD}{r^2}Z\psi\not{\nabla}_{\mathbb{S}^2}\psi\right).
\end{aligned}
\tag{4.21}
$$

The same way, we can also write the equation in terms of the null vector fields $\underline{L}$ and L,

$$
\begin{aligned}
0=&\underline{L}\left(-2f\left(L\psi\right)^2+\frac{fD}{2r^2}\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2+\frac{fD}{2}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right)\\
&+L\left(2f\left(\underline{L}\psi\right)^2-\frac{fD}{2r^2}\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2-\frac{fD}{2}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right)\\
&-\frac{df}{dr}D\left((L\psi)^2+(\underline{L}\psi)^2\right)+\frac{D}{2}\frac{d}{dr}\left(\frac{fD}{r^2}\right)\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2\\
&+\frac{D}{2}\frac{d}{dr}\left(fD\left[\mu^2+\frac{D'}{r}\right]\right)\psi^2+\not{\nabla}_{\mathbb{S}^2}\left(\frac{fD}{r^2}Z\psi\not{\nabla}_{\mathbb{S}^2}\psi\right).
\end{aligned}
\tag{4.22}
$$

We take $f\equiv-1$, so $f'=0$, then,

$$
\begin{aligned}
0=&\underline{L}\left(2\left(L\psi\right)^2-\frac{D}{2r^2}\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2-\frac{D}{2}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right)\\
&+L\left(-2\left(\underline{L}\psi\right)^2+\frac{D}{2r^2}\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2+\frac{D}{2}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right)\\
&-\frac{D}{2}\frac{d}{dr}\left(\frac{D}{r^2}\right)\left|\not{\nabla}_{\mathbb{S}^2}\psi\right|^2-\frac{D}{2}\frac{d}{dr}\left(D\left[\mu^2+\frac{D'}{r}\right]\right)\psi^2-\not{\nabla}_{\mathbb{S}^2}\left(\frac{D}{r^2}Z\psi\not{\nabla}_{\mathbb{S}^2}\psi\right).
\end{aligned}
\tag{4.23}
$$

The last term on the right hand side corresponds to divergences with respect to $\mathbb{S}^2$, which vanish after integration over $\mathbb{S}^2$, via the divergence theorem.

Now we integrate this equation over $\{r\geq\tilde{R}\}$, with $R<\tilde{R}<r_c$. We are interested in finding a relation for the energies of $\check{\phi}_\ell$ and ϕ for $\{r\geq\tilde{R}\}$ because, using Proposition 10, we would be

able to extend that relation to $\{r \leq \tilde{R}\}$. Our integration surface is:

$$\mathcal{Q}_{\tau_1}^{\tau_2} := \bigcup_{\tau_1 \leq \tau \leq \tau_2} \Sigma_\tau \cap \{\tilde{R} \leq r \leq r_c\}.$$

It is represented in Fig. 4.3. It is important to note that the hypersurface $\{r = \tilde{R}\}$ is timelike, not null, as $\{r = r_c\}$ is. Furthermore,

$$\mathcal{R}_{\tau_1}^{\tau_2} = \mathcal{P}_{\tau_1}^{\tau_2} \cup \mathcal{Q}_{\tau_1}^{\tau_2}.$$

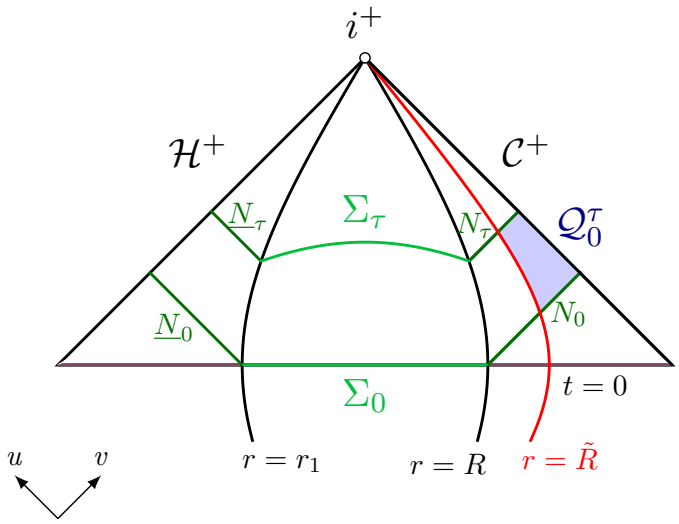


FIGURE 4.3: A 2-dimensional representation of the spacetime subset $\mathcal{R}$ of the RNdS spacetime, for values of $\tau \geq 0$. Two of the spacelike Cauchy hypersurfaces Σ_τ are represented by a green line, for a general value τ and for $\tau = 0$. The black lines are timelike hypersurfaces of constant r, one for each of the values close to the horizon that we have chosen, r_1 and R, and in red we have the timelike hypersurfaces of $\{r = \tilde{R}\}$. The blue area is the integration surface $\mathcal{Q}_0^\tau$. Each point in the picture represents a round 2-sphere of radius r.

We integrate equation (4.23) by parts over $d\sigma\, du\, dv$ on our new integration surface $\mathcal{Q}_{\tau_1}^{\tau_2}$. Since the hypersurfaces of constant τ, Σ_τ, are outgoing null hypersurfaces in the region we are integrating on, N_{τ_1} and N_{τ_2}, only the terms in $\underline{L}(\cdots)$ will contribute along those hypersurfaces.

$$\begin{aligned}
0 = &\int_{N_{\tau_2} \cap \{r \geq \tilde{R}\}} \left[2\left(L\psi\right)^2 - \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 - \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, dv \\
&- \int_{N_{\tau_1} \cap \{r \geq \tilde{R}\}} \left[2\left(L\psi\right)^2 - \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 - \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right] d\sigma\, dv \\
&+ \int_{v=\tau_1+\tilde{R}}^{v=\tau_2+\tilde{R}} \left[2\left(L\psi\right)^2 - \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 - \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right]\Big|_{r=\tilde{R}} d\sigma\, dv \\
&- \int_{u=\tau_1-\tilde{R}}^{u=\tau_2-\tilde{R}} \left[-2\left(\underline{L}\psi\right)^2 + \frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\psi^2\right]\Big|_{r=\tilde{R}} d\sigma\, du
\end{aligned}$$

$$
\begin{aligned}
&-\int_{\tau_1}^{\tau_2}\int_{\tilde{R}}^{r_c}\frac{D}{2}\left[\left(\frac{D'}{r^2}-\frac{2D}{r^3}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\left(D'\mu^2+\frac{(D')^2+DD'}{r}-\frac{D'}{r^2}\right)\psi^2\right]d\sigma\,du\,dv\\
&-\int_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}}2\left(\underline{L}\psi\right)^2 d\sigma\,du.
\end{aligned}
$$

where we used that $D(r_c)=0$. In this region, it is better to work in Bondi coordinates (u,r,θ,φ). We change coordinates using (2.9), then $du=du$, and $dv=du+\frac{2}{D}dr$, and the vector field Y defined in (2.12). For the bulk term, $du\,dv\equiv\frac{2}{D}du\,dr$. Then,

$$
\begin{aligned}
&\int_{\tau_1}^{\tau_2}\left[2\left(L\psi\right)^2+2\left(\underline{L}\psi\right)^2-\frac{D}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2-D\left(\mu^2+\frac{D'}{r}\right)\psi^2\right]\Big|_{r=\tilde{R}}d\sigma\,du\\
&-\int_{\tau_1}^{\tau_2}\int_{\tilde{R}}^{r_c}\left[\left(\frac{D'}{r^2}-\frac{2D}{r^3}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\left(D'\mu^2+\frac{(D')^2+DD'}{r}-\frac{D'}{r^2}\right)\psi^2\right]d\sigma\,du\,dr\\
&\qquad=-\int_{N_{\tau_2}\cap\{r\ge\tilde{R}\}}\left[D\left(Y\psi\right)^2-\frac{1}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2-\left(\mu^2+\frac{D'}{r}\right)\psi^2\right]d\sigma\,dr\\
&+\int_{N_{\tau_1}\cap\{r\ge\tilde{R}\}}\left[D\left(Y\psi\right)^2-\frac{1}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2-\left(\mu^2+\frac{D'}{r}\right)\psi^2\right]d\sigma\,dr-\int_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}}2\left(\underline{L}\psi\right)^2 d\sigma\,du.
\end{aligned}
\tag{4.24}
$$

As we have done before in Proposition 10, the bulk term has to be bounded by the initial energy, $E[\psi](\tau_1)$. In order to get this integrated energy estimate, we will try to bound the terms in this previous equation by the initial energy defined in (3.4). We start with the integral over N_{τ_2}, in double null coordinates,

$$
\begin{aligned}
&\int_{N_{\tau_2}\cap\{r\ge\tilde{R}\}}\left[2\left(L\psi\right)^2-\frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2-\frac{D}{2}\left(\mu^2+\frac{D'}{r}\right)\psi^2\right]d\sigma\,dv\\
&\le C\int_{N_{\tau_2}\cap\{r\ge\tilde{R}\}}\left[2(L\psi)^2+\frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\frac{D}{2}\left(\mu^2+\frac{D'}{r}\right)\psi^2\right]d\sigma\,dv\\
&\le C\int_{N_{\tau_2}}\left[2(L\psi)^2+\frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\frac{D}{2}\left(\mu^2+\frac{D'}{r}\right)\psi^2\right]d\sigma\,dv\\
&\le C\,E[\psi](\tau_2)\ \le C\,E[\psi](\tau_1).
\end{aligned}
$$

where we used the definition (3.4) and the energy boundedness (3.6). The same holds for the integral over N_{τ_1},

$$
\int_{N_{\tau_1}\cap\{r\ge\tilde{R}\}}\left[2\left(L\psi\right)^2-\frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2-\frac{D}{2}\left(\mu^2+\frac{D'}{r}\right)\psi^2\right]d\sigma\,dv\le C\,E[\psi](\tau_1).
$$

For the integral over $\mathcal{C}^+$ in (4.24), we can use the energy boundedness relation (3.5),

$$\int\limits_{\mathcal{C}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} 2\,(\underline{L}\psi)^2\,d\sigma\,du \leq C\,E[\psi](\tau_1) \leq C\,E[\psi](\tau_1).$$

This way we got a integrated energy boundedness condition for the region $\mathcal{Q}^{\tau_2}_{\tau_1}$:

$$\begin{aligned}&\int\limits_{\tau_1}^{\tau_2}\left[2\,(L\psi)^2+2\,(\underline{L}\psi)^2-\frac{D}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2-D\left(\mu^2+\frac{D'}{r}\right)\psi^2\right]\Big|_{r=\tilde{R}}d\sigma\,du\\ &-\int\limits_{\tau_1}^{\tau_2}\int\limits_{\tilde{R}}^{r_c}\left[\left(\frac{D'}{r^2}-\frac{2D}{r^3}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\left(D'\mu^2+\frac{(D')^2+DD'}{r}-\frac{D'}{r^2}\right)\psi^2\right]d\sigma\,du\,dr\\ &\qquad\leq C\,E[\psi](\tau_1).\end{aligned}\tag{4.25}$$

We will now restrict to $\psi=\psi_\ell$, for finite $\ell\leq L_0$, and observe that the spacetime integral on the left-hand side of (4.25) is non-negative definite, using the relation (2.15) in combination with $\ell(\ell+1)+\mu^2>0$.

$$\begin{aligned}&\int\limits_{\tau_1}^{\tau_2}2\left[(L\psi_\ell)^2+(\underline{L}\psi_\ell)^2-\frac{D}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi_\ell\right|^2-\frac{D}{2}\left(\mu^2+\frac{D'}{r}\right)\psi_\ell^2\right]\Big|_{r=\tilde{R}}d\sigma\,du\\ &-\int\limits_{\tau_1}^{\tau_2}\int\limits_{\tilde{R}}^{r_c}\left[\left(\frac{D'}{r^2}-\frac{2D}{r^3}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi_\ell\right|^2+\left(D'\mu^2+\frac{(D')^2+DD'}{r}-\frac{D'}{r^2}\right)\psi_\ell^2\right]d\sigma\,du\,dr\\ &\geq\int\limits_{\tau_1}^{\tau_2}2\left[(L\psi_\ell)^2+(\underline{L}\psi_\ell)^2-\left(\frac{\ell(\ell+1)D}{2r^2}+\frac{D}{2}\left(\mu^2+\frac{D'}{r}\right)\right)\psi_\ell^2\right]\Big|_{r=\tilde{R}}d\sigma\,du\\ &-\int\limits_{\tau_1}^{\tau_2}\int\limits_{\tilde{R}}^{r_c}\left(D'\mu^2+\frac{(D')^2+DD'}{r}+\frac{\ell(\ell+1)D'-D'}{r^2}-\frac{2\ell(\ell+1)D}{r^3}\right)\psi_\ell^2\,d\sigma\,du\,dr.\end{aligned}\tag{4.26}$$

Inside the boundary term $\{r=\tilde{R}\}$, we want the term ψ_ℓ^2 to be absorbed by $(L\psi_\ell)^2+(\underline{L}\psi_\ell)^2$. We rewrite the integrand by considering separately each ψ_ℓ. Using $\psi_\ell=r\phi_\ell=rw_\ell\check{\phi}_\ell$,

$$\begin{aligned}(L\psi_\ell)^2+(\underline{L}\psi_\ell)^2&=\left(L\left(rw_\ell\check{\phi}_\ell\right)\right)^2+\left(\underline{L}\left(rw_\ell\check{\phi}_\ell\right)\right)^2\\ &=\left(\frac{D}{2}\frac{d(rw_\ell)}{dr}\check{\phi}_\ell+rw_\ell\,L\check{\phi}_\ell\right)^2+\left(-\frac{D}{2}\frac{d(rw_\ell)}{dr}\check{\phi}_\ell+rw_\ell\,\underline{L}\check{\phi}_\ell\right)^2\end{aligned}$$

where we used that, for a surface of constant t, the null vector fields take the form $L=\frac{D}{2}\partial_r$ and $\underline{L}=-\frac{D}{2}\partial_r$. Consider the following,

$$\begin{aligned}L\check{\psi}_\ell&=a+b, & (L\check{\psi}_\ell)^2&=a^2+b^2+2ab,\\ \underline{L}\check{\psi}_\ell&=-a+c, & (\underline{L}\check{\psi}_\ell)^2&=a^2+c^2-2ac.\end{aligned}$$

By Young's inequality (2.17) with $\epsilon > 0$ arbitrarily small,

$$2ab \geq -\epsilon a^2 - \frac{1}{\epsilon} b^2,$$
$$-2ac \geq -\epsilon a^2 - \frac{1}{\epsilon} c^2.$$

Then,

$$(L\psi_\ell)^2 + (\underline{L}\psi_\ell)^2 \geq 2(1-\epsilon)a^2 + \left(1 - \frac{1}{\epsilon}\right) b^2 + \left(1 - \frac{1}{\epsilon}\right) c^2.$$

Substituting a, b and c,

$$(L\psi)^2 + (\underline{L}\psi)^2 \geq \frac{1-\epsilon}{2} D^2 \left(\frac{d(rw_\ell)}{dr}\right)^2 \frac{\psi_\ell^2}{r^2 w_\ell^2} + \left(1 - \frac{1}{\epsilon}\right) r^2 w_\ell^2 \left[\left(L\check{\phi}_\ell\right)^2 + \left(\underline{L}\check{\phi}_\ell\right)^2\right].$$

We are in the area close to r_c, so the stationary function $w_\ell(r)$ takes the form (2.51), its derivative is (4.6), and take the expression for $D(r)$ in (2.3),

$$\frac{D^2}{r^2 w_\ell^2} \left(\frac{d(rw_\ell)}{dr}\right)^2 \Bigg|_{r \uparrow r_c} \approx \frac{1}{\log^2(r_c - r)} + O(1).$$

Then we find:

$$(L\psi)^2 + (\underline{L}\psi)^2 \geq \frac{1-\epsilon}{2} \frac{C}{\log^2(r_c - r)} \psi_\ell^2 + \left(1 - \frac{1}{\epsilon}\right) r^2 w_\ell^2 \left[\left(L\check{\phi}_\ell\right)^2 + \left(\underline{L}\check{\phi}_\ell\right)^2\right].$$

for a constant $C > 0$. Inserting this back into the boundary term of the right-hand side of (4.26):

$$\begin{aligned}
&\int_{\tau_1}^{\tau_2} 2\left[(L\psi_\ell)^2 + (\underline{L}\psi_\ell)^2 - \left(\frac{\ell(\ell+1)D}{2r^2} + \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\right)\psi_\ell^2\right]\Bigg|_{r=\tilde{R}} d\sigma\, du \\
&\geq \int_{\tau_1}^{\tau_2} 2\left[\frac{1-\epsilon}{2}\frac{C}{\log^2(r_c - r)} - \frac{\ell(\ell+1)D}{2r^2} - \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\right]\psi_\ell^2\Big|_{r=\tilde{R}} d\sigma\, du \qquad (4.27) \\
&\quad + 2\left(1 - \frac{1}{\epsilon}\right) \int_{\tau_1}^{\tau_2} r^2 w_\ell^2 \left[\left(L\check{\phi}_\ell\right)^2 + \left(\underline{L}\check{\phi}_\ell\right)^2\right]\Bigg|_{r=\tilde{R}} d\sigma\, du.
\end{aligned}$$

This equation implies that, in order to absorb the term ψ_ℓ^2 of (4.26) by $(L\psi_\ell)^2 + (\underline{L}\psi_\ell)^2$, we have to compare:

$$\frac{C}{\log^2(r_c - \tilde{R})} \quad \text{v.s.} \quad \begin{cases} \ell(\ell+1)D(\tilde{R}), \\ \left(\mu^2 + \dfrac{D'}{r}\right) D(\tilde{R}). \end{cases}$$

The function $D(r)$ is linear in $(r_c - r)$. Consider the change of variable $z = (r_c - r)$. For $z \in \mathcal{R}$ satisfying $0 < z < 2$, the expansion of the logarithm as a polynomial series is (Olver, 1974):

$$\log z = \sum_{n=1}^{\infty} \left[\frac{(-1)^{n+1}}{n}(z-1)^n\right].$$

The leading order of the function $(\log z)^{-2}$ is then:

$$\frac{1}{\log^2 z} = \frac{C}{(z-1)^2} + O\left((z-1)^{-4}\right).$$

This implies that, for a small value of $z = r_c - \tilde{R}$, there exits a value $z_0 > 1$ such that

$$\frac{1}{\log^2 z} > C\, z,$$

for all $z < z_0$.

We can represent graphically the functions $(\log z)^{-2}$ and z in Fig. 4.4. We see that the

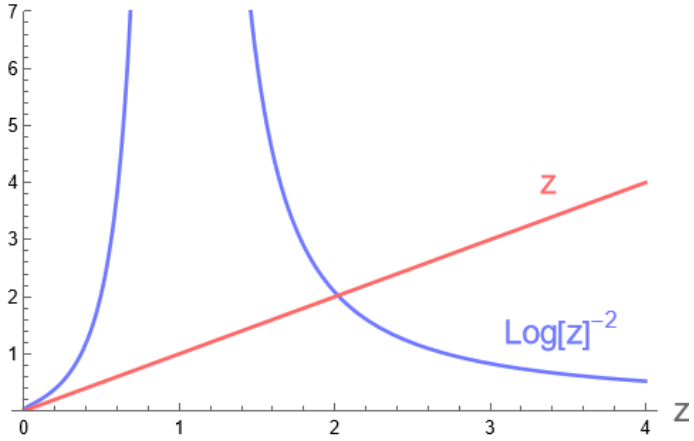


FIGURE 4.4: Plot of the functions $(\log z)^{-2}$, in blue, and z, in red, for an inertal of obtained through the use of the software `Wolfram Mathematica`.

function $(\log z)^{-2}$ goes to infinity at a value of $z = 1$. It is bigger than the function z for a small value of $z = r_c - \tilde{R}$. We can obtain this value numerically,

$$\frac{1}{\log^2 (r_c - \tilde{R})} > (r_c - \tilde{R}) \qquad \text{when} \quad (r_c - \tilde{R}) < 2.02075\,.$$

which agrees with value we got from the expansion.

We also want $(1-\epsilon)/2$ to be positive. It can be done by fixing the constant ϵ, e.g. taking $\epsilon = 1/2$. Then, for $(r_c - \tilde{R}) < 2$,

$$\int_{\tau_1}^{\tau_2} 2\left[\frac{1-\epsilon}{2}\frac{C}{\log^2 (r_c - r)} - \frac{\ell(\ell+1)D}{2r^2} - \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\right]\psi_\ell^2\Big|_{r=\tilde{R}}\, d\sigma\, du > 0.$$

Considering this positivity and equation (4.27),

$$\int_{\tau_1}^{\tau_2} 2\left[\frac{1-\epsilon}{2}\frac{C}{\log^2 (r_c - r)} - \frac{\ell(\ell+1)D}{2r^2} - \frac{D}{2}\left(\mu^2 + \frac{D'}{r}\right)\right]\psi_\ell^2\Big|_{r=\tilde{R}}\, d\sigma\, du$$

$$\leq 2\left(\frac{1}{\epsilon} - 1\right)\int_{\tau_1}^{\tau_2} r^2 w_\ell^2 \left[\left(L\check{\phi}_\ell\right)^2 + \left(\underline{L}\check{\phi}_\ell\right)^2\right]\Big|_{r=\tilde{R}}\, d\sigma\, du.$$

Combining the integrated energy boundedness condition for $\mathcal{Q}^{\tau_2}_{\tau_1}$, (4.25), with this previous inequality,

$$
\begin{aligned}
&-\int_{\tau_1}^{\tau_2}\int_{\tilde{R}}^{r_c}\left(D'\mu^2+\frac{(D')^2+DD'}{r}+\frac{\ell(\ell+1)D'-D'}{r^2}-\frac{2\ell(\ell+1)D}{r^3}\right)\psi_\ell^2\,d\sigma\,du\,dr\\
&\quad+\int_{\tau_1}^{\tau_2}2\left[\frac{1-\epsilon}{2}\frac{C}{\log^2(r_c-r)}-\frac{\ell(\ell+1)D}{2r^2}-\frac{D}{2}\left(\mu^2+\frac{D'}{r}\right)\right]\psi_\ell^2\Big|_{r=\tilde{R}}\,d\sigma\,du\\
&\qquad\le C\,E[\psi](\tau_1)+\frac{C}{\epsilon}\int_{\tau_1}^{\tau_2}r^2w_\ell^2\left[\left(L\check{\phi}_\ell\right)^2+\left(\underline{L}\check{\phi}_\ell\right)^2\right]\Big|_{r=\tilde{R}}\,d\sigma\,du.
\end{aligned}
$$

From the integrated energy estimate for $\check{\phi}_\ell$ (4.10), we already had control over the term $\left(L\check{\phi}_\ell\right)^2+\left(\underline{L}\check{\phi}_\ell\right)^2$, which is bounded by $\check{E}\left[\check{\phi}_\ell\right](\tau_1)$, and with (4.19) we find that is also bounded by the energy of ϕ_ℓ.

$$
\int_{\tau_1}^{\tau_2}2r^{-q}\left(\left(\underline{L}\check{\phi}_\ell\right)^2+\left(L\check{\phi}_\ell\right)^2\right)\Big|_{r=\tilde{R}}\,d\sigma\,du\le C\,E\left[\phi_\ell\right](\tau_1).
$$

Finally, we find the integrated energy estimate for the region $\mathcal{Q}^{\tau_2}_{\tau_1}$,

$$
\begin{aligned}
&-\int_{\tau_1}^{\tau_2}\int_{\tilde{R}}^{r_c}\left(D'\mu^2+\frac{(D')^2+DD'}{r}+\frac{\ell(\ell+1)D'-D'}{r^2}-\frac{2\ell(\ell+1)D}{r^3}\right)\psi_\ell^2\,d\sigma\,du\,dr\\
&\quad+\int_{\tau_1}^{\tau_2}2\left[\frac{1-\epsilon}{2}\frac{C}{\log^2(r_c-r)}-\frac{\ell(\ell+1)D}{2r^2}-\frac{D}{2}\left(\mu^2+\frac{D'}{r}\right)\right]\psi_\ell^2\Big|_{r=\tilde{R}}\,d\sigma\,du\\
&\qquad\le C\,E\left[\phi_\ell\right](\tau_1).
\end{aligned}
\tag{4.28}
$$

The value of the stationary function $w_\ell(r)$ changes through $\mathcal{P}^{\tau_2}_{\tau_1}$, as we have seen, so we are going to consider each one of those areas. Consider the Hardy inequality,

$$
\int_a^b x^q f^2\,dx+\frac{2a^{q+1}}{q+1}f^2(a)\le\frac{4}{(q+1)^2}\int_a^b x^{q+2}\left(\frac{df}{dx}\right)^2\,dx+\frac{2b^{q+1}}{q+1}f^2(b).
$$

We are going to apply it for every region, for $f\equiv\check{\phi}_\ell$ and $q=0$.

(i) $\boldsymbol{\{R\le r\le\tilde{R}\}}$ Consider Bondi coordinates (u,r,θ,φ). Take $x\equiv r$.

$$
\begin{aligned}
\int_{\tau_1}^{\tau_2}\int_R^{\tilde{R}}\check{\phi}_\ell^2\,d\sigma\,du\,dr+\int_{\tau_1}^{\tau_2}\check{\phi}_\ell^2\Big|_{r=R}\,d\sigma\,du&\le C\int_{\tau_1}^{\tau_2}\int_R^{\tilde{R}}r^2\left|Y\check{\phi}_\ell\right|^2\,d\sigma\,du\,dr+C\int_{\tau_1}^{\tau_2}\check{\phi}_\ell^2\Big|_{r=\tilde{R}}\,d\sigma\,du\\
&\le C_{\tilde{R}}\int_{\tau_1}^{\tau_2}\int_{N_\tau}r^2\left|L\check{\phi}_\ell\right|^2\,d\sigma\,du\,dr+C_{\tilde{R}}\int_{\tau_1}^{\tau_2}\psi_\ell^2\Big|_{r=\tilde{R}}\,d\sigma\,du,
\end{aligned}
$$

where $C_{\tilde{R}} \approx C/(r_c - \tilde{R})^2$. This second inequality appears because there exists a constant $C > 0$ such that $w_\ell(r) \geq C$ in $r \leq \tilde{R}$, and since we are inside $\{R \leq r \leq \tilde{R}\}$, that implies $\left|\check{\phi}_\ell\right|^2 \sim |\psi_\ell|^2$.

Use (4.28) to bound the term $\{r = \tilde{R}\}$ by the initial energy, and the bulk term on the right-hand side can be bounded using (4.19). Then,

$$\int_{\tau_1}^{\tau_2}\int_{R}^{\tilde{R}} \check{\phi}_\ell^2 \, d\sigma \, du \, dr + \int_{\tau_1}^{\tau_2} \check{\phi}_\ell^2 \Big|_{r=R} \, d\sigma \, du \leq C \, E\left[\phi_\ell\right](\tau_1). \tag{4.29}$$

(ii) $\boldsymbol{\{r_1 \leq r \leq R\}}$ Consider coordinates (t, r, θ, φ). Take $x \equiv r$.

$$\int_{\tau_1}^{\tau_2}\int_{r_1}^{R} \check{\phi}_\ell^2 \, d\sigma \, dt \, dr + \int_{\tau_1}^{\tau_2} \check{\phi}_\ell^2 \Big|_{r=r_1} \, d\sigma \, dt \leq C \int_{\tau_1}^{\tau_2}\int_{r_1}^{R} r^2 \left|X\check{\phi}_\ell\right|^2 \, d\sigma \, dt \, dr + C \int_{\tau_1}^{\tau_2} \check{\phi}_\ell^2 \Big|_{r=R} \, d\sigma \, dt.$$

Use the previous result (4.29) to bound the term $\{r = R\}$ by the initial energy. For bounding the bulk term on the right-hand side, consider the integrated energy boundedness condition for the renormalized energy (4.10), and (4.19). Then,

$$\int_{\tau_1}^{\tau_2}\int_{r_1}^{R} \check{\phi}_\ell^2 \, d\sigma \, dt \, dr + \int_{\tau_1}^{\tau_2} \check{\phi}_\ell^2 \Big|_{r=r_1} \, d\sigma \, dt \leq C \, E\left[\phi_\ell\right](\tau_1). \tag{4.30}$$

(iii) $\boldsymbol{\{r_+ \leq r \leq r_1\}}$ Consider Eddington–Finkelstein coordinates (v, r, θ, φ). Take $x \equiv r - r_+$.

$$\begin{aligned}\int_{\tau_1}^{\tau_2}\int_{r_+}^{r_1} \check{\phi}_\ell^2 \, d\sigma \, dv \, dr + \int_{\tau_1}^{\tau_2} \check{\phi}_\ell^2 \Big|_{r=r_+} \, d\sigma \, dv &\leq C \int_{\tau_1}^{\tau_2}\int_{r_+}^{r_1} (r - r_+)^2 \left|\underline{Y}\check{\phi}_\ell\right|^2 \, d\sigma \, dv \, dr + C \int_{\tau_1}^{\tau_2} \check{\phi}_\ell^2 \Big|_{r=r_1} \, d\sigma \, dv \\ &\leq C \int_{\tau_1}^{\tau_2}\int_{r_+}^{r_1} \left|\underline{L}\check{\phi}_\ell\right|^2 \, d\sigma \, dv \, dr + C \int_{\tau_1}^{\tau_2} \check{\phi}_\ell^2 \Big|_{r=r_1} \, d\sigma \, dv.\end{aligned}$$

Use the previous result (4.30) to bound the term $\{r = R\}$ by the initial energy. For bounding the bulk term on the right-hand side, consider the integrated energy boundedness condition for the renormalized energy (4.10), and (4.19). Then,

$$\int_{\tau_1}^{\tau_2}\int_{r_+}^{r_1} \check{\phi}_\ell^2 \, d\sigma \, dv \, dr + \int_{\tau_1}^{\tau_2} \check{\phi}_\ell^2 \Big|_{r=r_+} \, d\sigma \, dv \leq C \, E\left[\phi_\ell\right](\tau_1).$$

All these results together with Proposition 10, give us the following estimate:

$$\begin{aligned}&\int_{\tau_1}^{\tau_2}\int_{r_+}^{r_1} \left[\left(Z\check{\phi}_\ell\right)^2 + \left(T\check{\phi}_\ell\right)^2 + \check{\phi}_\ell^2\right] d\sigma \, dv \, dr + \int_{\tau_1}^{\tau_2}\int_{r_1}^{R} \left[\left(Z\check{\phi}_\ell\right)^2 + \left(T\check{\phi}_\ell\right)^2 + \check{\phi}_\ell^2\right] d\sigma \, dt \, dr \\ &\qquad + \int_{\tau_1}^{\tau_2}\int_{R}^{\tilde{R}} \left[\left(Z\check{\phi}_\ell\right)^2 + \left(T\check{\phi}_\ell\right)^2 + \check{\phi}_\ell^2\right] d\sigma \, du \, dr \leq C \, E\left[\phi_\ell\right](\tau_1).\end{aligned} \tag{4.31}$$

Looking at the integrands on the left-hand side,

$$\left|T\check{\phi}_\ell\right|^2 = \left|T\left(\frac{\phi_\ell}{w_\ell}\right)\right|^2 = \frac{1}{w_\ell^2}\left|T\phi_\ell\right|^2,$$
$$\left|Z\check{\phi}_\ell\right|^2 = \left|Z\left(\frac{\phi_\ell}{w_\ell}\right)\right|^2 = \left|\frac{Z\phi_\ell}{w_\ell} - \frac{D}{w_\ell^2}\frac{dw_\ell}{dr}\phi_\ell\right|^2 \geq \frac{1-\epsilon}{w_\ell^2}\left|Z\phi_\ell\right|^2 - \left(1-\frac{1}{\epsilon}\right)\frac{1}{w_\ell^4}\left(\frac{dw_\ell}{dr}\right)^2\left|\phi_\ell\right|^2,$$

where we applied Young's inequality (2.17) with $\epsilon > 0$ arbitrarily small. Since they are all in $r \leq \tilde{R}$, there exists a constant $C > 0$ such that $w_\ell(r) \geq C$, then:

$$\left|\check{\phi}_\ell\right|^2 \sim \left|\phi_\ell\right|^2, \qquad \left|T\check{\phi}_\ell\right|^2 \sim \left|T\phi_\ell\right|^2, \qquad \left|Z\check{\phi}_\ell\right|^2 \geq C(1-\epsilon)\left|Z\phi_\ell\right|^2 - C\left(1-\frac{1}{\epsilon}\right)\left|\phi_\ell\right|^2.$$

Substituting back into (4.31),

$$\int_{\tau_1}^{\tau_2}\int_{r_+}^{r_1}\left[(Z\phi_\ell)^2 + (T\phi_\ell)^2 + \phi_\ell^2\right]d\sigma\, dv\, dr + \int_{\tau_1}^{\tau_2}\int_{r_1}^{R}\left[(Z\phi_\ell)^2 + (T\phi_\ell)^2 + \phi_\ell^2\right]d\sigma\, dt\, dr$$
$$+ \int_{\tau_1}^{\tau_2}\int_{R}^{\tilde{R}}\left[(Z\phi_\ell)^2 + (T\phi_\ell)^2 + \phi_\ell^2\right]d\sigma\, du\, dr \leq C\, E\left[\phi_\ell\right](\tau_1).$$

Sum ℓ for from $\ell = 0$ to $\ell = L_0$, and we arrive at the integrated energy estimate (4.20) for $\ell \leq L_0$. □

4.4 Integrated energy estimates for $\ell > L_0$

In this section, we consider the high angular frequency part $\phi_{>L_0}$, where $L_0 \in \mathbb{N}_0$ will be taken sufficiently large. Since L_0 is large, the term $\mu^2\phi$ from the Klein-Gordon equation (2.31) does not play an important role.

Proposition 12. *Let $\phi \in C^\infty(\mathcal{R}) \cap C^1(\hat{\mathcal{R}})$ be a solution to (2.31) in the spacetime $(\mathcal{M}_{ext}, g)$. Let $\tilde{R}$ be close to the cosmological horizon. Then there exists a suitably large $L_0 \in \mathbb{N}_0$ and a constant $C = C(M, Q, \Lambda, \mu, \tilde{R}, L_0)$ such that for all $0 \leq \tau_1 \leq \tau_2 \leq \infty$:*

$$\int_{\tau_1}^{\tau_2}\int_{r_+}^{r_1}\left[(Z\phi_{>L_0})^2 + (T\phi_{>L_0})^2 + \phi_{>L_0}^2\right]d\sigma\, dv\, dr + \int_{\tau_1}^{\tau_2}\int_{r_1}^{R}\left[(Z\phi_{>L_0})^2 + (T\phi_{>L_0})^2 + \phi_{>L_0}^2\right]d\sigma\, dt\, dr$$
$$+ \int_{\tau_1}^{\tau_2}\int_{R}^{\tilde{R}}\left[(Z\phi_{>L_0})^2 + (T_{>L_0})^2 + \phi_{>L_0}^2\right]d\sigma\, du\, dr \leq C\, E[\phi](\tau_1). \tag{4.32}$$

Proof. Let $f : [r_+, r_c] \to \mathbb{R}$ be a differentiable function, which will be specified later. We multiply both sides of the Klein-Gordon equation in terms of the function ψ (3.3) with $\frac{D}{2}\frac{df}{dr}\psi$ to obtain:

$$\begin{aligned} 0 =& \frac{D}{2}\frac{df}{dr}\psi\left[-4\underline{L}L\psi + \frac{D}{r^2}\not{\Delta}_{\mathbb{S}^2}\psi - D\left(\mu^2 + \frac{D'}{r}\right)\psi\right] \\ =& -2D\frac{df}{dr}\psi\underline{L}L\psi + \frac{D^2}{2r^2}\frac{df}{dr}\psi\not{\Delta}_{\mathbb{S}^2}\psi - \frac{D^2}{2}\frac{df}{dr}\left(\mu^2 + \frac{D'}{r}\right)\psi^2. \end{aligned}$$

Applying the Leibniz rule and the definitions of the vector fields $\underline{L}$ and L in (2.11),

$$\begin{aligned} 0 =& \underline{L}\left(-2\frac{df}{dr_*}\psi L\psi\right) - L\left(\frac{1}{2}\frac{d^2f}{dr_*^2}\psi^2\right) + 2\frac{df}{dr_*}\underline{L}\psi L\psi + \frac{1}{4}\frac{d^3f}{dr_*^3}\psi^2 \\ & - \frac{D}{2}\frac{df}{dr_*}\left(\frac{1}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \left[\mu^2 + \frac{D'}{r}\right]\psi^2\right) + \nabla\!\!\!/_{\mathbb{S}^2}\left(\frac{D}{2r^2}\frac{df}{dr_*}\psi\nabla\!\!\!/_{\mathbb{S}^2}\psi\right) \end{aligned}$$

Changing the vector fields to Z and T,

$$\begin{aligned} 0 =& Z\left(\frac{df}{dr_*}\psi L\psi - \frac{1}{4}\frac{d^2f}{dr_*^2}\psi^2\right) + T\left(-\frac{df}{dr_*}\psi L\psi - \frac{1}{4}\frac{d^2f}{dr_*^2}\psi^2\right) + 2\frac{df}{dr_*}\underline{L}\psi L\psi \\ & + \frac{1}{4}\frac{d^3f}{dr_*^3}\psi^2 - \frac{D}{2}\frac{df}{dr_*}\left(\frac{1}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \left[\mu^2 + \frac{D'}{r}\right]\psi^2\right) + \nabla\!\!\!/_{\mathbb{S}^2}\left(\frac{D}{2r^2}\frac{df}{dr_*}\psi\nabla\!\!\!/_{\mathbb{S}^2}\psi\right). \end{aligned}$$

Combine this equation with (4.21) by summing them,

$$\begin{aligned} 0 =& Z\left(f(L\psi)^2 + f(\underline{L}\psi)^2 - \frac{fD}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 - \frac{fD}{2}\left[\mu^2 + \frac{D'}{r}\right]\psi^2 + \frac{df}{dr_*}\psi L\psi - \frac{1}{4}\frac{d^2f}{dr_*^2}\psi^2\right) \\ & + T\left(-f\left(L\psi\right)^2 + f(\underline{L}\psi)^2 - \frac{df}{dr_*}\psi L\psi - \frac{1}{4}\frac{d^2f}{dr_*^2}\psi^2\right) - \frac{df}{dr_*}\left((L\psi)^2 + (\underline{L}\psi)^2\right) \\ & + \frac{1}{2}\frac{d}{dr_*}\left(\frac{fD}{r^2}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{1}{2}\frac{d}{dr_*}\left(fD\left[\mu^2 + \frac{D'}{r}\right]\right)\psi^2 + 2\frac{df}{dr_*}\underline{L}\psi L\psi + \frac{1}{4}\frac{d^3f}{dr_*^3}\psi^2 \\ & - \frac{D}{2}\frac{df}{dr_*}\left(\frac{1}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \left[\mu^2 + \frac{D'}{r}\right]\psi^2\right) + \nabla\!\!\!/_{\mathbb{S}^2}\left(\frac{D}{2r^2}\frac{df}{dr_*}\psi\nabla\!\!\!/_{\mathbb{S}^2}\psi + \frac{fD}{r^2}Z\psi\nabla\!\!\!/_{\mathbb{S}^2}\psi\right). \end{aligned}$$

The last term on the right hand side corresponds to divergences with respect to $\mathbb{S}^2$, which vanish after integration over $\mathbb{S}^2$, via the divergence theorem. Using the Leibniz rule,

$$\begin{aligned} 0 =& Z\left(f(L\psi)^2 + f(\underline{L}\psi)^2 - \frac{fD}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 - \frac{fD}{2}\left[\mu^2 + \frac{D'}{r}\right]\psi^2 + \frac{df}{dr_*}\psi L\psi - \frac{1}{4}\frac{d^2f}{dr_*^2}\psi^2\right) \\ & + T\left(-f\left(L\psi\right)^2 + f(\underline{L}\psi)^2 - \frac{df}{dr_*}\psi L\psi - \frac{1}{4}\frac{d^2f}{dr_*^2}\psi^2\right) - \frac{df}{dr_*}\left(Z\psi\right)^2 + \frac{1}{4}\frac{d^3f}{dr_*^3}\psi^2 \\ & + \frac{f}{2}\frac{d}{dr_*}\left(\frac{D}{r^2}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{f}{2}\frac{d}{dr_*}\left(D\left[\mu^2 + \frac{D'}{r}\right]\right)\psi^2 + \nabla\!\!\!/_{\mathbb{S}^2}\left(\frac{D}{2r^2}\frac{df}{dr_*}\psi\nabla\!\!\!/_{\mathbb{S}^2}\psi + \frac{fD}{r^2}Z\psi\nabla\!\!\!/_{\mathbb{S}^2}\psi\right). \end{aligned} \tag{4.33}$$

We need for all the terms to be positive, in order to have an proper estimate. For the term $(Z\psi)^2$ to be positive, we need:

$$\frac{df}{dr_*} \leq 0. \tag{4.34}$$

Note that in the term $\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2$ we find an equivalent to the defining equation for the photon sphere $\{r = r_{PS}\}$ (2.27),

$$\frac{d}{dr_*}\left(\frac{D(r)}{r^2}\right)\Big|_{r=r_{PS}} = \frac{r^2DD' - 2rD^2}{r^4}\Big|_{r=r_{PS}} = 0,$$

Since we are considering the angular momentum $\ell > L_0$ to be quite large, the term $\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2$ will absorb the terms ψ^2. However, for the value of the photon sphere, the term $\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2$ goes to zero. This implies that the terms ψ^2 have to remain positive on the photon sphere, giving us another condition for $f(r)$,

$$\left[\frac{1}{4}\frac{d^3f}{dr_*^3} + \frac{f}{2}\frac{d}{dr_*}\left(D\left[\mu^2 + \frac{D'}{r}\right]\right)\right]\Big|_{r=r_{PS}} \geq 0. \tag{4.35}$$

As we know, the function $D(r)$ is always positive in (r_+, r_c).We need to consider a function $f(r)$ so that (4.34) and (4.35) are fulfilled. Define the value $r^*_{PS} := r_*(r_{PS})$. Consider the function $f(r_*) = \arctan(r^*_{PS} + r_*)$, its derivative with respect to r_* is:

$$\frac{df}{dr_*} = \frac{-1}{1 + \left(r^*_{PS} + r_*\right)^2} \leq 0.$$

The condition (4.34) is fulfilled for all $r_+ \leq r \leq r_c$. The other derivatives are:

$$\frac{d^2f}{dr_*^2} = \frac{-2\left(r^*_{PS} + r_*\right)}{\left(1 + \left(r^*_{PS} + r_*\right)^2\right)^2}, \qquad \frac{d^3f}{dr_*^3} = \frac{2 - 6\left(r^*_{PS} + r_*\right)^2}{\left(1 + \left(r^*_{PS} + r_*\right)^2\right)^3}.$$

Evaluating on r^*_{PS}:

$$f(r^*_{PS}) = 0, \qquad \frac{d^3f}{dr_*^3}(r^*_{PS}) = 2.$$

Condition (4.35) is fulfilled, but only the term $\frac{d^3f}{dr_*^3}$ is non-negative on the photon sphere.

Applying the relation (2.15), we can estimate for $\phi_{>L_0}$, with L_0 suitable large:

$$\begin{aligned}\int_{\mathbb{S}^2} \frac{f}{2}\frac{d}{dr_*}\left(\frac{D}{r^2}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi_{>L_0}\right|^2 d\sigma + \int_{\mathbb{S}^2}\left[\frac{1}{4}\frac{d^3f}{dr_*^3} + \frac{f}{2}\frac{d}{dr_*}\left(D\left[\mu^2 + \frac{D'}{r}\right]\right)\right]\psi^2_{>L_0}d\sigma \\ \geq C\int_{\mathbb{S}^2}\psi^2_{>L_0}d\sigma.\end{aligned} \tag{4.36}$$

with the constant C depending on L_0.

Considering (4.33), and changing the vector fields X and T for the null vector fields $\underline{L}$ and L,

$$\begin{aligned}0 = \underline{L}\left(-2f\left(L\psi\right)^2 - 2\frac{df}{dr_*}\psi L\psi + \frac{fD}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{fD}{2}\left[\mu^2 + \frac{D'}{r}\right]\psi^2\right) \\ + L\left(2f(\underline{L}\psi)^2 - \frac{1}{2}\frac{d^2f}{dr_*^2}\psi^2 - \frac{fD}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 - \frac{fD}{2}\left[\mu^2 + \frac{D'}{r}\right]\psi^2\right) - \frac{df}{dr_*}\left(Z\psi\right)^2 + \frac{1}{4}\frac{d^3f}{dr_*^3}\psi^2 \\ + \frac{f}{2}\frac{d}{dr_*}\left(\frac{D}{r^2}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2 + \frac{f}{2}\frac{d}{dr_*}\left(D\left[\mu^2 + \frac{D'}{r}\right]\right)\psi^2 + \nabla\!\!\!/_{\mathbb{S}^2}\left(\frac{D}{2r^2}\frac{df}{dr_*}\psi\nabla\!\!\!/_{\mathbb{S}^2}\psi + \frac{fD}{r^2}Z\psi\nabla\!\!\!/_{\mathbb{S}^2}\psi\right).\end{aligned}$$

Integrating by parts over $d\sigma\, du\, dv$ on the integration surface $\mathcal{R}^{\tau_2}_{\tau_1}$,

$$
\begin{aligned}
0 = &\int\limits_{\Sigma_{\tau_2}\cap\{r_1\le r\le R\}} \left[-f\frac{4}{D}\left((L\psi)^2+(\underline{L}\psi)^2\right)+\frac{1}{D}\frac{d^2f}{dr_*^2}\psi^2-4\frac{df}{dr}\psi L\psi+\frac{2f}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2\right] d\sigma\, dr\\
&-\int\limits_{\Sigma_{\tau_1}\cap\{r_1\le r\le R\}} \left[-f\frac{4}{D}\left((L\psi)^2+(\underline{L}\psi)^2\right)+\frac{1}{D}\frac{d^2f}{dr_*^2}\psi^2-4\frac{df}{dr}\psi L\psi+\frac{2f}{r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2\right] d\sigma\, dr\\
&+\int\limits_{N_{\tau_2}} \left[-2f\,(L\psi)^2-2\frac{df}{dr_*}\psi L\psi+\frac{fD}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\frac{fD}{2}\left(\mu^2+\frac{D'}{r}\right)\psi^2\right] d\sigma\, dv\\
&-\int\limits_{N_{\tau_1}} \left[-2f\,(L\psi)^2-2\frac{df}{dr_*}\psi L\psi+\frac{fD}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\frac{fD}{2}\left(\mu^2+\frac{D'}{r}\right)\psi^2\right] d\sigma\, dv\\
&+\int\limits_{\underline{N}_{\tau_2}} \left[2f(\underline{L}\psi)^2-\frac{1}{2}\frac{d^2f}{dr_*^2}\psi^2-\frac{fD}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2-\frac{fD}{2}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right] d\sigma\, du\\
&-\int\limits_{\underline{N}_{\tau_1}} \left[2f(\underline{L}\psi)^2-\frac{1}{2}\frac{d^2f}{dr_*^2}\psi^2-\frac{fD}{2r^2}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2-\frac{fD}{2}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right] d\sigma\, du\\
&+\int\limits_{\mathcal{C}^+\cap\{\tau_1\le\tau\le\tau_2\}} \left[2f(\underline{L}\psi)^2\right] d\sigma\, du+\int\limits_{\mathcal{H}^+\cap\{\tau_1\le\tau\le\tau_2\}} \left[-2f\,(L\psi)^2\right] d\sigma\, dv\\
&+\int_{\tau_1}^{\tau_2}\int\limits_{\Sigma_\tau}\left[-\frac{df}{dr_*}(Z\psi)^2+\frac{1}{4}\frac{d^3f}{dr_*^3}\psi^2+\frac{f}{2}\frac{d}{dr_*}\left(\frac{D}{r^2}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\frac{f}{2}\frac{d}{dr_*}\left(D\left[\mu^2+\frac{D'}{r}\right]\right)\psi^2\right] d\sigma\, du\, dv,
\end{aligned}
$$

for $f(r_*)=\arctan(r^*_{PS}+r_*)$, where we used that $D(r_c)=0=D(r_+)$. For the terms $\Sigma_\tau\cap\{r_1\le r\le R\}$ the most convenient coordinate system is (t,r,θ,φ), so we applied a change of coordinates over the differential forms such that, $du=2dt-\frac{2}{D}dr$ and $dv=2dt+\frac{2}{D}dr$.

The boundary terms $\mathcal{C}^+$ and $\mathcal{H}^+$ can be estimated using (3.2). To have control over the spacetime integral of $(L\psi_{>L_0})^2+(\underline{L}\psi_{>L_0})^2$, we apply (4.22) with $f(r)=r^{-1}$,

$$
\begin{aligned}
0=&\underline{L}\left(-\frac{2}{r}(L\psi)^2+\frac{D}{2r^3}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\frac{D}{2r}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right)\\
&+L\left(\frac{2}{r}(\underline{L}\psi)^2-\frac{D}{2r^3}\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2-\frac{D}{2r}\left[\mu^2+\frac{D'}{r}\right]\psi^2\right)+\frac{D}{r^2}\left((L\psi)^2+(\underline{L}\psi)^2\right)\\
&+\frac{D}{2}\frac{d}{dr}\left(\frac{D}{r^3}\right)\left|\nabla\!\!\!/_{\mathbb{S}^2}\psi\right|^2+\frac{D}{2}\frac{d}{dr}\left(\frac{D}{f}\left[\mu^2+\frac{D'}{r}\right]\right)\psi^2+\nabla\!\!\!/_{\mathbb{S}^2}\left(\frac{D}{r^3}Z\psi\nabla\!\!\!/_{\mathbb{S}^2}\psi\right).
\end{aligned}
$$

Thus, we find the final result (4.32). □

Appendix A

Differential equations with regular singularities

In this appendix we will discuss some important aspects and definitions that we need on this thesis. More detailed explanations and further information on ODEs can be found in Chapter 5 of (Olver, 1974).

A.1 Existence theorems for Linear Differential Equations

Theorem A.1.1 (Fuchs, 1866)**.** *Let $f(z)$ and $g(z)$ be holomorphic in a simply connected domain Z. Then the equation*

$$\frac{d^2w}{dr^2} + f(z)\frac{dw}{dz} + g(z)w = 0. \tag{A.1}$$

has an infinity of solutions which are holomorphic in Z. If the values of w and dw/dz are prescribed at any point, then the solution is unique.[1]

Definition A.1.1 (Fundamental pair)**.** Let $w_1(z)$ and $w_2(z)$ be a pair of solutions of (A.1) with the property that any other solution can be expressed in the form

$$w(z) = Aw_1(z) + Bw_2(z),$$

where A and B are constants. Then $w_1(z)$ and $w_2(z)$ are said to comprise a fundamental pair.

We suppose that Z is a disk $|z-a| < r$, and that z_0 is a point of Z at which the values

$$a_0 = w(z_0), \qquad a_1 = \frac{dw}{dz}(z_0),$$

are prescribed. Let r be the distance of the nearest of the singularities of $f(z)$ and $g(z)$ from $z = z_0$, and

$$f(z) = \sum_{s=0}^{\infty} f_s(z-z_0)^s, \qquad g(z) = \sum_{s=0}^{\infty} g_s(z-z_0)^s,$$

be the Taylor expansions of $f(z)$ and $g(z)$ within $|z-z_0| < r$.

[1] The proof of this theorem can be found in pages 145-146 of (Olver, 1974).

Theorem A.1.1 shows that all holomorphic solutions of (A.1) can be expanded in series of the form

$$w(z) = \sum_{s=0}^{\infty} a_s (z - z_0)^s,$$

also convergent within $|z - z_0| < r$. Substituting in (A.1) and equating coefficients, we find that a_0 and a_1 may be prescribed arbitrarily; higher coefficients are then determined recursively by

$$\begin{aligned} -s(s-1)a_s = (s-1)f_0 a_{s-1} + (s-2)f_1 a_{s-2} + \cdots \\ + f_{s-2}a_1 + g_0 a_{s-2} + g_1 a_{s-3} + \cdots + g_{s-2}a_0 \qquad (s \geq 2). \end{aligned}$$

A.2 Nature of the solutions in the neighbourhood of a regular singularity

We can classify the singularities of our functions as:

- If the functions $f(z)$ and $g(z)$ are both analytic at $z = z_0$, then this point is said to be an *ordinary point* of the differential equation (A.1).
- If $z = z_0$ is not an ordinary point, but both $(z - z_0)f(z)$ and $(z - z_0)^2 g(z)$ are analytic there, then z_0 is said to be a *regular singularity*, or *singularity of the first kind*.
- Lastly, if z_0 is neither an ordinary point nor a regular singularity, then it is said to be an *irregular singularity*, or *singularity of the second kind*.

When the singularities of $f(z)$ and $g(z)$ at z_0 are no worse than poles, z_0 is said to be a *singularity of rank* $k - 1$, where k is the least integer such that both $(z - z_0)^k f(z)$ and $(z - z_0)^{2k} g(z)$ are analytic. Thus a regular singularity is of rank zero. If either $f(z)$ or $g(z)$ has an essential singularity at z_0, then the rank may be said to be infinite.

In A.1 we showed that, in the neighbourhood of an ordinary point, the differential equation has linearly independent pairs of holomorphic solutions. In the present section we construct convergent series solutions in the neighbourhood of a regular singularity.

Without loss of generality the regular singularity can be taken at the origin. Thus we assume that in a neighbourhood $|z| < r$ there exist convergent series expansions

$$zf(z) = \sum_{s=0}^{\infty} f_s z^s, \qquad z^2 g(z) = \sum_{s=0}^{\infty} g_s z^s, \tag{A.2}$$

where at least one of the coefficients f_0, g_0, and g_1 is non-zero.

The form that we may reasonably expect the solutions to take can be found by approximating $f(z)$ and $g(z)$ by means of the leading terms in (A.2); thus

$$\frac{d^2 w}{dr^2} + \frac{f_0}{z}\frac{dw}{dz} + \frac{g_0}{z^2} w = 0.$$

Exact solutions of this equation are given by $w = z^\alpha$ where a is a root of the quadratic equation

$$\alpha(\alpha - 1) + f_0 \alpha + g_0 = 0. \tag{A.3}$$

which we call *indicial equation*. The two possible values of α are called the *exponents* or *indices* of the singularity. Accordingly, as a possible solution of (A.1) we try the series

$$w(z) = z^{\alpha} \sum_{s=0}^{\infty} a_s z^s. \tag{A.4}$$

Substituting in the given differential equation by means of (A.2) and (A.4), and formally equating coefficients of $z^{\alpha+s-2}$, we derive

$$Q(\alpha+s)a_s = -\sum_{j=0}^{s-1} \{(\alpha+j)\, f_{s-j} + g_{s-j}\}\, a_j \qquad (s = 1, 2, \ldots) \tag{A.5}$$

where $Q(\alpha)$ denotes the left-hand side of (A.3). Equation (A.5) determines $a_1, a_2, \ldots$ recursively in terms of an arbitrarily assigned (non-zero) value of a_0. The procedure runs into difficulty if, and only if, $Q(\alpha+s)$ vanishes for a positive integer value of s.

Accordingly, when the roots of the indicial equation are distinct and do not differ by an integer, two series of the form (A.4) can be found that formally satisfy the differential equation. In other cases only one solution of this type is available, unless the right-hand side of (A.5) vanishes at the same value of the positive integer s for which $Q(\alpha+s) = 0$.

Theorem A.2.1 (Frobenius, 1873)**.** *With the notation and conditions of this section, the series* (A.4) *converges and defines a solution of the differential equation* (A.1) *when* $|z| < r$*, provided that the other exponent is not of the form* $\alpha + s$*, where* s *is a positive integer.*[2]

If α is a non-negative integer, then the solution with exponent α is analytic at $z = 0$. When α is a negative integer the solution has a pole, and when α is non-integral, there is a branch point. Again, provided that the exponent difference is not an integer, the theorem can be applied twice and the solutions obtained comprise a fundamental pair, at least one of which has a branch point at the singularity.

A.2.1 Second solution when the exponents differ by an integer or zero

Suppose that α and β are the roots of the indicial equation (A.3) and $\alpha - \beta = n$, where n is a positive integer or zero. Theorem A.2.1 furnishes the solution

$$w_1(z) = z^{\alpha} \sum_{s=0}^{\infty} a_s z^s. \tag{A.6}$$

To find an independent second solution we use a standard substitution for depressing the order of a differential equation having a known solution, given by

$$w_2(z) = w_1(z) v(z).$$

Inserting this solution into our equation (A.1), then

$$v''(z) + \left(2\frac{w_1'(z)}{w_1(z)} + f(z)\right) v'(z) = 0.$$

[2]Proof: pages 149-150 of (Olver, 1974).

Regarding this as a first-order differential equation in $v'(z)$, we obtain:

$$v(z) = \int \frac{1}{(w_1(z))^2} \exp\left[-\int f(z')\,dz'\right] dz.$$

Inserting the Taylor expansion of $f(z)$ (A.2) into this integral:

$$v(z) = \int \frac{1}{z^{2\alpha}(a_0 + a_1 + \cdots)^2} \exp\left[-f_0 \log z - f_1 z - \frac{1}{2} f_2 z^2 - \cdots\right] dz.$$

From the indicial equation (A.3) we have $f_0 = 1 - \alpha - \beta = 1 + n - 2\alpha$. Hence

$$v(z) = \int \frac{\phi(z)}{z^{n+1}}\,dz.$$

where $\phi(z)$ is analytic at $z = 0$. Let the Maclaurin expansion of $\phi(z)$ be denoted by

$$\phi(z) = \sum_{s=0}^{\infty} \phi_s z^s,$$

where the ϕ_s are expressible in terms of the a_s and f_s. In particular,

$$\phi_0 = \frac{1}{a_0^2}.$$

Integrating $z^{-n-1}\phi(z)$ and multiplying the result by $w_1(z)$, we obtain

$$w_2(z) = w_1(z)\left(-\sum_{s=0}^{n-1} \frac{\phi_s}{(n-s)z^{n-s}} + \phi_n \log z + \sum_{s=n+1}^{\infty} \frac{\phi_s z^{s-n}}{s-n}\right).$$

When $n = 0$, that is, when the exponents coincide, $w_2(z)$ has the form

$$w_2(z) = \phi_0 w_1(z) \log z + z^{\alpha+1} \sum_{s=0}^{\infty} b_s z^s. \tag{A.7}$$

Alternatively, when n is a positive integer, $w_2(z)$ takes the form

$$w_2(z) = \phi_n w_1(z) \log z + z^{\beta} \sum_{s=0}^{\infty} c_s z^s.$$

The leading coefficient in the last sum is

$$c_0 = -\frac{a_0 \phi_0}{n} = -\frac{1}{n a_0},$$

and is always non-zero.

Bibliography

Angelopoulos, Y., S. Aretakis, and D. Gajic (2023). "Price's law and precise late-time asymptotics for subextremal Reissner–Nordström black holes". In: *Annales Henri Poincaré* 24.9, pp. 3215–3287.

Costa, J.L. and P.M. Girão (2020). "Higher order linear stability and instability of Reissner–Nordström's Cauchy horizon". In: *Analysis and Mathematical Physics* 10.3, p. 40.

Dafermos, M. and I. Rodnianski (2007). "The wave equation on Schwarzschild-de Sitter spacetimes". In: *arXiv preprint arXiv:0709.2766*.

— (2013). "Lectures on black holes and linear waves". In: *Clay Math. Proc* 17, pp. 97–205.

Dias, O.J.C. et al. (2018). "Strong cosmic censorship in de Sitter space". In: *Physical Review D* 97.10, p. 104060.

Gajic, D. (2023). "Late-time asymptotics for geometric wave equations with inverse-square potentials". In: *Journal of Functional Analysis* 285.7, p. 110058.

Hawking, S.W. and G.F.R. Ellis (1973). *The large scale structure of space-time*. Cambridge university press.

José, J.V. and E.J. Saletan (1998). *Classical dynamics: a contemporary approach*. Cambridge University Press.

Klein, C. (2023). "The effect of quantum fields on black-hole interiors". PhD thesis. Dissertation, Leipzig, Universität Leipzig, 2023.

Landsman, K. (2021). *Foundations of General Relativity: From Einstein to Black Holes*. Radboud University Press.

Luk, J. and S.J. Oh (2019). "Strong cosmic censorship in spherical symmetry for two-ended asymptotically flat initial data I. The interior of the black hole region". In: *Annals of Mathematics* 190.1, pp. 1–111.

Mokdad, M. (2017). "Reissner–Nordstrøm–de Sitter manifold: photon sphere and maximal analytic extension". In: *Classical and Quantum Gravity* 34.17, p. 175014.

Olver, F.W.J. (1974). *Introduction to asymptotics and special functions, Acad*. Academic Press: New York, San Fransisco, London.

Sbierski, J. (2015). "Characterisation of the energy of Gaussian beams on Lorentzian manifolds: with applications to black hole spacetimes". In: *Analysis & PDE* 8.6, pp. 1379–1420.

Sogge, C. D. (2008). *Lectures on non-linear wave equations*. International Press Boston.

Wald, R.M. (2010). *General relativity*. University of Chicago Press.